Cubos de Dados de Observação da Terra e Análise de Séries Temporais de Imagens

Karine Reis Ferreira
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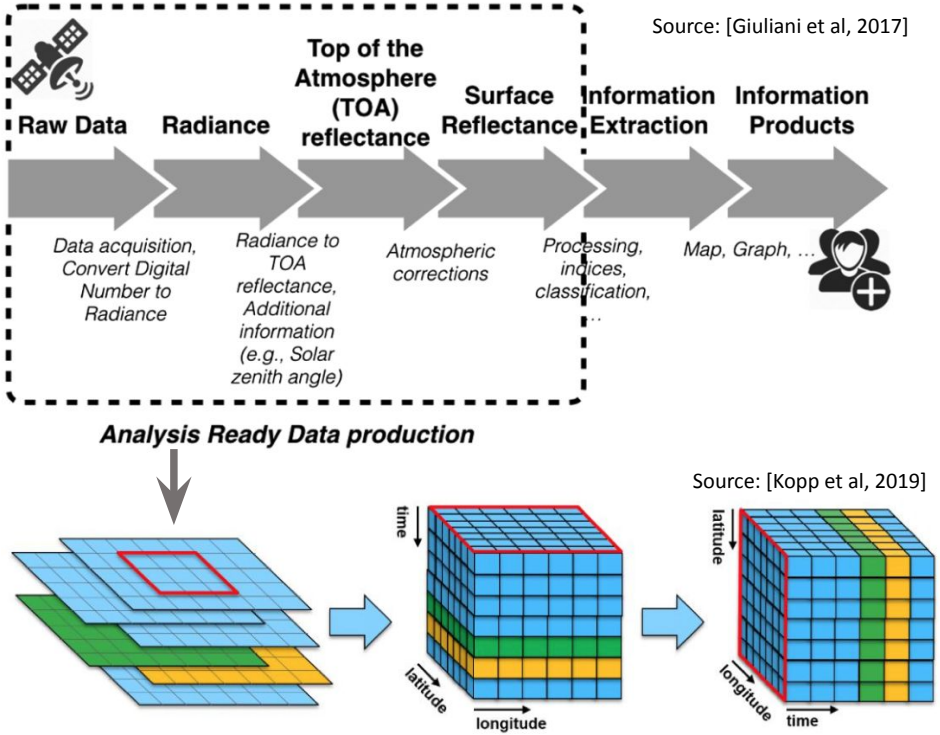
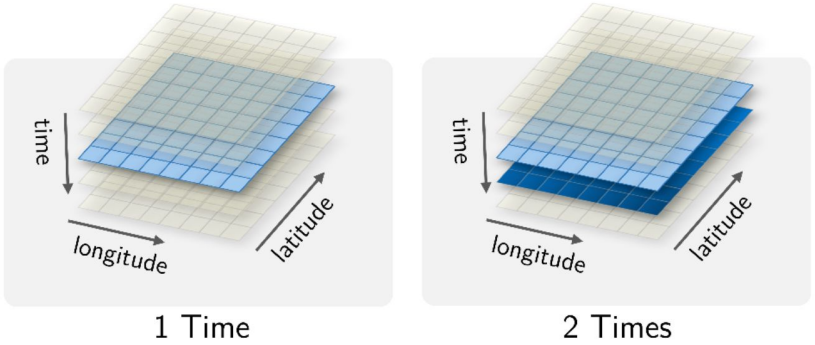
Brazilian National Institute for Space Research (INPE)

07/11/2026

What are data cubes?

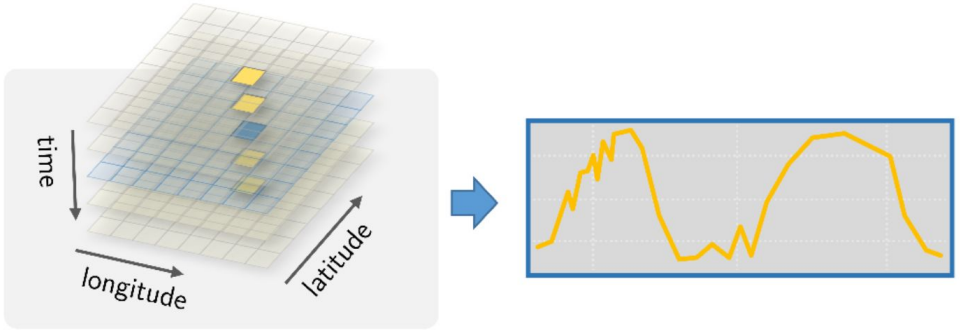


What are data cubes?



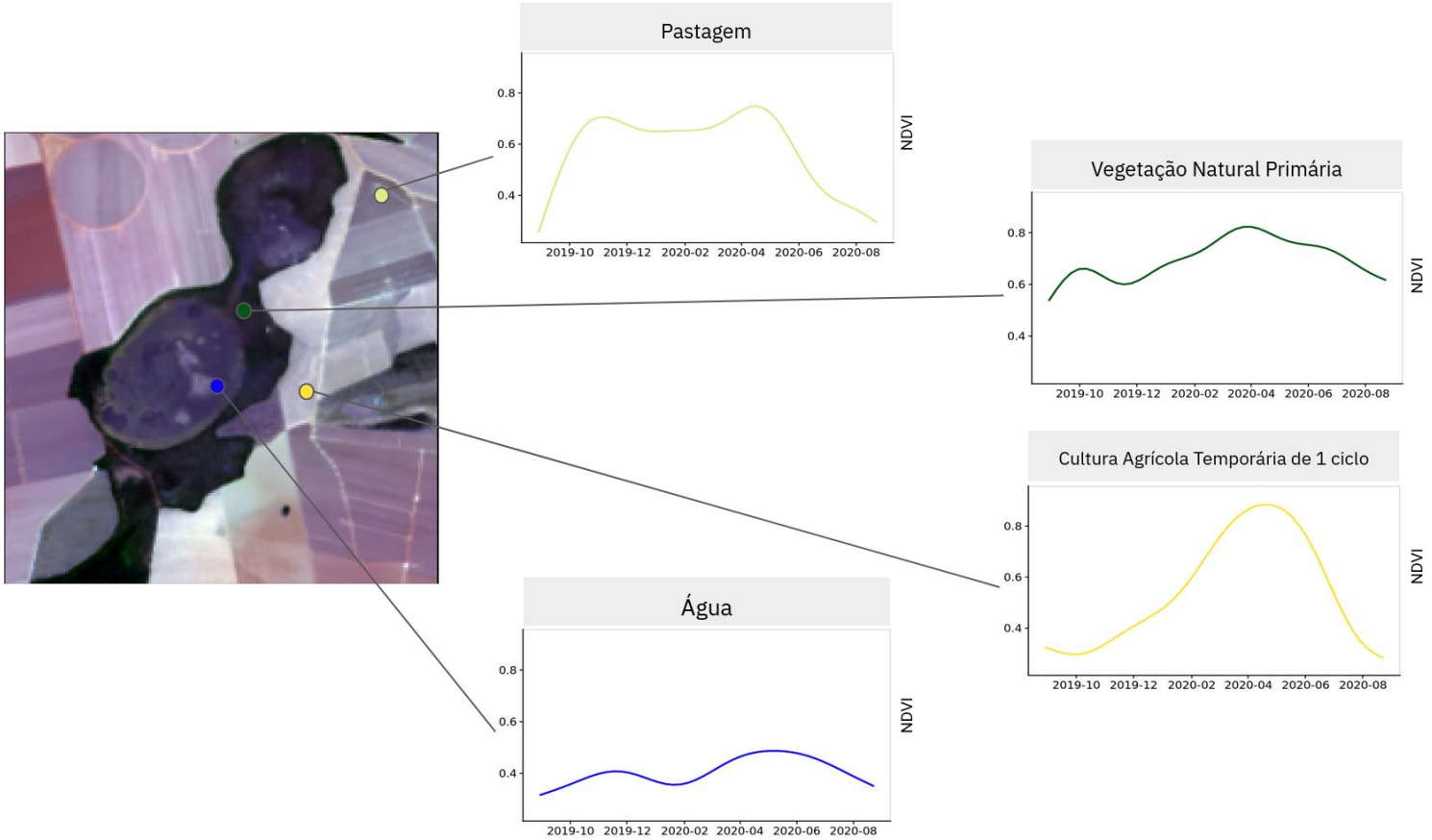
1 Time 2 Times

Image selection

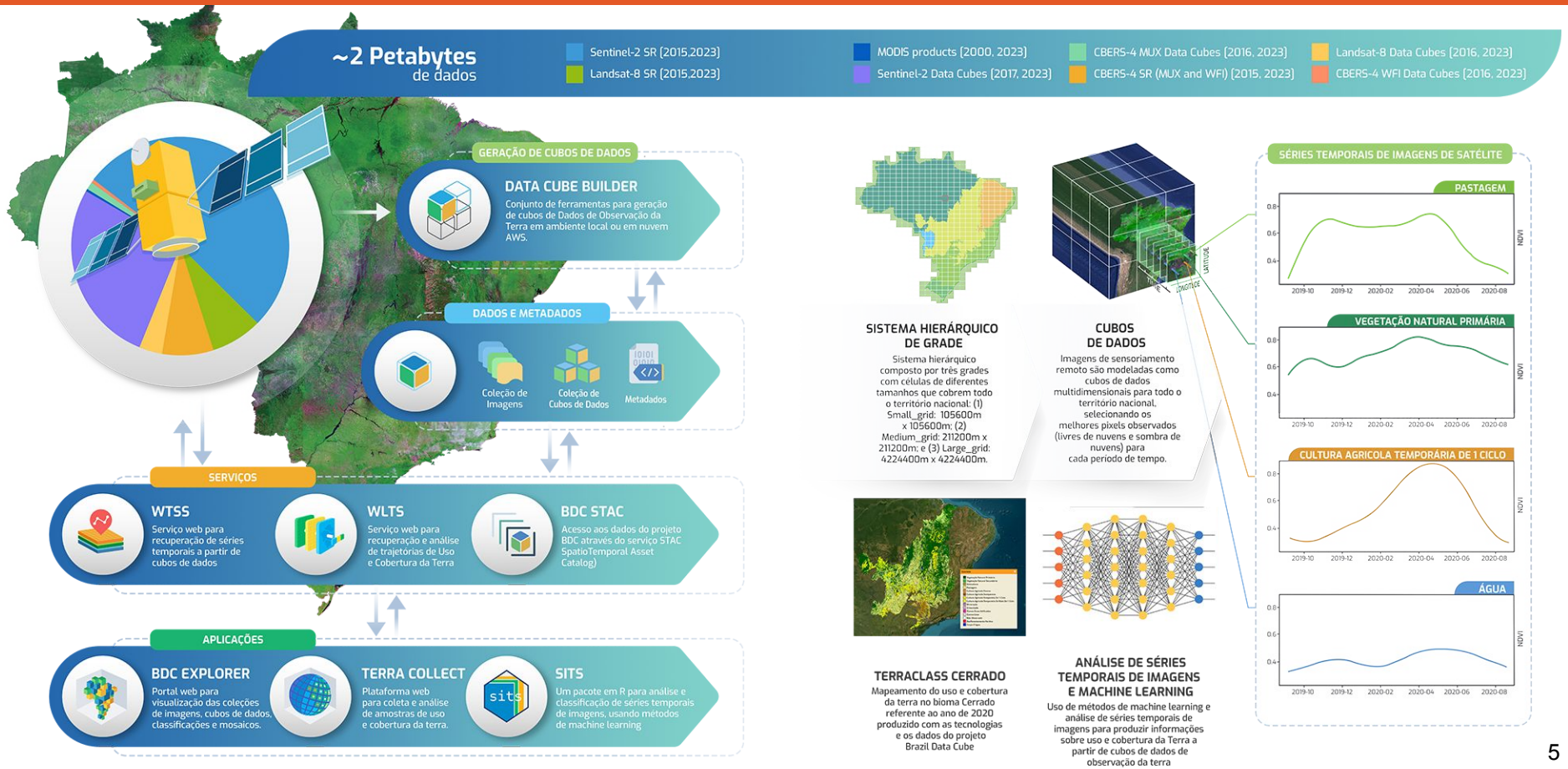


Time series extraction

What are data cubes?



Brazil Data Cube



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Dados Geoespaciais

Acervos de dados disponibilizados na BIG

Imagens

CBERS-4A
CBERS-4
CBERS-2B
CBERS-2
Amazonia-1
Sentinel-2
Sentinel-1
Sentinel-3
Landsat
GOES
Cubos de Dados

Meteorológicos

MERGE
SAMeT
Radiação Solar
Projeções de mudanças climáticas regionais

Topográficos

Topodata

Imagens

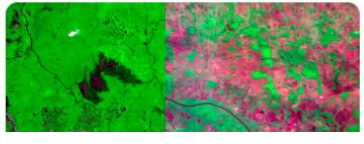
Satélite CBERS-4A

O satélite CBERS-4A (China-Brazil Earth Resources Satellite) foi lançado em 20 de dezembro de 2019. Esse satélite é equipado com três câmeras imageadoras, sendo um importante instrumento para monitoramento ambiental. A câmera Multiespectral e Pancromática de Ampla Varredura (Wide Swath Panchromatic and Multispectral – WPM) produz imagens pancromáticas com 2 metros de resolução espacial e multiespectrais de 8 metros. A câmera Multiespectral (Regular Multispectral Camera – MUX) produz imagens com resolução espacial de 16,5 metros. O tempo de revisita das câmeras WPM e MUX é de 31 dias. A... Read more

Satélite CBERS-4

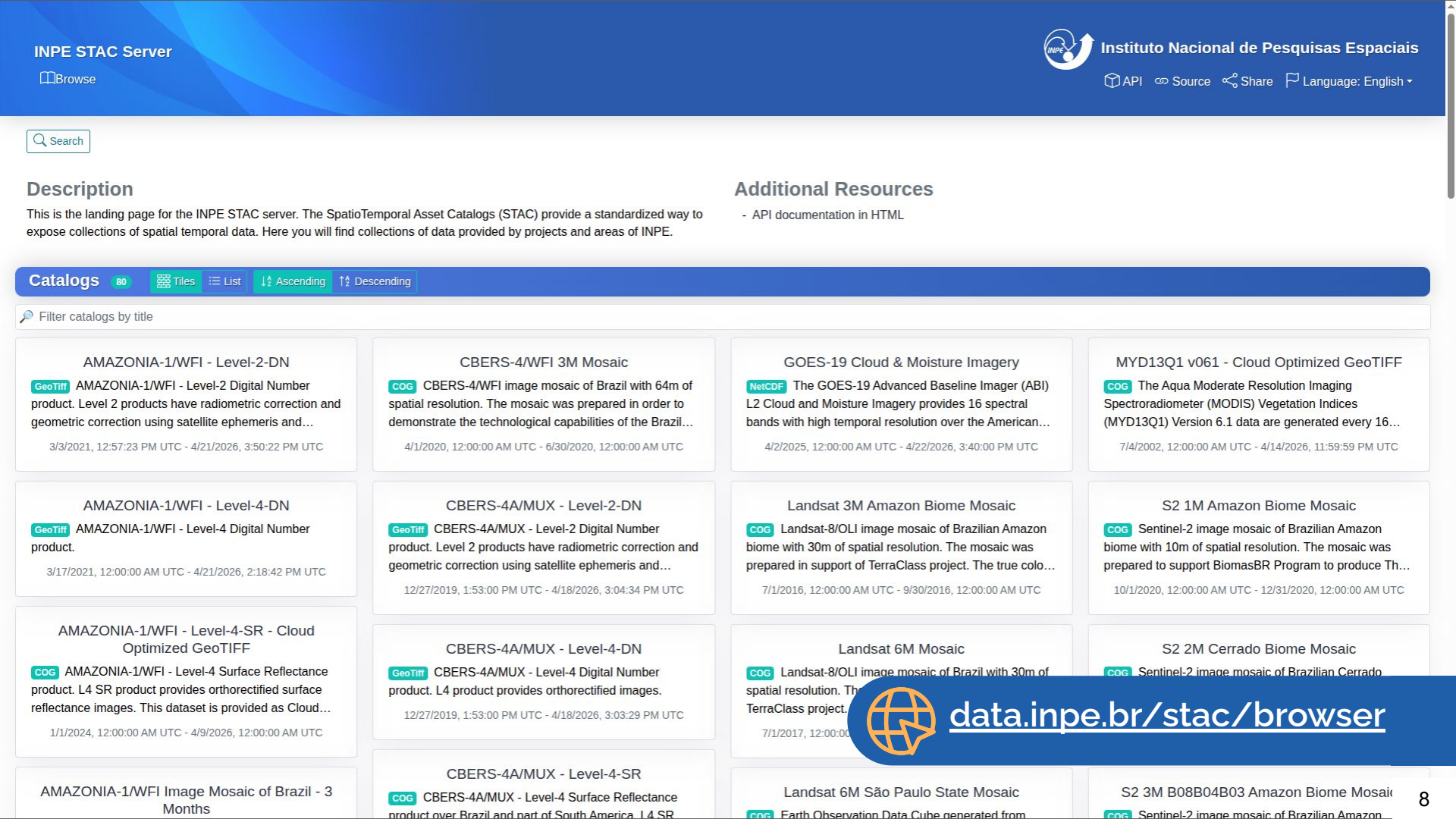
O satélite CBERS-4 (China-Brazil Earth Resources Satellite) foi lançado em 7 de dezembro de 2014. Esse satélite é equipado com três câmeras imageadoras, sendo um importante instrumento para monitoramento ambiental. A câmera Multiespectral e Pancromática de Ampla Varredura (Panchromatic and Multispectral Camera – PAN) produz imagens pancromáticas com 5 metros de resolução espacial e multiespectrais de 10 metros. A câmera Multiespectral (Regular Multispectral Camera – MUX) produz imagens com resolução espacial de 20 metros e uma taxa de revisita de até 26 dias. A terceira câmera, Imageadora de Campo Largo (Wide Field Imaging Camera – WFI), produz imagens... Read more

data.inpe.br/dados



Satélite CBERS-2B

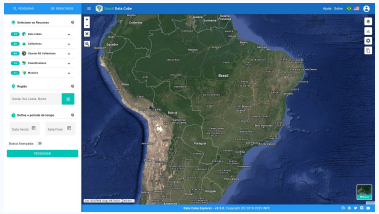
O satélite CBERS-2B foi lançado em 19 de setembro de 2007 tendo operado até 16 de abril de 2010. Foi o terceiro satélite do Programa CBERS (China-Brazil Earth Resources Satellite ou Satélite Sino-Brasileiro de Recursos Terrestres), em cooperação com a China. Esse satélite continha três câmeras imageadoras. A câmera CCD (Charged-Coupled Device) produziu imagens com 20 metros de resolução espacial com uma revisita de 26 dias. A câmera WFI (Wide Field Imaging Camera – Imageadora de Campo Largo) produziu imagens com 260 metros de resolução espacial, com características idênticas a do mesmo sensor no CBERS-2. A câmera HRC


INPE STAC Server
Browse
Instituto Nacional de Pesquisas Espaciais
API
Source
Share
Language: English
Search
Description
This is the landing page for the INPE STAC server. The SpatioTemporal Asset Catalogs (STAC) provide a standardized way to expose collections of spatial temporal data. Here you will find collections of data provided by projects and areas of INPE.
Additional Resources
- API documentation in HTML
Catalogs 80
Tiles
List
Ascending
Descending
Filter catalogs by title
AMAZONIA-1/WFI - Level-2-DN
GeoTiff AMAZONIA-1/WFI - Level-2 Digital Number product. Level 2 products have radiometric correction and geometric correction using satellite ephemeris and…
3/3/2021, 12:57:23 PM UTC - 4/21/2026, 3:50:22 PM UTC
CBERS-4/WFI 3M Mosaic
COG CBERS-4/WFI image mosaic of Brazil with 64m of spatial resolution. The mosaic was prepared in order to demonstrate the technological capabilities of the Brazil…
4/1/2020, 12:00:00 AM UTC - 6/30/2020, 12:00:00 AM UTC
GOES-19 Cloud & Moisture Imagery
NetCDF The GOES-19 Advanced Baseline Imager (ABI) L2 Cloud and Moisture Imagery provides 16 spectral bands with high temporal resolution over the American…
4/2/2025, 12:00:00 AM UTC - 4/22/2026, 3:40:00 PM UTC
MYD13Q1 v061 - Cloud Optimized GeoTIFF
COG The Aqua Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices (MYD13Q1) Version 6.1 data are generated every 16…
7/4/2002, 12:00:00 AM UTC - 4/14/2026, 11:59:59 PM UTC
AMAZONIA-1/WFI - Level-4-DN
GeoTiff AMAZONIA-1/WFI - Level-4 Digital Number product.
3/17/2021, 12:00:00 AM UTC - 4/21/2026, 2:18:42 PM UTC
CBERS-4A/MUX - Level-2-DN
GeoTiff CBERS-4A/MUX - Level-2 Digital Number product. Level 2 products have radiometric correction and geometric correction using satellite ephemeris and…
12/27/2019, 1:53:00 PM UTC - 4/18/2026, 3:04:34 PM UTC
Landsat 3M Amazon Biome Mosaic
COG Landsat-8/OLI image mosaic of Brazilian Amazon biome with 30m of spatial resolution. The mosaic was prepared in support of TerraClass project. The true colo…
7/1/2016, 12:00:00 AM UTC - 9/30/2016, 12:00:00 AM UTC
S2 1M Amazon Biome Mosaic
COG Sentinel-2 image mosaic of Brazilian Amazon biome with 10m of spatial resolution. The mosaic was prepared to support BiomasBR Program to produce Th…
10/1/2020, 12:00:00 AM UTC - 12/31/2020, 12:00:00 AM UTC
AMAZONIA-1/WFI - Level-4-SR - Cloud Optimized GeoTIFF
COG AMAZONIA-1/WFI - Level-4 Surface Reflectance product. L4 SR product provides orthorectified surface reflectance images. This dataset is provided as Cloud…
1/1/2024, 12:00:00 AM UTC - 4/9/2026, 12:00:00 AM UTC
CBERS-4A/MUX - Level-4-DN
GeoTiff CBERS-4A/MUX - Level-4 Digital Number product. L4 product provides orthorectified images.
12/27/2019, 1:53:00 PM UTC - 4/18/2026, 3:03:29 PM UTC
Landsat 6M Mosaic
COG Landsat-8/OLI image mosaic of Brazil with 30m of spatial resolution. Th… TerraClass project.
7/1/2017, 12:00:00
S2 2M Cerrado Biome Mosaic
COG Sentinel-2 image mosaic of Brazilian Cerrado
data.inpe.br/stac/browser
AMAZONIA-1/WFI Image Mosaic of Brazil - 3 Months
CBERS-4A/MUX - Level-4-SR
COG CBERS-4A/MUX - Level-4 Surface Reflectance product over Brazil and part of South America. L4 SR
Landsat 6M São Paulo State Mosaic
COG Earth Observation Data Cube generated from
S2 3M B08B04B03 Amazon Biome Mosaic
COG Sentinel-2 image mosaic of Brazilian Amazon

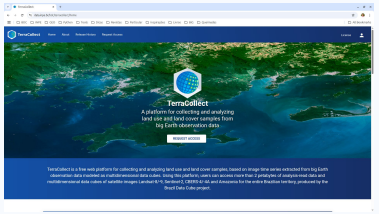
SpatioTemporal Asset Catalog (STAC)

Aplicações

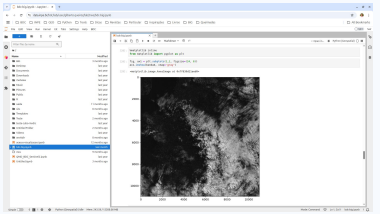
STAC Browser\



DataCube Explorer



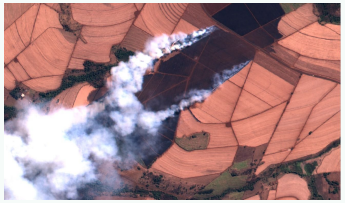
TerraCollect



Python & R

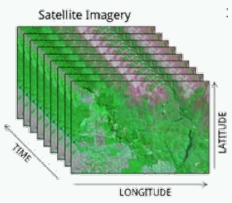
BDC - STAC

Coleções de Dados



CBERS-4A/WPM

Sentinel-2



Cubos Dados

GOES

Outras coleções

Interoperability

Copernicus Data Space Ecosystem (CDSE)

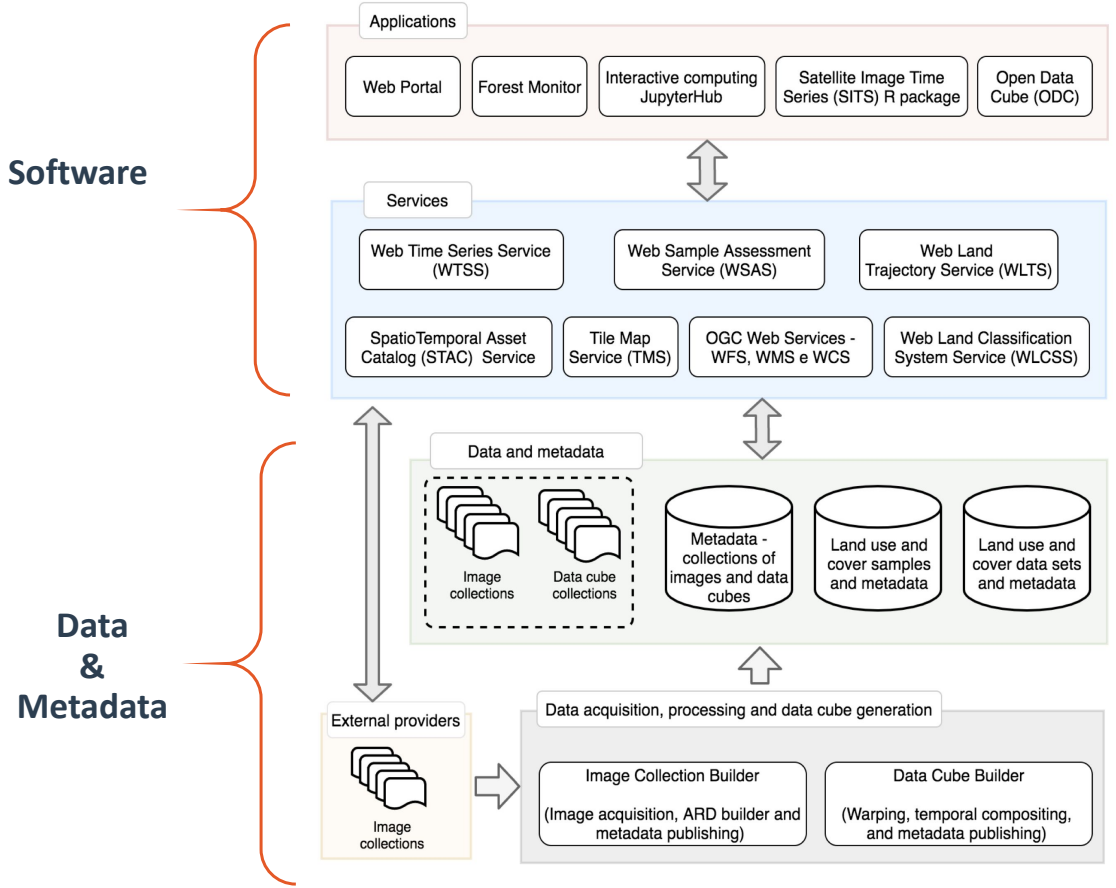
SpatioTemporal Asset Catalog

A specification that defines a common structure, in JSON and GEOJSON files, for describing and cataloging geographic data.

Source: https://stacspec.org/en

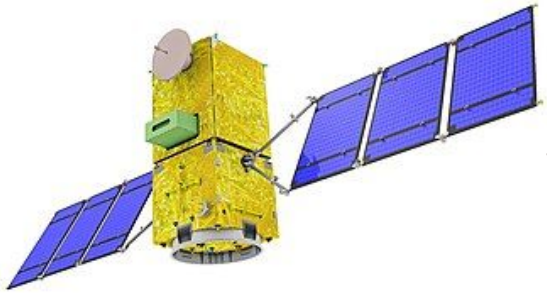
BIG/BDC Platform: General Reference Architecture

Open Data And Software Technologies

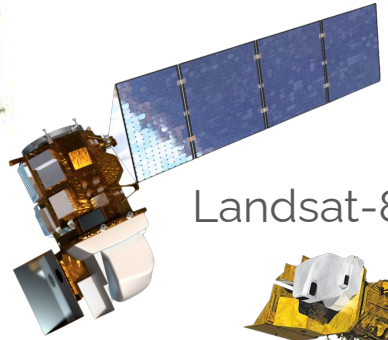


Fonte: Adapted from Ferreira et al. (2020)

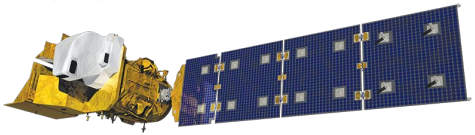
CBERS-4 e 4A



AMAZONIA-1



Landsat-8

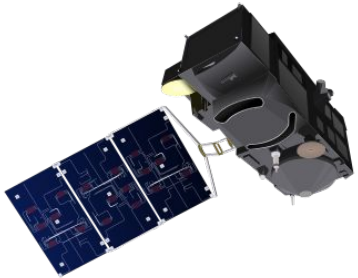


Landsat-9

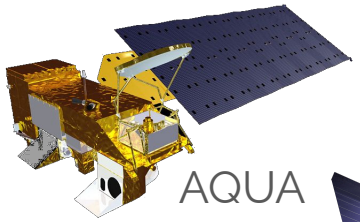
Earth Surface Imaging Products

Sentinel-2A,2B e 2C

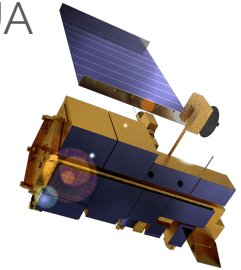
Sentinel-1A e 1C



Sentinel-3

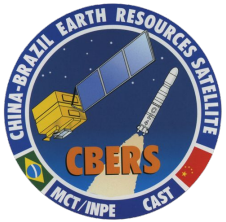


AQUA



TERRA

CBERS: launch history

1999	2003	2007	2013	2014	2019
					
CBERS-1	**CBERS-2**	**CBERS-2B**	**CBERS-3**	**CBERS-4**	**CBERS-4A**
14 OUT 1999	21 OUT 2003	19 SET 2007	09 DEZ 2013	07 DEZ 2014	20 DEZ 2019

CBERS-4 Cameras			
Característica	**PAN**	**MUX**	**WFI**
Spectral Bands	0,51-0,85μm (PAN) 0,52-0,59μm (G) 0,63-0,69μm (R) 0,77-0,89μm (NIR)	0,45-0,52μm (B) 0,52-0,59μm (G) 0,63-0,69μm (R) 0,77-0,89μm (NIR)	0,45-0,52μm (B) 0,52-0,59μm (G) 0,63-0,69μm (R) 0,77-0,89μm (NIR)
Spatial Resolution	5 m / 10 m	20 m	64 m
Swath	60 km	120 km	866 km
Revisit	52 days	26 days	5 days
Quantization	8 bits	8 bits	10 bits

CBERS-4/WFI

CBERS-4/WFI
Spatial Resolution: 64 m
Revisit: 5 days
Local: West Bahia

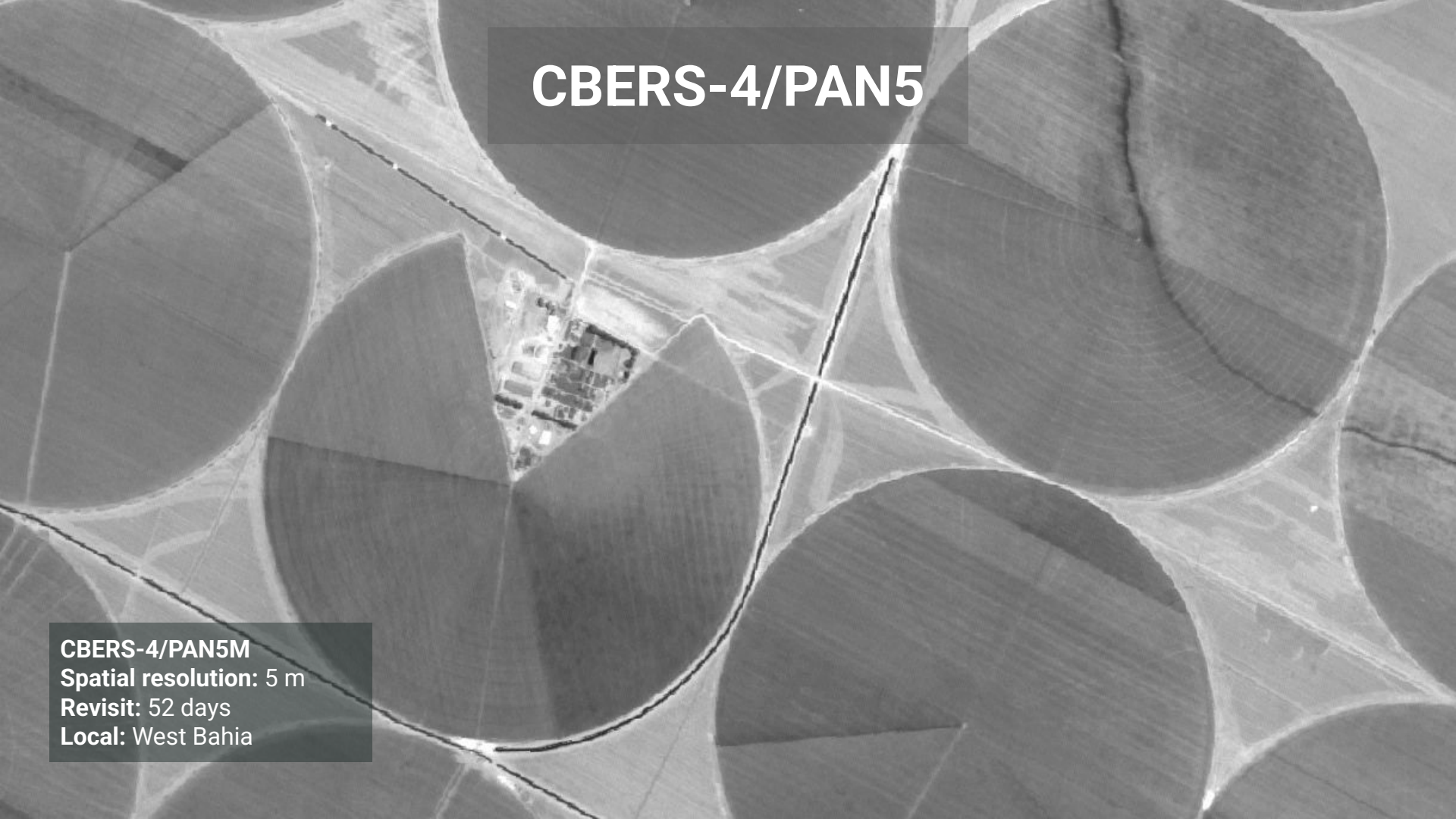
CBERS-4/MUX

CBERS-4/MUX
Spatial resolution: 20 m
Revisit: 26 days
Local: West Bahia

CBERS-4/PAN10

CBERS-4/PAN10M
Spatial resolution: 10 m
Revisit: 52 days
Local: West Bahia

CBERS-4/PAN5



CBERS-4/PAN5M
Spatial resolution: 5 m
Revisit: 52 days
Local: West Bahia

CBERS-4: available products

CBERS-4/MUX - Level-2-DN

GeoTiff CBERS-4/MUX - Level-2 Digital Number. Level 2 products have radiometric correction and geometric correction using satellite ephemeris an…

12/8/2014, 12:56:32 PM UTC - 6/8/2026, 12:09:37 PM UTC

CBERS-4/MUX - Level-4-DN

GeoTiff CBERS-4/MUX - Level-4 Digital Number product. L4 product provides orthorectified images.

12/9/2014, 1:53:55 PM UTC - 6/8/2026, 12:04:51 PM UTC

CBERS-4/MUX - Level-4-SR - Cloud Optimized GeoTIFF

COG CBERS-4/MUX - Level-4 Surface Reflectance product over Brazil and part of South America. L4 SR product provides orthorectified…

1/1/2016, 12:00:00 AM UTC - 5/31/2026, 12:00:00 AM UTC

CBERS-4/MUX - Level-4-SR - Data Cube - LCF 2 months

COG Earth Observation Data Cube generated from CBERS-4/MUX Level-4 SR product over Brazil extension. This dataset is provided in Clou…

1/1/2016, 12:00:00 AM UTC - 4/30/2026, 12:00:00 AM UTC

CBERS-4/PAN10M - Level-2-DN

GeoTiff CBERS-4/PAN10M - Level-2 Digital Number. Level 2 products have radiometric correction and geometric correction using satellit…

12/9/2014, 1:53:35 PM UTC - 6/8/2026, 1:48:30 PM UTC

CBERS-4/PAN10M - Level-4-DN

GeoTiff CBERS-4/PAN10M - Level-4 Digital Number. L4 product provides orthorectified images.

12/9/2014, 1:53:35 PM UTC - 6/8/2026, 1:43:43 PM UTC

CBERS-4/PAN5M - Level-2-DN

GeoTiff CBERS-4/PAN5M - Level-2 Digital Number. Level 2 products have radiometric correction and geometric correction using satellit…

12/9/2014, 1:53:35 PM UTC - 6/8/2026, 1:48:28 PM UTC

CBERS-4/PAN5M - Level-4-DN

GeoTiff CBERS-4/PAN5M - Level-4 Digital Number. L4 product provides orthorectified images.

12/9/2014, 1:53:35 PM UTC - 6/8/2026, 1:42:56 PM UTC

CBERS-4/WFI - Level-2-DN

GeoTiff CBERS4/WFI - Level-2 Digital Number product. Level 2 products have radiometric correction and geometric correction using satellit…

12/14/2014, 12:43:57 PM UTC - 6/8/2026, 1:48:43 PM UTC

CBERS-4/WFI - Level-4-DN

GeoTiff CBERS4/WFI - Level-4 Digital Number product. L4 product provides orthorectified images.

12/9/2014, 1:53:35 PM UTC - 6/8/2026, 1:45:42 PM UTC

CBERS-4/WFI - Level-4-SR - Cloud Optimized GeoTIFF

COG CBERS-4/WFI Surface Reflectance product over Brazil and part of South America. L4 SR product provides orthorectified surface reflectanc…

1/1/2016, 12:00:00 AM UTC - 5/31/2026, 12:00:00 AM UTC

CBERS-4/WFI - Level-4-SR - Data Cube - LCF 16 days

COG Earth Observation Data Cube generated from CBERS-4/WFI Level-4 SR product over Brazil extension. This dataset is provided in Cloud…

1/1/2016, 12:00:00 AM UTC - 5/8/2026, 12:00:00 AM UTC

CBERS-4/WFI 3M Mosaic

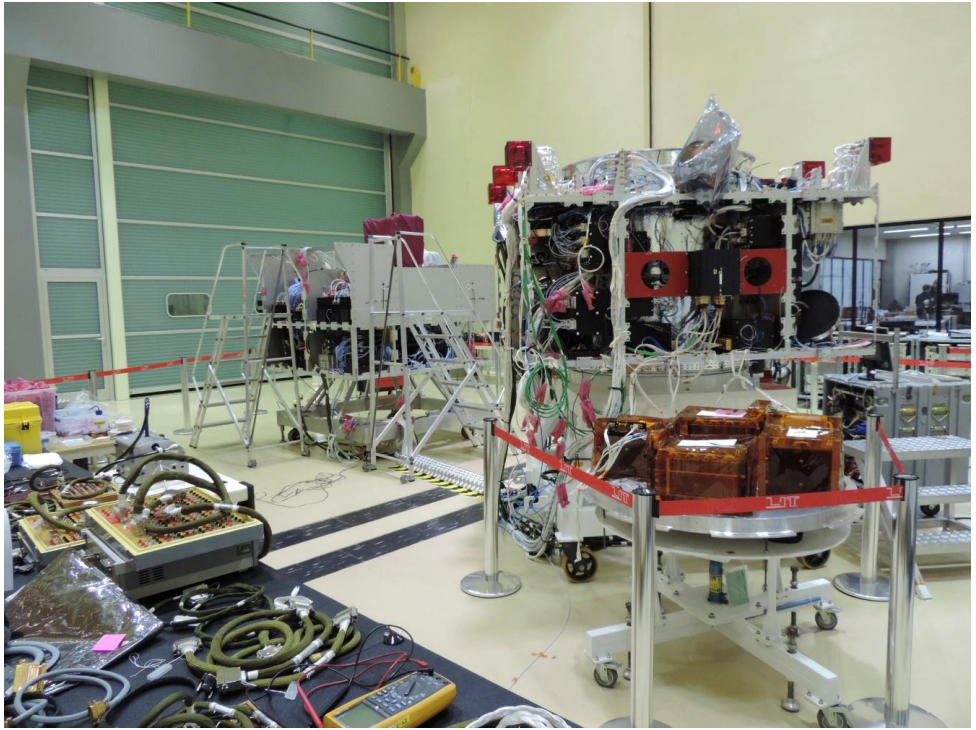
COG CBERS-4/WFI image mosaic of Brazil with 64m of spatial resolution. The mosaic was prepared in order to demonstrate the technological…

4/1/2020, 12:00:00 AM UTC - 6/30/2020, 12:00:00 AM UTC

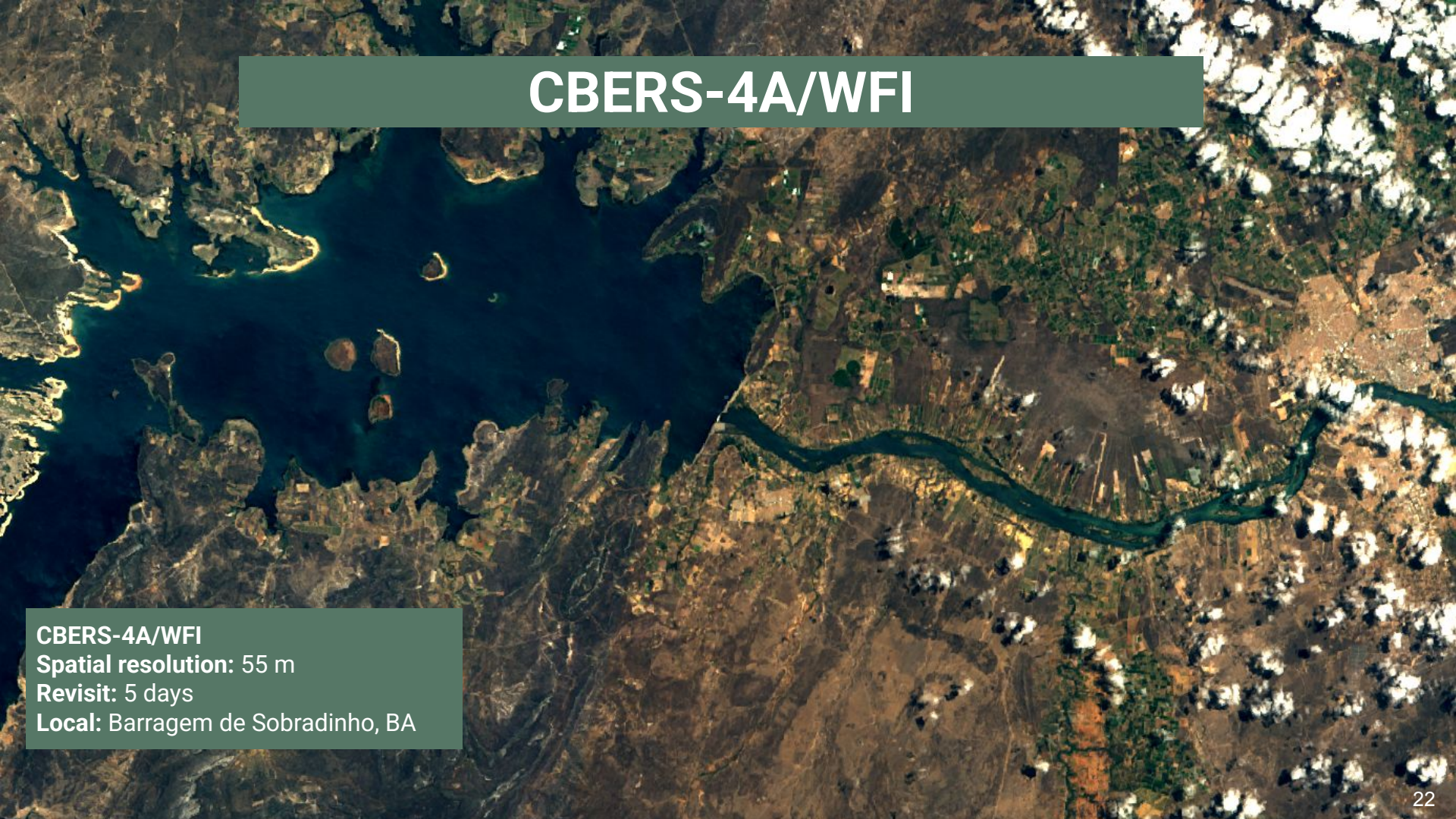
data.inpe.br/dados/cbers-4

Câmeras CBERS-4A			
Characteristics	**WPM**	**MUX**	**WFI**
Spectral Bands	0,45-0,52μm (B) 0,52-0,59μm (G) 0,63-0,69μm (R) 0,77-0,89μm (NIR) 0,45-0,90μm (PAN)	0,45-0,52μm (B) 0,52-0,59μm (G) 0,63-0,69μm (R) 0,77-0,89μm (NIR)	0,45-0,52μm (B) 0,52-0,59μm (G) 0,63-0,69μm (R) 0,77-0,89μm (NIR)
Spatial Resolution	2 m e 8 m	16 m	55 m
Swath	92 km	95 km	684 km
Revisit	31 days	31 days	5 days
Quantization	10 bits	8 bits	10 bits

CBERS-4A: integration and tests at LIT



CBERS-4A/WFI



CBERS-4A/WFI
Spatial resolution: 55 m
Revisit: 5 days
Local: Barragem de Sobradinho, BA

CBERS-4A/MUX

CBERS-4A/MUX
Spatial resolution: 16 m
Revisit: 31 days
Local: Barragem de Sobradinho, BA

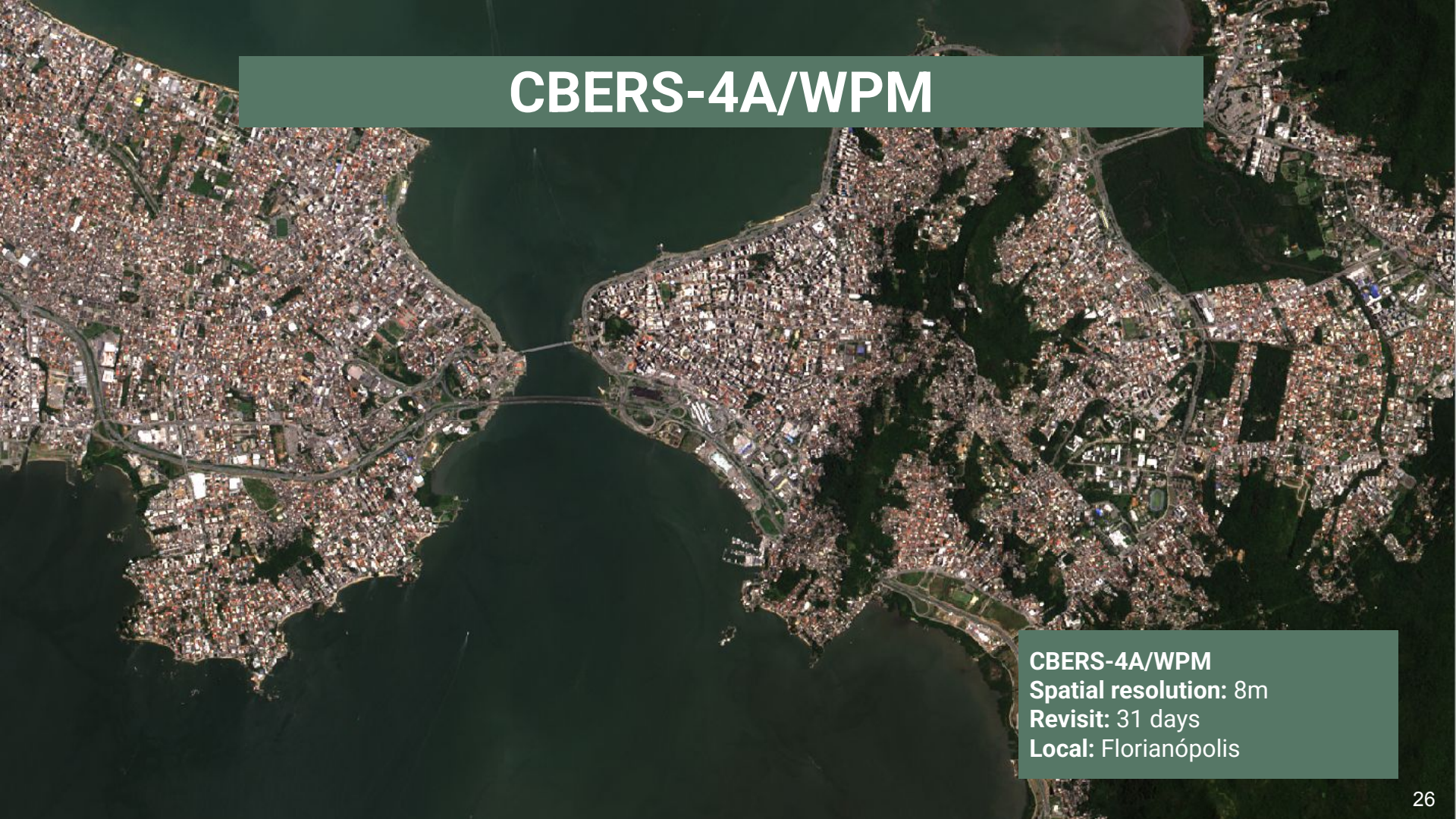
CBERS-4A/WPM

CBERS-4A/WPM
Spatial resolution: 8m
Revisit: 31 days
Local: Barragem de Sobradinho, BA

CBERS-4A/WPM

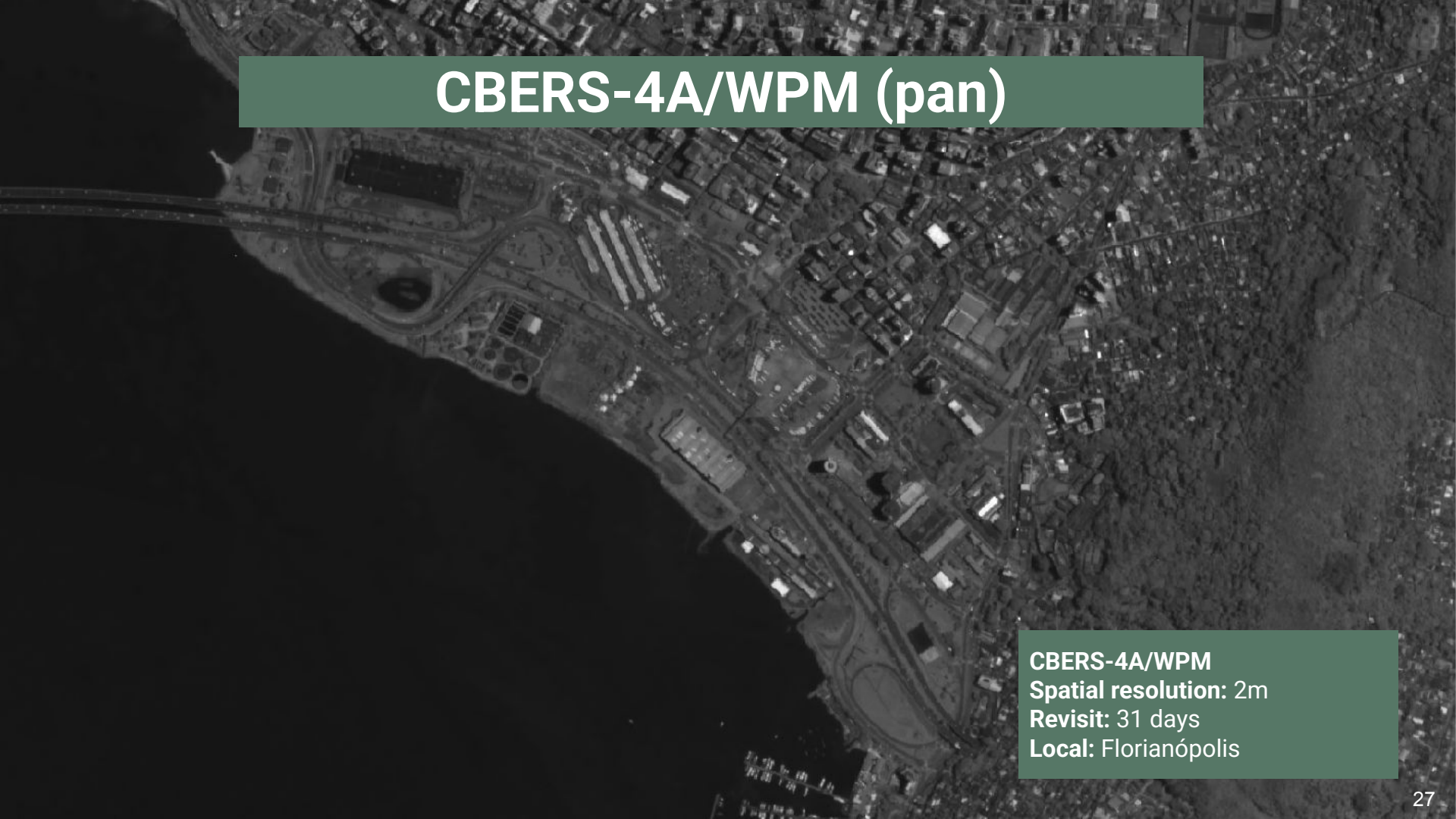
CBERS-4A/WPM
Spatial resolution: 2m
Revisit: 31 days
Local: Guarda Mor, MG

CBERS-4A/WPM



CBERS-4A/WPM
Spatial resolution: 8m
Revisit: 31 days
Local: Florianópolis

CBERS-4A/WPM (pan)



CBERS-4A/WPM
Spatial resolution: 2m
Revisit: 31 days
Local: Florianópolis

CBERS-4A/WPM-fusionado

CBERS-4A/WPM
Spatial resolution: 2m
Revisit: 31 days
Local: Florianópolis

CBERS-4A/WPM
Spatial resolution: 2m
Local: Porto Alegre, RS
Date: 09 May de 2024
Source: Laércio Namikawa

CBERS-4A/WPM
Spatial Resolution: 2m
Local: Porto Alegre, RS
Date: 09 May 2024
Sourece: Laércio Namikawa

CBERS-4A/WPM
Spatial Resolution: 2m
Local: Porto Alegre, RS
Date: 09 May 2024
Source: Laércio Namikawa

CBERS-4A/WPM
Spatial Resolution: 2m
Local: Porto Alegre, RS
Date: 09 May 2024
Source: Laércio Namikawa

CBERS-4A/WPM
Spatial resolution: 2m
Local: Porto Alegre, RS
Date: 09 May 2024
Source: Laércio Namikawa

CBERS-4A/WPM
Spatial resolution: 2m
Local: Porto Alegre, RS
Date: 09 May 2024
Source: Laércio Namikawa

CBERS-4A: available products

CBERS-4A/MUX - Level-2-DN

GeoTiff CBERS-4A/MUX - Level-2 Digital Number product. Level 2 products have radiometric correction and geometric correction using satellite ephemeris and attitude data (syste…

12/27/2019, 1:53:00 PM UTC - 6/8/2026, 2:45:27 PM UTC

CBERS-4A/MUX - Level-4-DN

GeoTiff CBERS-4A/MUX - Level-4 Digital Number product. L4 product provides orthorectified images.

12/27/2019, 1:53:00 PM UTC - 6/8/2026, 1:13:47 PM UTC

CBERS-4A/MUX - Level-4-SR - Cloud Optimized GeoTIFF

COG CBERS-4A/MUX - Level-4 Surface Reflectance product over Brazil and part of South America. L4 SR product provides orthorectified surface reflectance images. This…

1/1/2026, 12:00:00 AM UTC - 5/31/2026, 12:00:00 AM UTC

CBERS-4A/WFI - Level-2-DN

COG CBERS-4A/WFI Level-2 Digital Number product. Level 2 products have radiometric correction and geometric correction using satellite ephemeris and attitude data (syste…

12/27/2019, 1:53:00 PM UTC - 6/8/2026, 2:49:43 PM UTC

CBERS-4A/WFI - Level-4-DN

COG CBERS-4A/WFI Level-4 Digital Number product. L4 product provides orthorectified images.

7/4/2020, 1:41:30 PM UTC - 6/8/2026, 2:48:00 PM UTC

CBERS-4A/WFI - Level-4-SR - Cloud Optimized GeoTIFF

COG CBERS-4A/WFI - Level-4 Surface Reflectance product over Brazil and part of South America. L4 SR product provides orthorectified surface reflectance images. This…

1/1/2020, 12:00:00 AM UTC - 5/31/2026, 12:00:00 AM UTC

CBERS-4A/WFI 3M Paraíba State Mosaic

COG CBERS-4A/WFI image mosaic of Brazil Paraíba State with 55m of spatial resolution. The mosaic was prepared in order to demonstrate the technological capabilities of the…

7/1/2020, 12:00:00 AM UTC - 9/30/2020, 12:00:00 AM UTC

CBERS-4A/WPM - Level-2-DN

GeoTiff CBERS-4A/WPM - Level-2 Digital Number product. Level 2 products have radiometric correction and geometric correction using satellite ephemeris and attitude data (syste…

12/27/2019, 1:53:00 PM UTC - 6/8/2026, 1:16:34 PM UTC

CBERS-4A/WPM - Level-4-DN

GeoTiff CBERS-4A/WPM - Level-4 Digital Number product. L4 product provides orthorectified images.

12/27/2019, 1:53:00 PM UTC - 6/8/2026, 1:13:46 PM UTC

CBERS-4A/WPM - Multispectral and Panchromatic Bands Fusioned

COG This collection contains 2 meter high-resolution, RGB products, generated using the Principal Components Fusion (PCA) method, with values coded between 1 and 255, with …

3/2/2023, 12:00:00 AM UTC - 5/18/2026, 12:00:00 AM UTC

data.inpe.br/dados/cbers-4a

AMAZONIA-1: LIT tests

Amazonia-1 Camera	
Characteristics	**WFI**
Spectral Bands	0,45-0,52μm (B) 0,52-0,59μm (G) 0,63-0,69μm (R) 0,77-0,89μm (NIR)
Spatial Resolution	64 m
Swath	850 km
Revisit	5 days
Quantization	10 bits

AMAZONIA-1/WFI

Amazonia-1/WFI
Spatial resolution: 64 m
Revisit: 5 days
Local: Ilha do Mel, PR
Source: Emiliano Castejon

AMAZONIA-1/WFI

Amazonia-1/WFI
Spatial resolution: 64 m
Revisit: 5 days
Local: Búzios, RJ
Source: Emiliano Castejon

Amazonia-1: available products

AMAZONIA-1/WFI - Level-2-DN

GeoTiff AMAZONIA-1/WFI - Level-2 Digital Number product. Level 2 products have radiometric correction and geometric correction using satellite ephemeris and attitude data (system correction).

3/3/2021, 12:57:23 PM UTC - 6/8/2026, 6:50:15 PM UTC

AMAZONIA-1/WFI - Level-4-DN

GeoTiff AMAZONIA-1/WFI - Level-4 Digital Number product.

3/17/2021, 12:00:00 AM UTC - 6/8/2026, 2:03:33 PM UTC

AMAZONIA-1/WFI - Level-4-SR - Cloud Optimized GeoTIFF

COG AMAZONIA-1/WFI - Level-4 Surface Reflectance product. L4 SR product provides orthorectified surface reflectance images. This dataset is provided as Cloud Optimized GeoTIFF (COG).

1/1/2024, 12:00:00 AM UTC - 5/31/2026, 12:00:00 AM UTC

AMAZONIA-1/WFI Image Mosaic of Brazil - 3 Months

COG AMAZONIA-1/WFI image mosaic of Brazil with 64m of spatial resolution. The mosaic was prepared in order to demonstrate the technological capabilities of the Brazil Data Cube project tools. The false color composition is based on the WFI bands 3, 4 and 1…

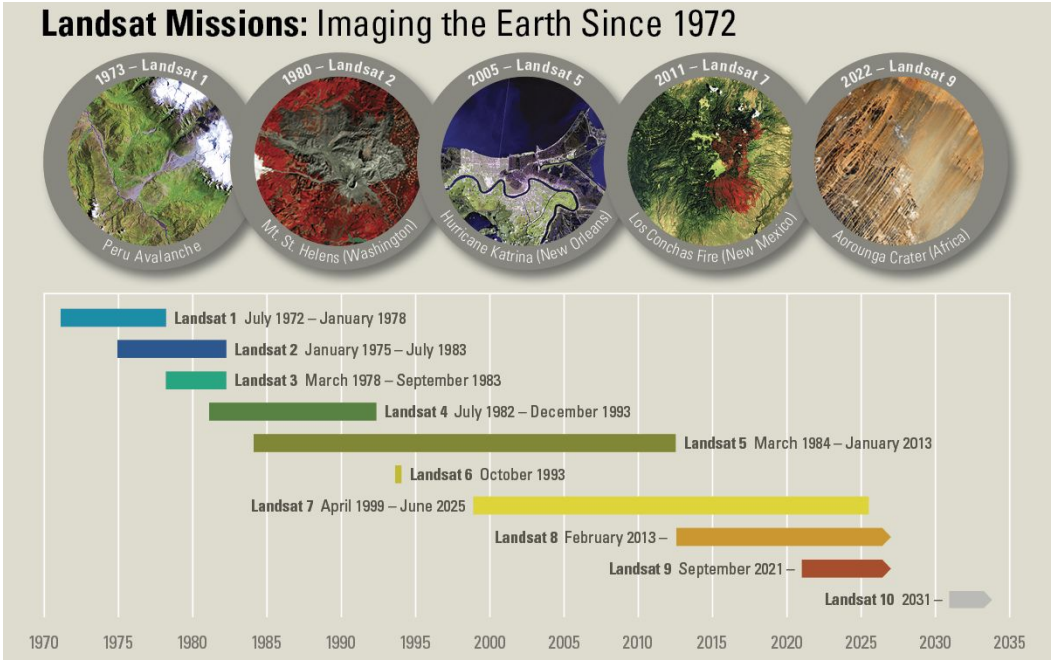
5/1/2024, 12:00:00 AM UTC - 10/31/2024, 12:00:00 AM UTC

data.inpe.br/dados/amazonia-1

The Landsat Program

The Landsat Program constitutes the longest-running series of Earth-observation satellite missions, having continuously acquired imagery from 1972 to the present day, with plans for the launch of additional satellites.

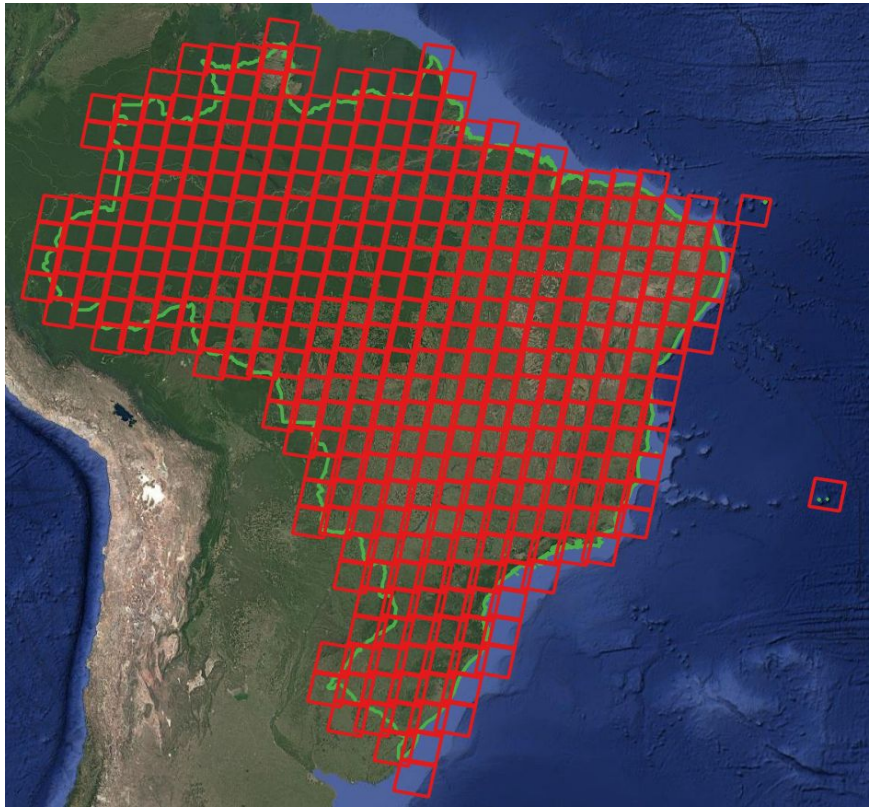
The data are distributed primarily through the Earth Explorer platform, though central distribution points—each with its own archiving strategy—are also used.



Fonte: https://www.usgs.gov/media/images/landsat-missions-timeline

Landsat-8 and Landsat-9 Cameras		
Characteristics	**OLI**	**TIRS**
Spectral Bands	0.43-0.45 (COASTAL) 0.45-0.51 (B) 0.53-0.59 (G) 0.64-0.67 (R) 0.85-0.88 (NIR) 1.57-1.65 (SWIR1) 2.11-2.29 (SWIR2) 0.50-0.68 (PAN) 1.36-1.38 (CIRRUS)	10.6-11.19 (TIRS1) 11.50-12.51 (TIRS2)
Spatial Resolution	15 m / 30 m	100* m (resample to 30 m)
Swath	185 km	185 km
Revisit	16 days	16 days
Quantization	12 bits	12 bits

Landsat data



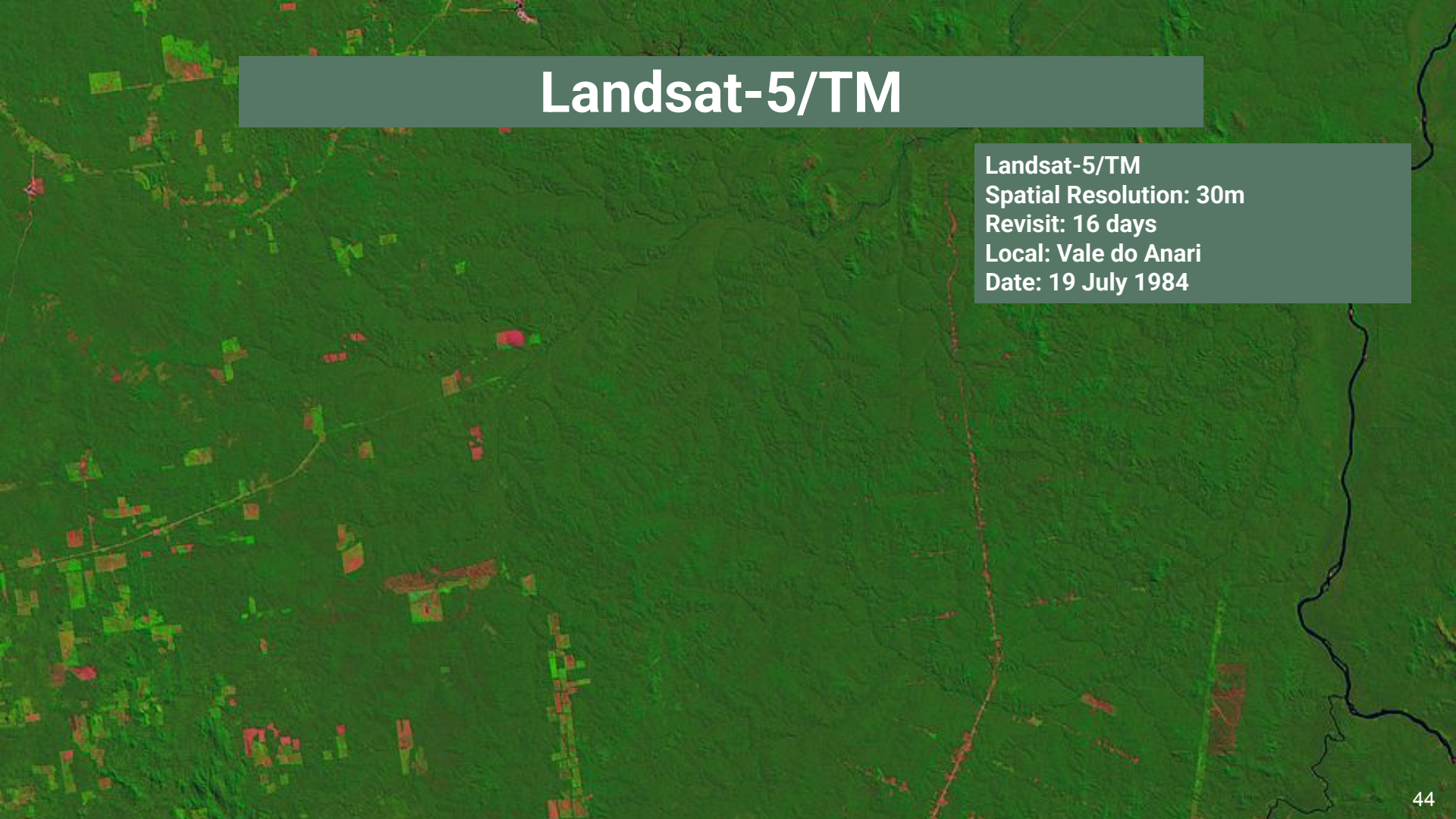
Spatial coverage: Brasil

Period:

- L2 (SR): 1990-(today)

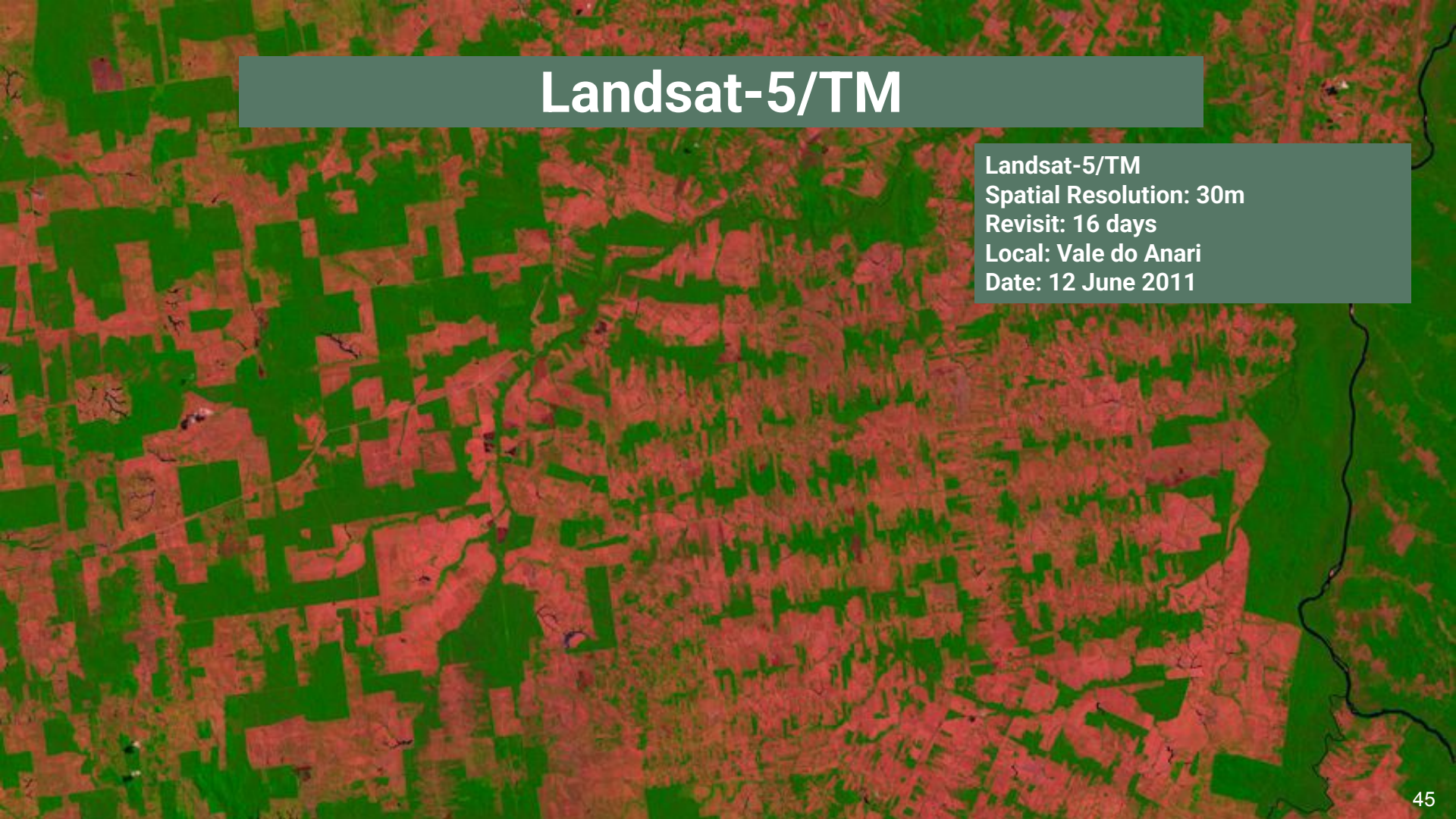
Revisit: ~8 days (Landsat-8 and Landsat-9)

Landsat-5/TM



Landsat-5/TM
Spatial Resolution: 30m
Revisit: 16 days
Local: Vale do Anari
Date: 19 July 1984

Landsat-5/TM

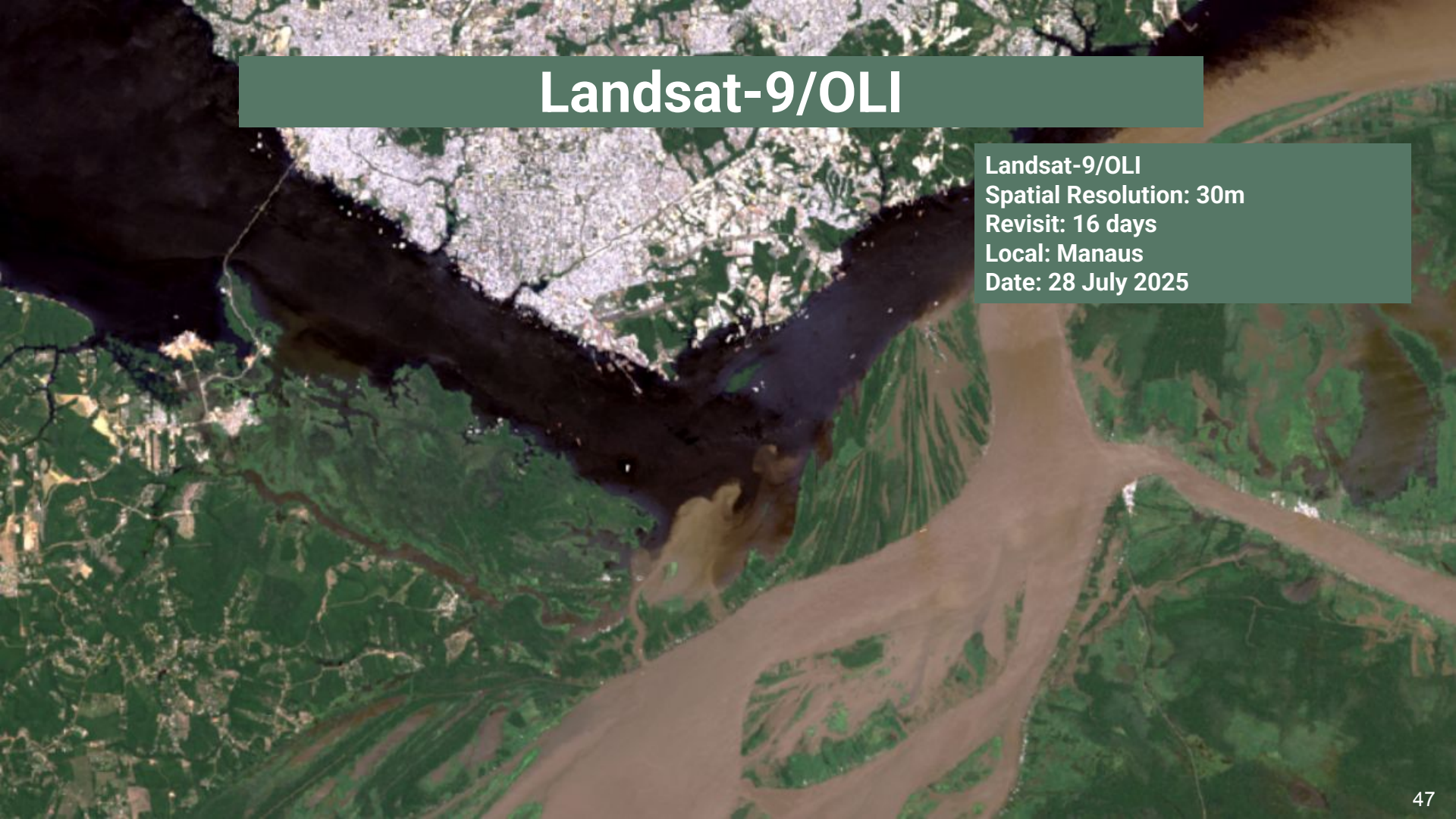


Landsat-5/TM
Spatial Resolution: 30m
Revisit: 16 days
Local: Vale do Anari
Date: 12 June 2011

Landsat-8/OLI

Landsat-8/OLI
Spatial Resolution: 30m
Revisit: 16 days
Local: West of Bahia
Date: 21 June 2025

Landsat-9/OLI



Landsat-9/OLI
Spatial Resolution: 30m
Revisit: 16 days
Local: Manaus
Date: 28 July 2025

Landsat: available products

Landsat 3M Amazon Biome Mosaic

COG Landsat-8/OLI image mosaic of Brazilian Amazon biome with 30m of spatial resolution. The mosaic was prepared in support of TerraClass…

7/1/2016, 12:00:00 AM UTC - 9/30/2016, 12:00:00 AM UTC

Landsat Collection 2 - Level-2

Text COG XML JSON Landsat Collection 2 Level-2 Science Products (https://www.usgs.gov/landsat-missions/landsat…

1/7/1985, 12:00:00 AM UTC - 6/6/2026, 12:00:00 AM UTC

Landsat INPE Data Archive

ZIP PNG Landsat Historical Data Archive of the Image Generation Laboratory (LGI) at National Institute for Space Reaserch (INPE). This datas…

5/21/1973, 12:00:00 AM UTC - 12/31/2021, 12:00:00 AM UTC

Landsat 6M Mosaic

COG Landsat-8/OLI image mosaic of Brazil with 30m of spatial resolution. The mosaic was prepared in support of TerraClass project. The…

7/1/2017, 12:00:00 AM UTC - 6/30/2018, 12:00:00 AM UTC

Landsat Collection 2 - Level-2 - Data Cube - LCF 16 days

COG Earth Observation Data Cube generated from Landsat Level-2 product over Brazil extension. This dataset is provided in Cloud…

1/1/1990, 12:00:00 AM UTC - 5/24/2026, 12:00:00 AM UTC

Landsat TSIRF bimonthly

COG Brazil, cloud-free, and reconstructed historical Landsat spectral bands are provided at a spatial resolution of 30 meters and temporal…

1/1/1997, 12:00:00 AM UTC - 12/31/2024, 11:59:59 PM UTC

Landsat 6M São Paulo State Mosaic

COG Earth Observation Data Cube generated from Landsat Level-2 product over Sao Paulo State extension. This dataset is provided in Clo…

5/1/2013, 12:00:00 AM UTC - 10/31/2023, 12:00:00 AM UTC

Landsat TSIRF yearly

COG Brazil, cloud-free, and reconstructed historical Landsat spectral bands are provided at a spatial resolution of 30 meters and temporal…

1/1/1997, 12:00:00 AM UTC - 12/31/2024, 11:59:59 PM UTC

data.inpe.br/dados/landsat

The Copernicus Program

Cooperation between the European Commission and the European Space Agency (ESA) to develop satellite-based Earth observation services;

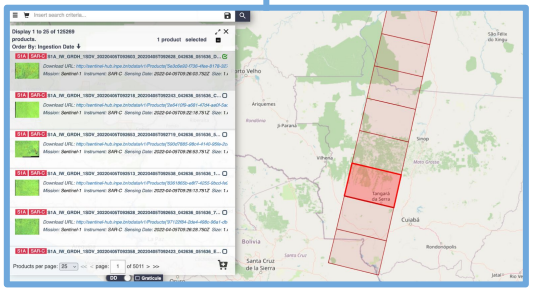
A family of orbital missions called **Sentinel**, featuring instruments such as radars and multispectral sensors, based on a long-term strategy;

Data access is provided through central distribution points, or hubs, which employ different archiving strategies.

Source: https://www.gov.br/inpe/pt-br/assuntos/ultimas-noticias/programa-copernicus-repositorio-espelho-de-dados-sentinel-no-inpe

Sentinel data mirroring at INPE

2022



Implementation of the mirroring system

Available products:

- Sentinel-2 L1C e L2A
- Sentinel-1 GRD

2023

Expansion of storage infrastructure and Migration of the STAC BIG mirroring system (Copernicus Data Space)

- Inclusion of Sentinel-3 OLCI L1B
- Production and provision of new products designed for visualization and dissemination (COG)

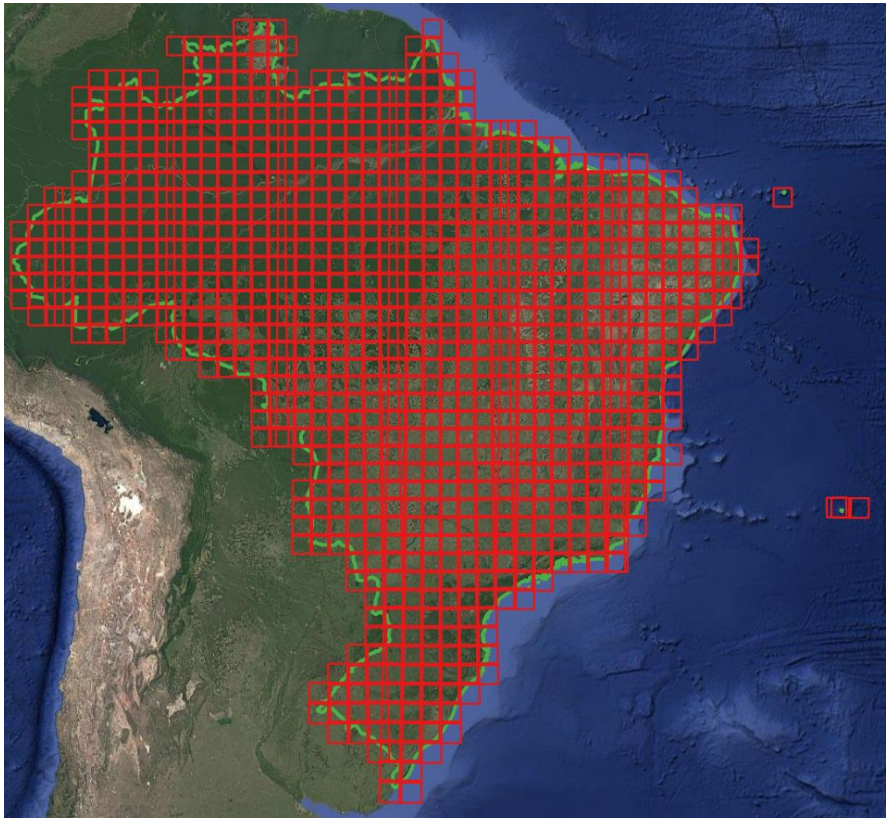
2024

Consolidation of the infrastructure for data access and sharing services

Under development:

- Data usage monitoring and statistics system.
- Incorporation of Sentinel-5P

Sentinel-2: Data coverage



Spatial coverage: Brasil

Period:

- L1C (TOA): 2021-2026 (today)
- L2A (SR): 2021-2026 (today)

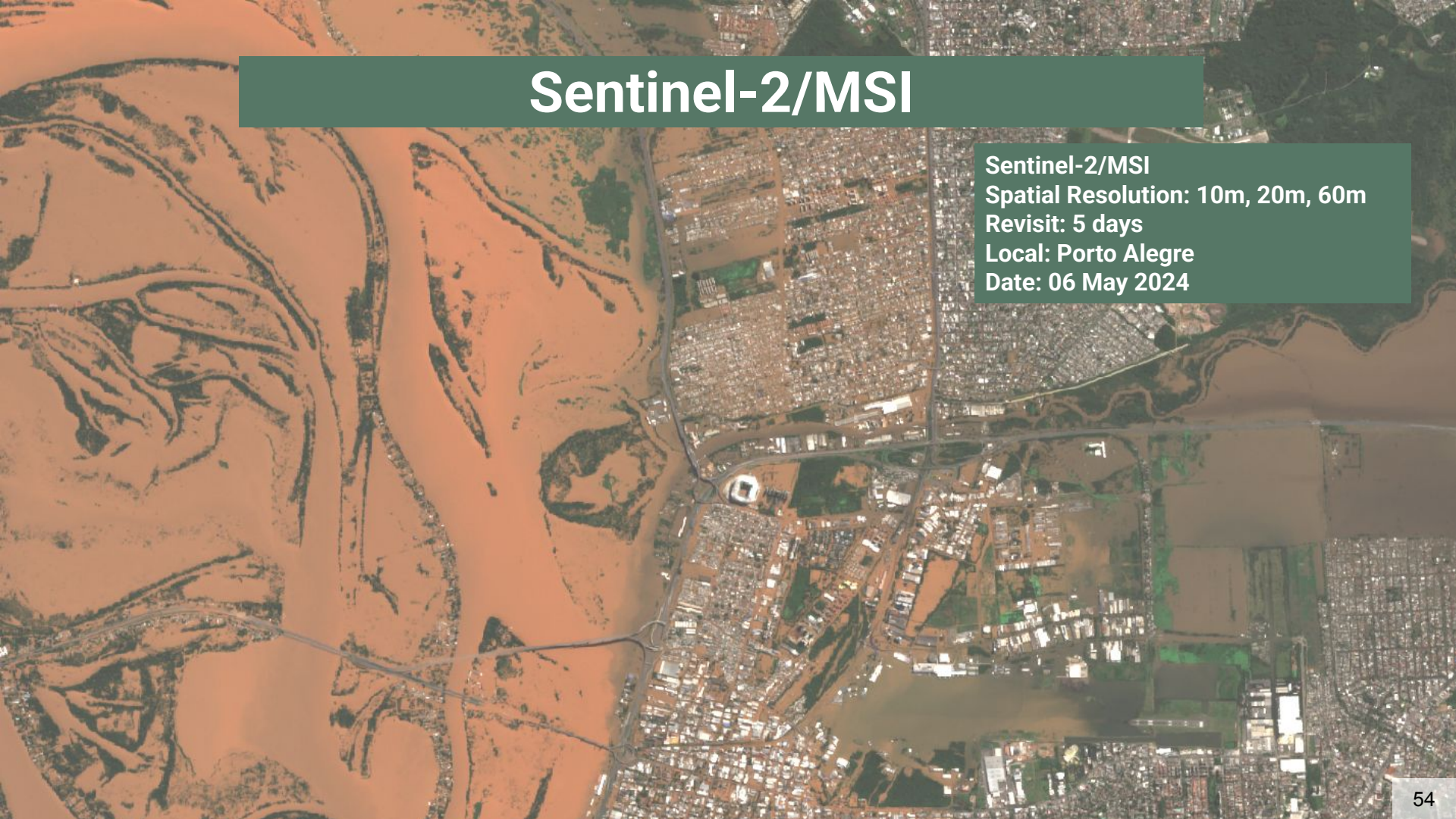
Revisit: < 5 days (Sentinel-2A, Sentinel-2B, Sentinel-2C)

Sentinel-2 Cameras		
Characteristics	**MSI**	
Spectral Bands	0.4325-0.4532 (B01: Coastal) 0.4594-0.5254 (B02: B) 0.5418-0.5778 (B03: G) 0.6491-0.6801 (B04: R) 0.6966-0.7116 (B05: Rededge 1) 0.7330-0.7480 (B06: Rededge 2) 0.7728-0.7928 (B07: Rededge 3)	0.7798-0.8858 (B08: NIR) 0.8542-0.8752 (B8A: NIR) 0.9351-0.9551 (B9: SWIR) 1.3580-1.1389 (B10: SWIR) 1.5682-1.16592 (B11: SWIR) 2.1149-2.22899 (B12: SWIR)
Spatial Resolution	10 m / 20 m / 60 m	
Swath	290 km	
Revisit	< 5 days 2A, 2B e 2C in operation (10 days individually satellite)	
Quantization	12 bits	

Sentinel-2/MSI

Sentinel-2/MSI
Spatial Resolution: 10m, 20m, 60m
Revisita: 5 days
Local: Porto Alegre
Date: 21 April 2024

Sentinel-2/MSI



Sentinel-2/MSI
Spatial Resolution: 10m, 20m, 60m
Revisit: 5 days
Local: Porto Alegre
Date: 06 May 2024

Sentinel-2: available products

S2 1M Amazon Biome Mosaic

COG Sentinel-2 image mosaic of Brazilian Amazon biome with 10m of spatial resolution. The mosaic was prepared to support BiomasBR Program to produce The European Unio…

10/1/2020, 12:00:00 AM UTC - 12/31/2020, 12:00:00 AM UTC

S2 3M Mosaic

COG Sentinel-2 image mosaic of Brazil with 10m of spatial resolution. It uses True color composition based on the MSI bands 04, 03 and 02 assigned to RGB channels. It's input i…

1/1/2018, 12:00:00 AM UTC - 12/31/2025, 11:59:59 PM UTC

Sentinel-2 - Level-1C

ZIP Copernicus Sentinel-2/MSI Level-1C product over Brazil. Level-1C product provides orthorectified Top-Of-Atmosphere (TOA) reflectance images.

7/16/2017, 2:47:29 PM UTC - 6/8/2026, 3:17:19 PM UTC

S2 2M Cerrado Biome Mosaic

COG Sentinel-2 image mosaic of Brazilian Cerrado biome with 10m of spatial resolution. The mosaic was prepared in support of TerraClass project. The false color composition is…

11/1/2023, 12:00:00 AM UTC - 8/31/2024, 12:00:00 AM UTC

S2 3M Paraiba State Mosaic

COG Sentinel-2 image mosaic of Brazilian Paraíba State with 10m of spatial resolution. The mosaic was prepared to support the partnership between the INPE's Health…

11/1/2019, 12:00:00 AM UTC - 1/31/2020, 12:00:00 AM UTC

Sentinel-2 - Level-2A - Cloud Optimized GeoTIFF

GeoTiff COG Copernicus Sentinel-2/MSI Level-2A product over Brazil. Level-2A product provides orthorectified surface reflectance images (Bottom-Of-Atmosphere - BOA). This…

1/2/2018, 1:12:39 PM UTC - 6/8/2026, 3:17:19 PM UTC

S2 3M B08B04B03 Amazon Biome Mosaic

COG Sentinel-2 image mosaic of Brazilian Amazon biome with 10m of spatial resolution. The mosaic was prepared in support of TerraClass project. The false color composition is…

6/1/2024, 12:00:00 AM UTC - 8/31/2024, 12:00:00 AM UTC

S2 4M Cerrado Biome Mosaic

COG Sentinel-2 image mosaic of Brazilian Cerrado biome with 10m of spatial resolution. The mosaic was prepared in support of TerraClass project. The false color composition is…

6/1/2022, 12:00:00 AM UTC - 9/30/2022, 12:00:00 AM UTC

Sentinel-2/MSI - Level-2A - Data Cube - LCF 16 days

COG Earth Observation Data Cube generated from Copernicus Sentinel-2/MSI Level-2A product over Brazil. This dataset is provided in Cloud Optimized GeoTIFF (COG) file…

1/1/2017, 12:00:00 AM UTC - 5/24/2026, 12:00:00 AM UTC

S2 3M B11B8AB04 Amazon Biome Mosaic

COG Sentinel-2 image mosaic of Brazilian Amazon biome with 10m of spatial resolution. The mosaic was prepared in support of TerraClass project. The false color composition is…

6/1/2022, 12:00:00 AM UTC - 8/31/2024, 12:00:00 AM UTC

S2 6M Yanomami Indigenous Territory Mosaic

COG Sentinel-2 image mosaic of Brazilian Yanomami Indigenous Territory with 10m of spatial resolution. The mosaic was prepared to support the partnership between th…

4/1/2019, 12:00:00 AM UTC - 4/1/2022, 12:00:00 AM UTC

data.inpe.br/dados/sentinel-2

Data Cubes

CBERS

CBERS/WFI	8 dias
CBERS-4/MUX	2 meses
CBERS-4/WFI	16 dias

Sentinel-2

Sentinel-2	16 dias

TERRA/AQUA (MODIS)

MOD13Q1	16 dias
MYD13Q1	16 dias

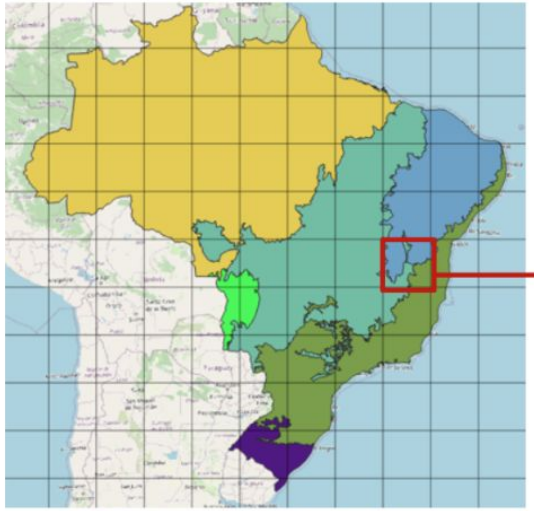
Landsat

Landsat SR	16 dias
Landsat TSIRF	2 meses
Landsat TSIRF	1 ano

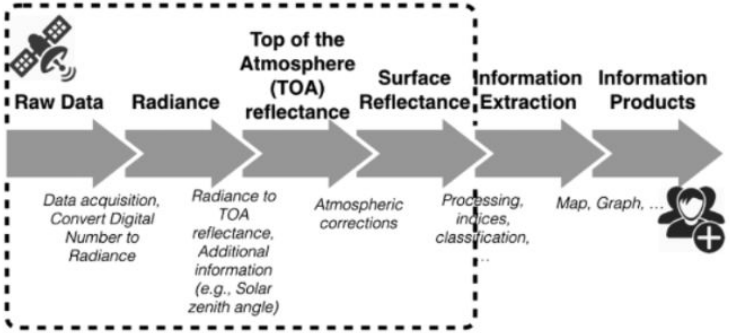
Brazil Data Cube

ARD and EO Data Cubes

2 Petabytes (PB)
Imagens Sen. Remoto

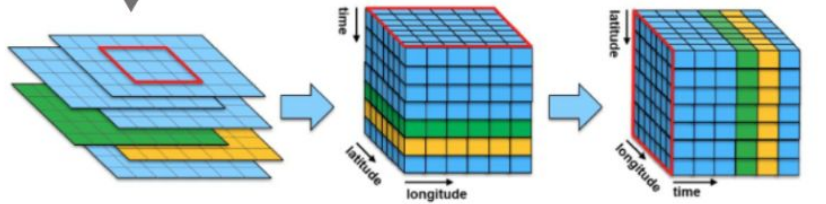


BDC Hierarchical Tiling System



Analysis Ready Data production

Source: [Giuliani et al, 2017]

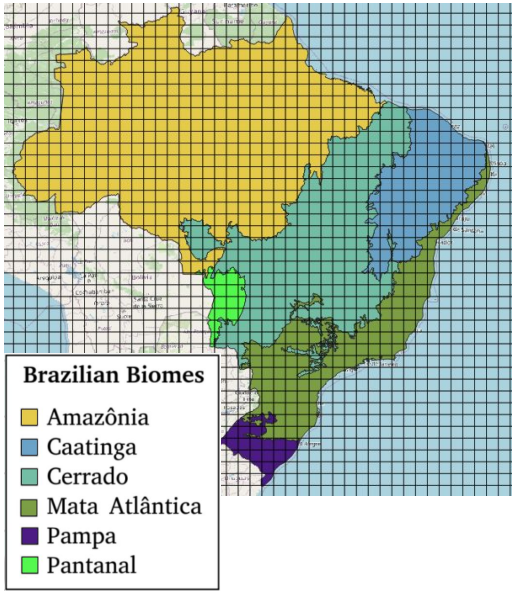


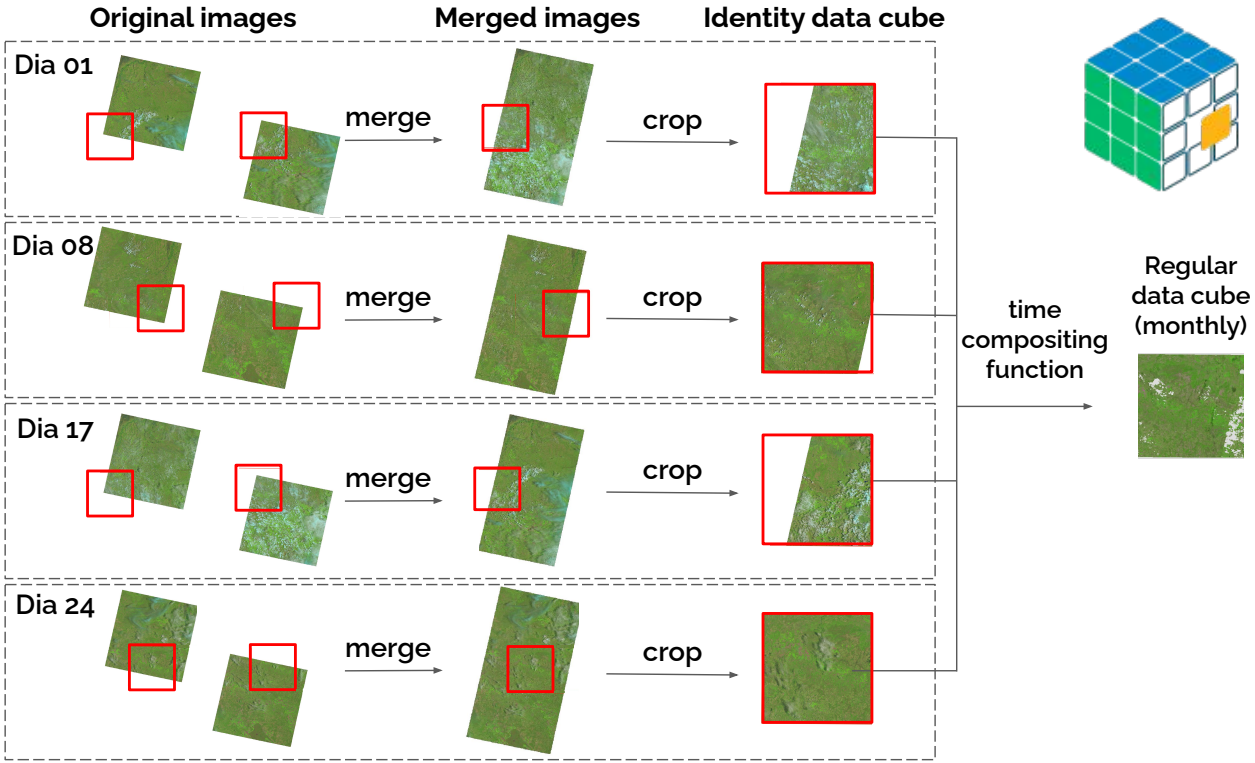
Collections of ARD satellite imagens

Multidimensional Data cubes

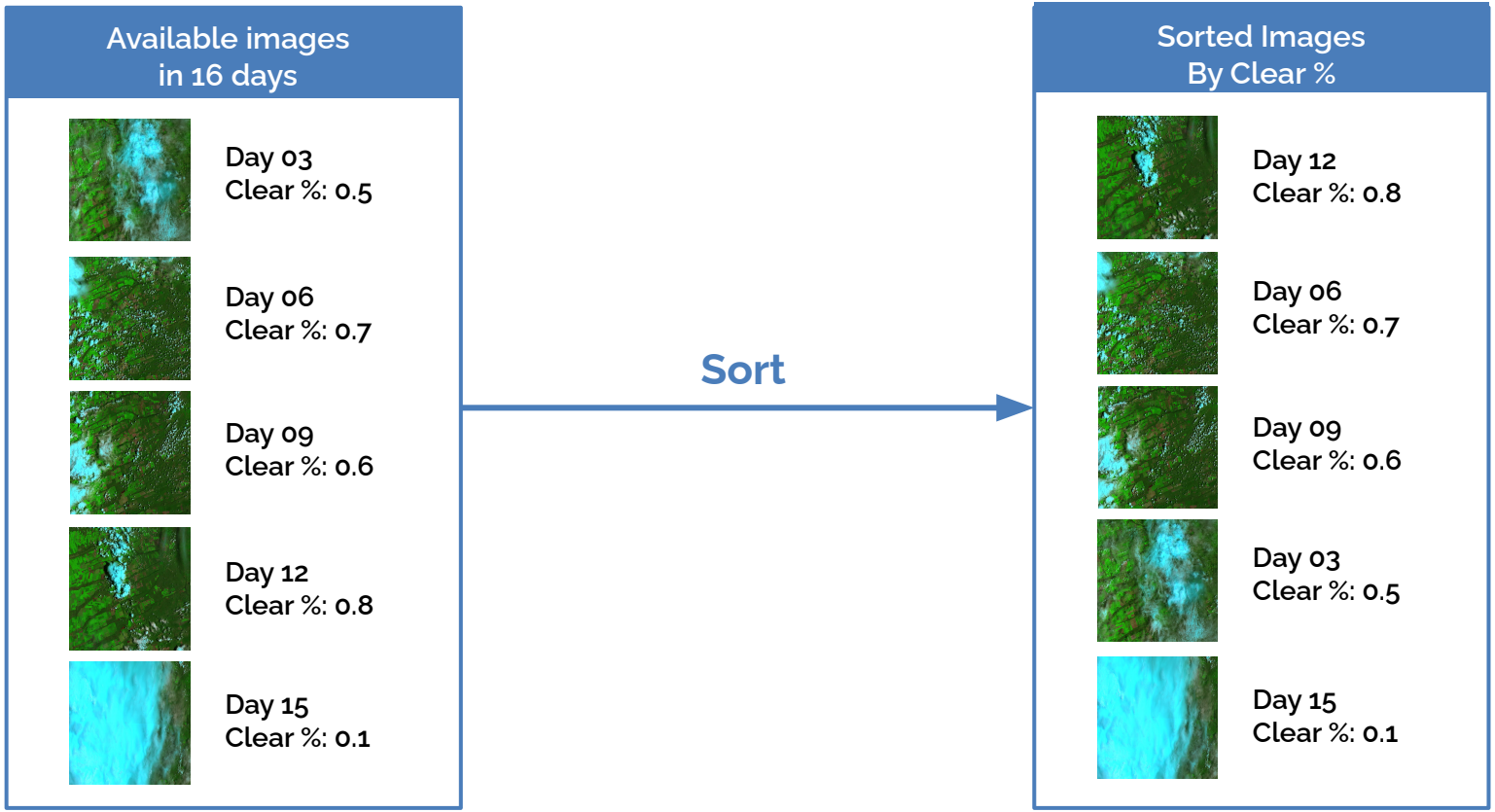
Source: [Kopp et al, 2019]

Data Cube Processing

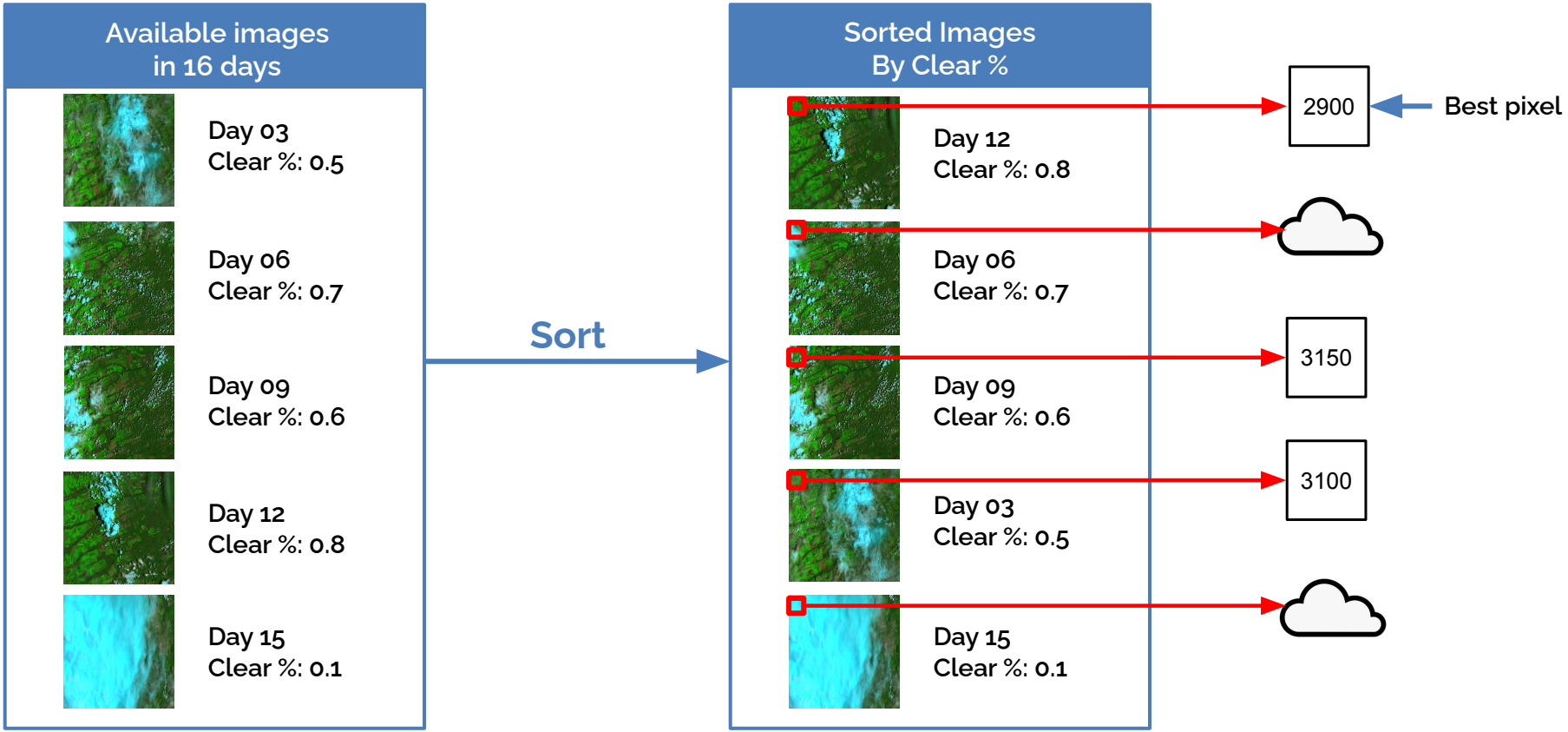




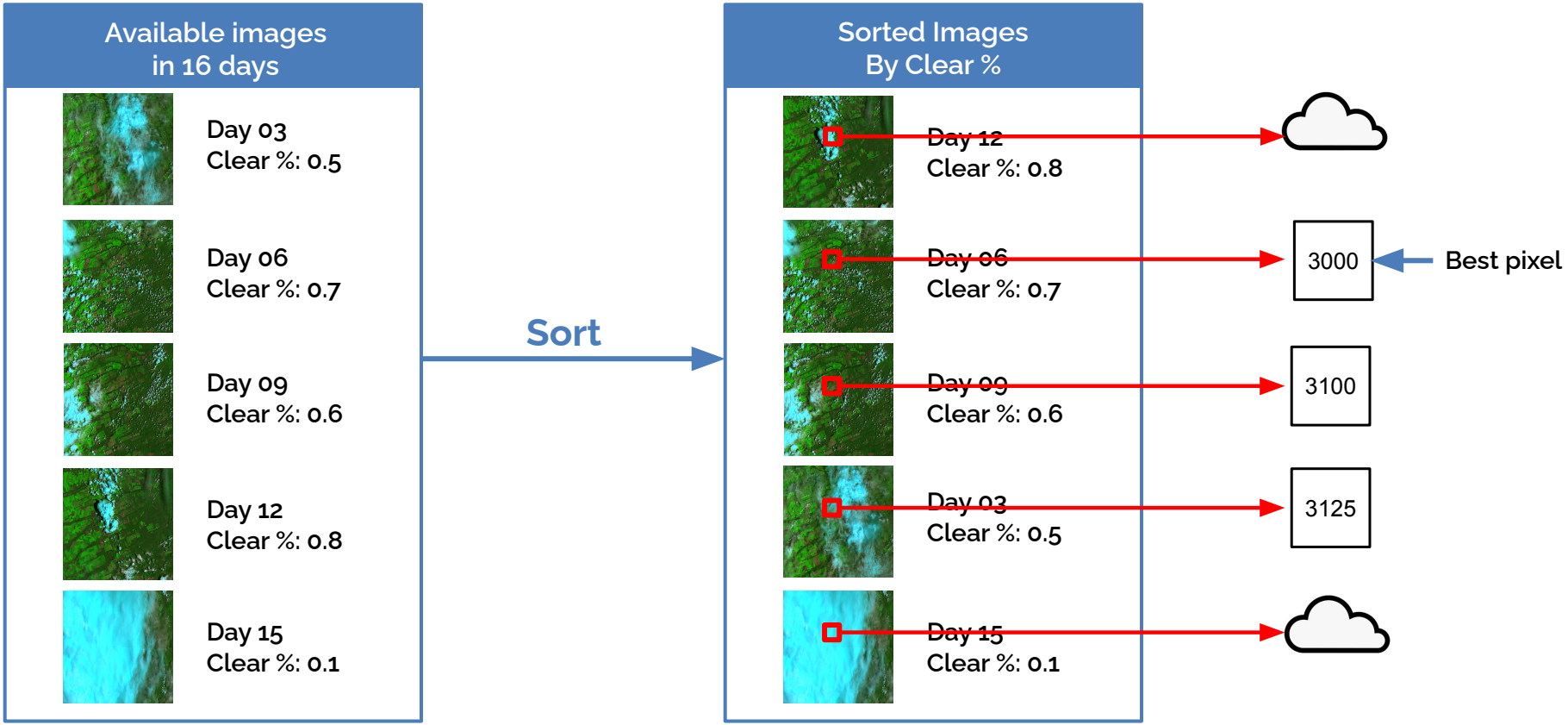
Temporal Composition: Least Cloud Cover First (LCF)



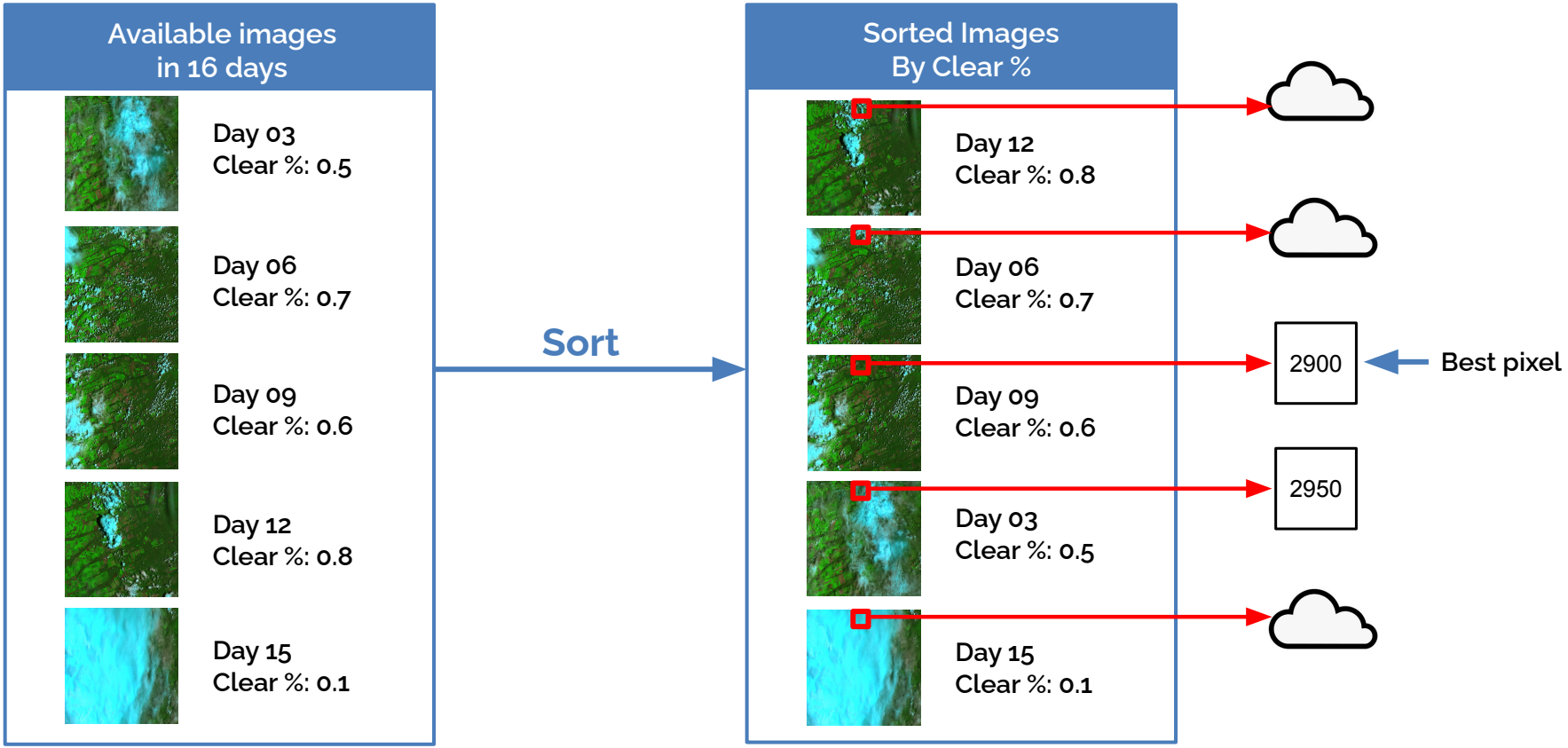
Temporal Composition: Least Cloud Cover First (LCF)



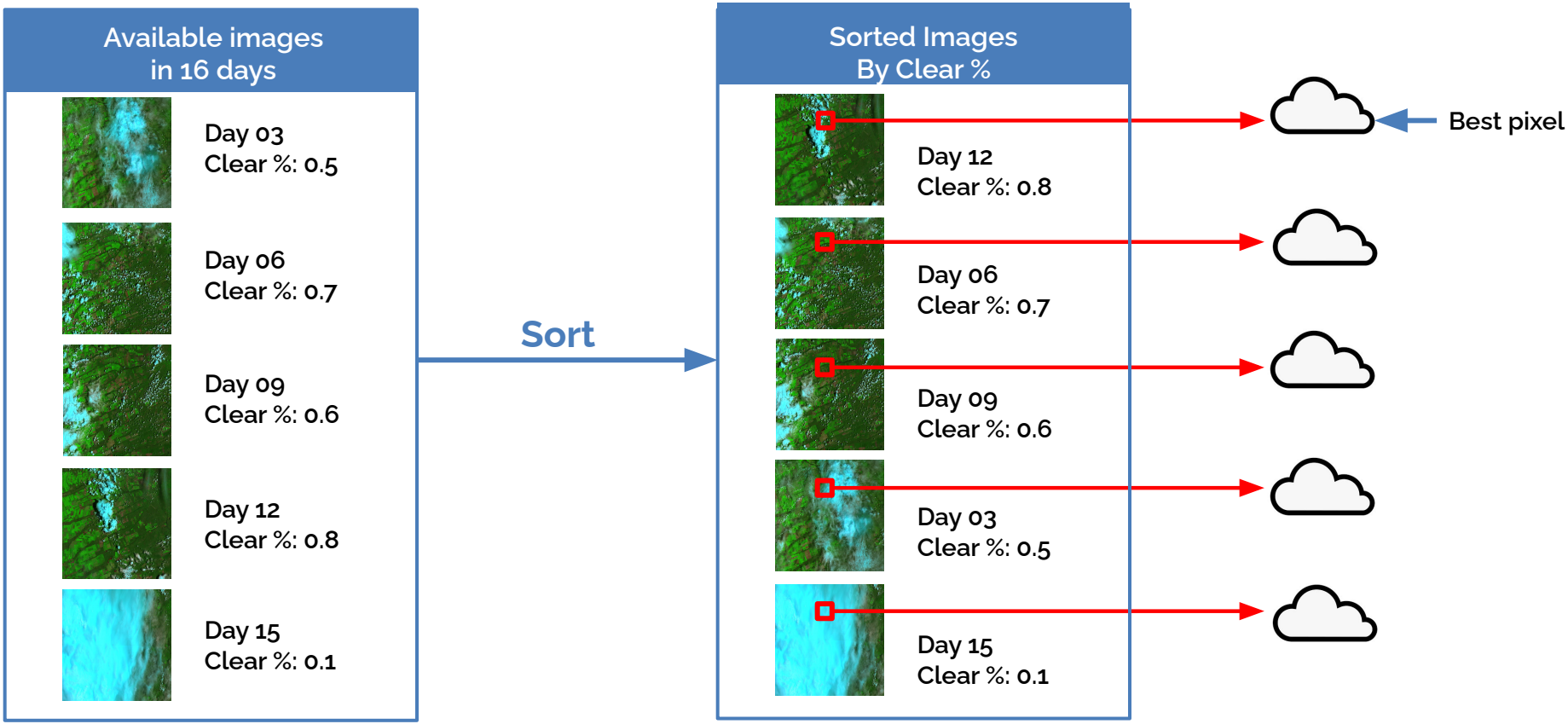
Temporal Composition: Least Cloud Cover First (LCF)



Temporal Composition: Least Cloud Cover First (LCF)

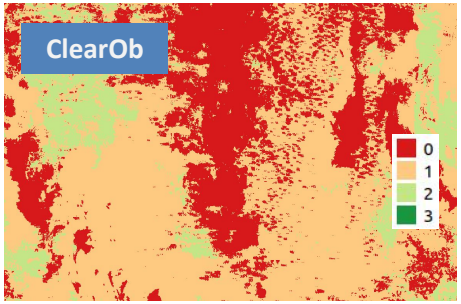


Temporal Composition: Least Cloud Cover First (LCF)

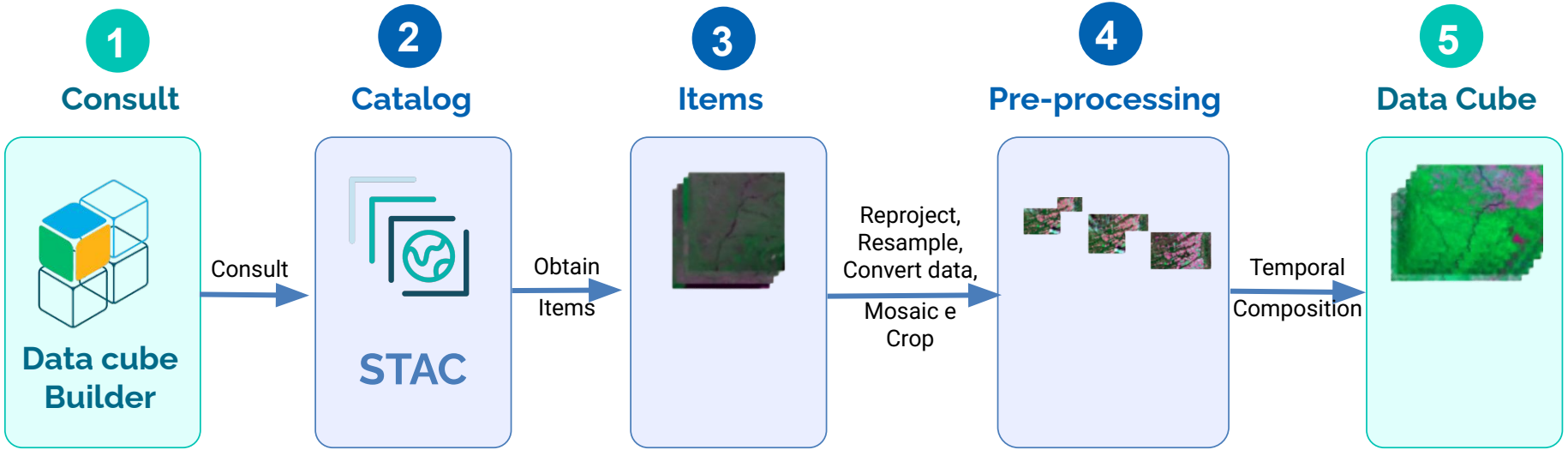


Data Cube Bands

	Attributes	Description
Spectral Bands	Surface reflectance bands	Images processed as surface reflectance level
Extras Bands	Provenance	Day of the year indicating the source image sensing date
	ClearOb	Number of Clear Observations (No cloud, or Cloud Shadow) in the temporal compositing period.
	TotalOb	Total Number of Observation in the temporal compositing period.
	Quality (Cloud Mask)	Cloud, Cloud Shadow and clear observation mask
Spectral Indices	NDVI	Normalized Difference Vegetation Index
	EVI	Enhanced Vegetation Index
	NBR	Normalized Burn Ratio Index



Production and cataloging of data cubes



Data Cubes: available products

CBERS-4/MUX - Level-4-SR - Data Cube - LCF 2 months

COG Earth Observation Data Cube generated from CBERS-4/MUX Level-4 SR product over Brazil extension. This dataset is provided in Cloud Optimized GeoTIFF (COG) file format.…

1/1/2016, 12:00:00 AM UTC - 4/30/2026, 12:00:00 AM UTC

CBERS-4/WFI - Level-4-SR - Data Cube - LCF 16 days

COG Earth Observation Data Cube generated from CBERS-4/WFI Level-4 SR product over Brazil extension. This dataset is provided in Cloud Optimized GeoTIFF (COG) file format.…

1/1/2016, 12:00:00 AM UTC - 5/8/2026, 12:00:00 AM UTC

CBERS/WFI - Level-4-SR - Data Cube - LCF 8 days

COG Earth Observation Data Cube generated from CBERS-4/WFI and CBERS-4A/WFI Level-4 SR products over Brazil extension. This dataset is provided in Cloud Optimized…

1/1/2020, 12:00:00 AM UTC - 5/24/2026, 12:00:00 AM UTC

Landsat 6M São Paulo State Mosaic

COG Earth Observation Data Cube generated from Landsat Level-2 product over Sao Paulo State extension. This dataset is provided in Cloud Optimized GeoTIFF (COG) file format.…

5/1/2013, 12:00:00 AM UTC - 10/31/2023, 12:00:00 AM UTC

Landsat Collection 2 - Level-2 - Data Cube - LCF 16 days

COG Earth Observation Data Cube generated from Landsat Level-2 product over Brazil extension. This dataset is provided in Cloud Optimized GeoTIFF (COG) file format. Th…

1/1/1990, 12:00:00 AM UTC - 5/24/2026, 12:00:00 AM UTC

MOD11A2 v061 - Cloud Optimized GeoTIFF

COG The Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Land Surface Temperature/Emissivity 8-Day (MOD11A2) Version 6.1…

2/18/2000, 12:00:00 AM UTC - 6/1/2026, 11:59:59 PM UTC

MOD13Q1 v061 - Cloud Optimized GeoTIFF

COG The Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices (MOD13Q1) Version 6.1 data are generated every 16 days at 250 meter…

2/18/2000, 12:00:00 AM UTC - 5/24/2026, 11:59:59 PM UTC

MYD11A2 v061 - Cloud Optimized GeoTIFF

COG The Aqua Moderate Resolution Imaging Spectroradiometer (MODIS) Land Surface Temperature/Emissivity 8-Day (MYD11A2) Version 6.1…

7/4/2002, 12:00:00 AM UTC - 6/1/2026, 11:59:59 PM UTC

MYD13Q1 v061 - Cloud Optimized GeoTIFF

COG The Aqua Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices (MYD13Q1) Version 6.1 data are generated every 16 days at 250 meter…

7/4/2002, 12:00:00 AM UTC - 6/1/2026, 11:59:59 PM UTC

Sentinel-2/MSI - Level-2A - Data Cube - LCF 16 days

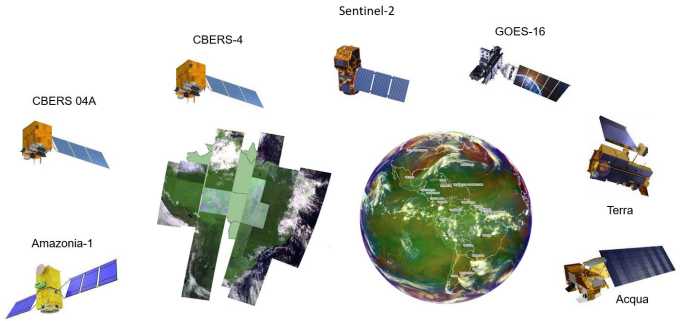
COG Earth Observation Data Cube generated from Copernicus Sentinel-2/MSI Level-2A product over Brazil. This dataset is provided in Cloud Optimized GeoTIFF (COG) file…

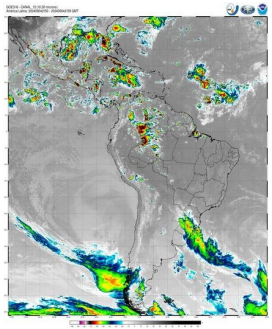
1/1/2017, 12:00:00 AM UTC - 5/24/2026, 12:00:00 AM UTC

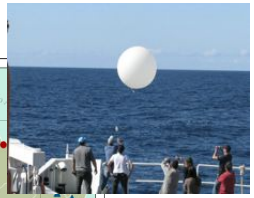
data.inpe.br/dados/cubos

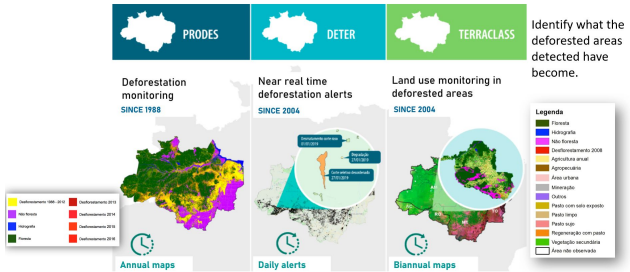
Base de Informações Georreferenciadas (BIG)

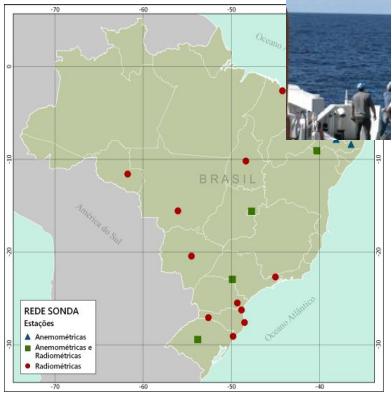
Objective: Integration and modernization of access to INPE's geospatial databases + analytical processing





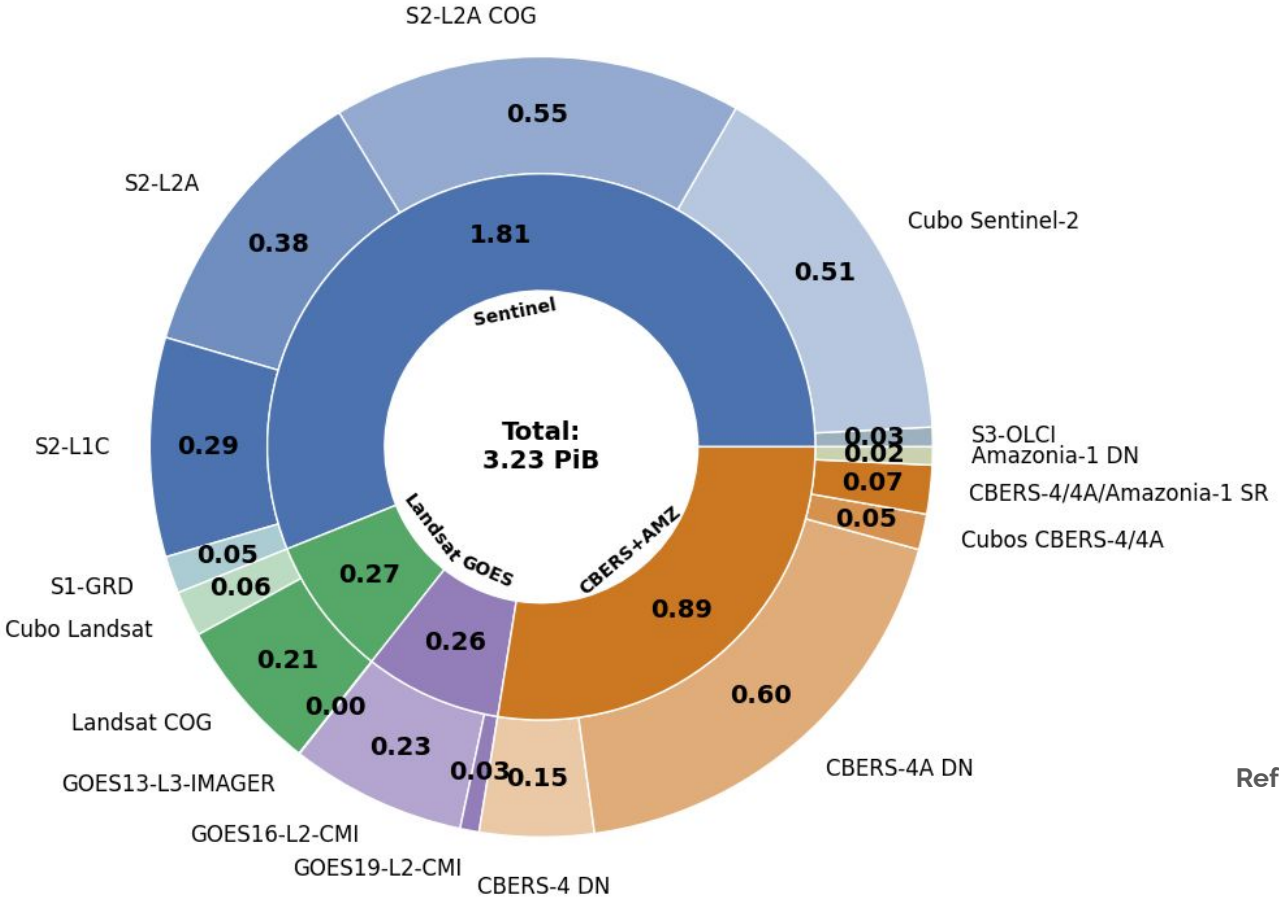






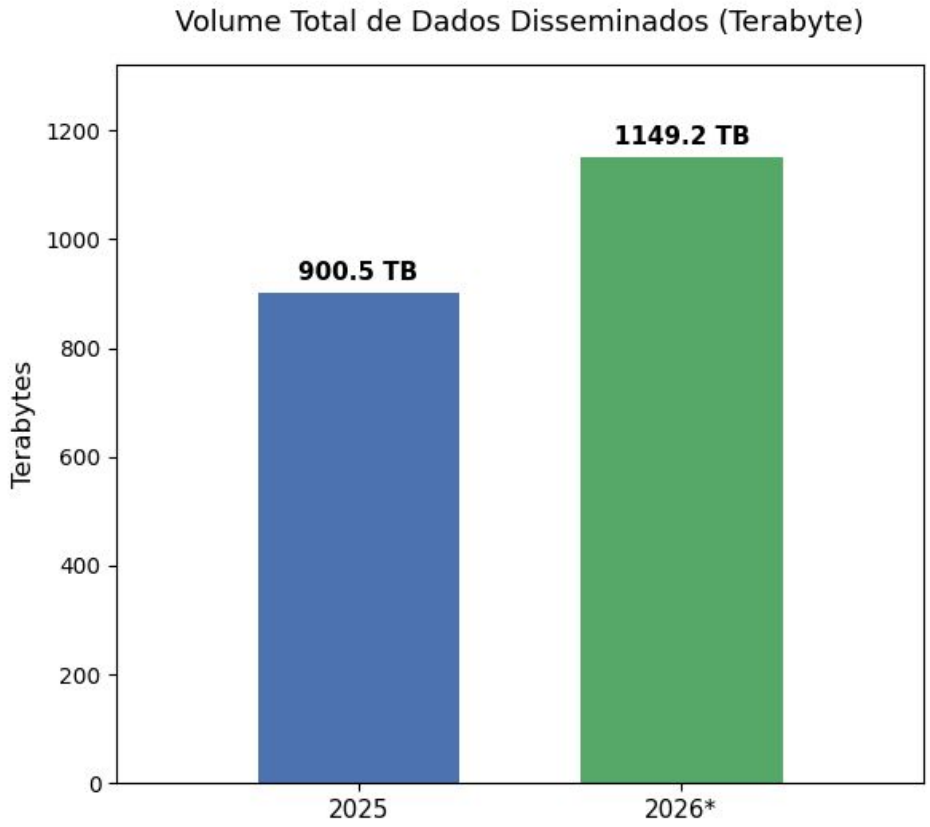
Fonte: http://sonda.ccst.inpe.br/

BIG/BDC Platform: volume of integrated data



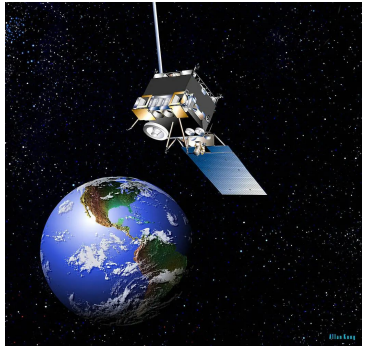
Referência: maio de 2026

BIG/BDC Platform: data access volume

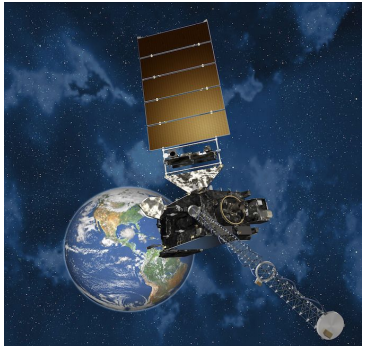


~1.1 Petabyte
em 2026*

*Values calculated up to
August 2026



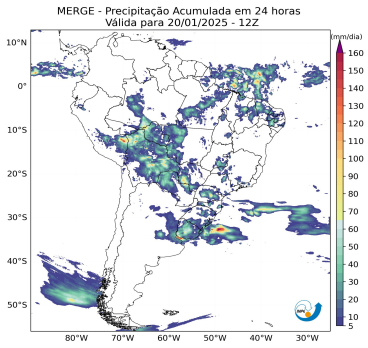
GOES-13



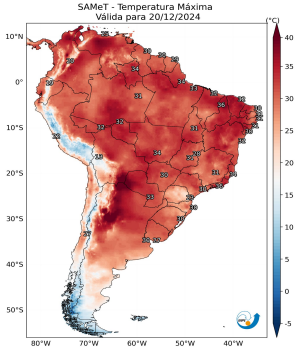
GOES-16

GOES-19

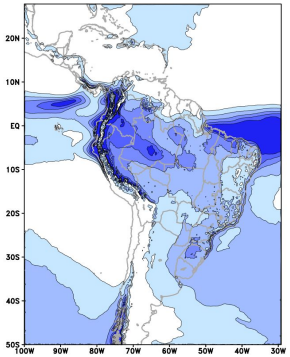
Environmental Satellites and Meteorological Products



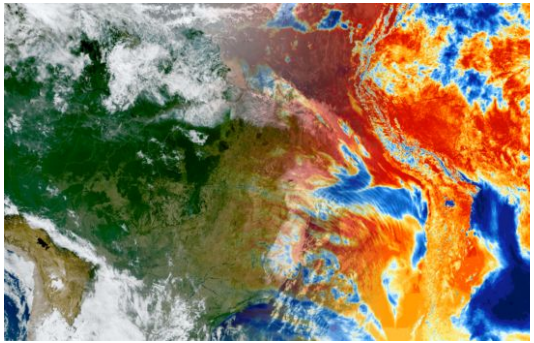
MERGE
Diário e Horário



SAMeT
Diário e Horário

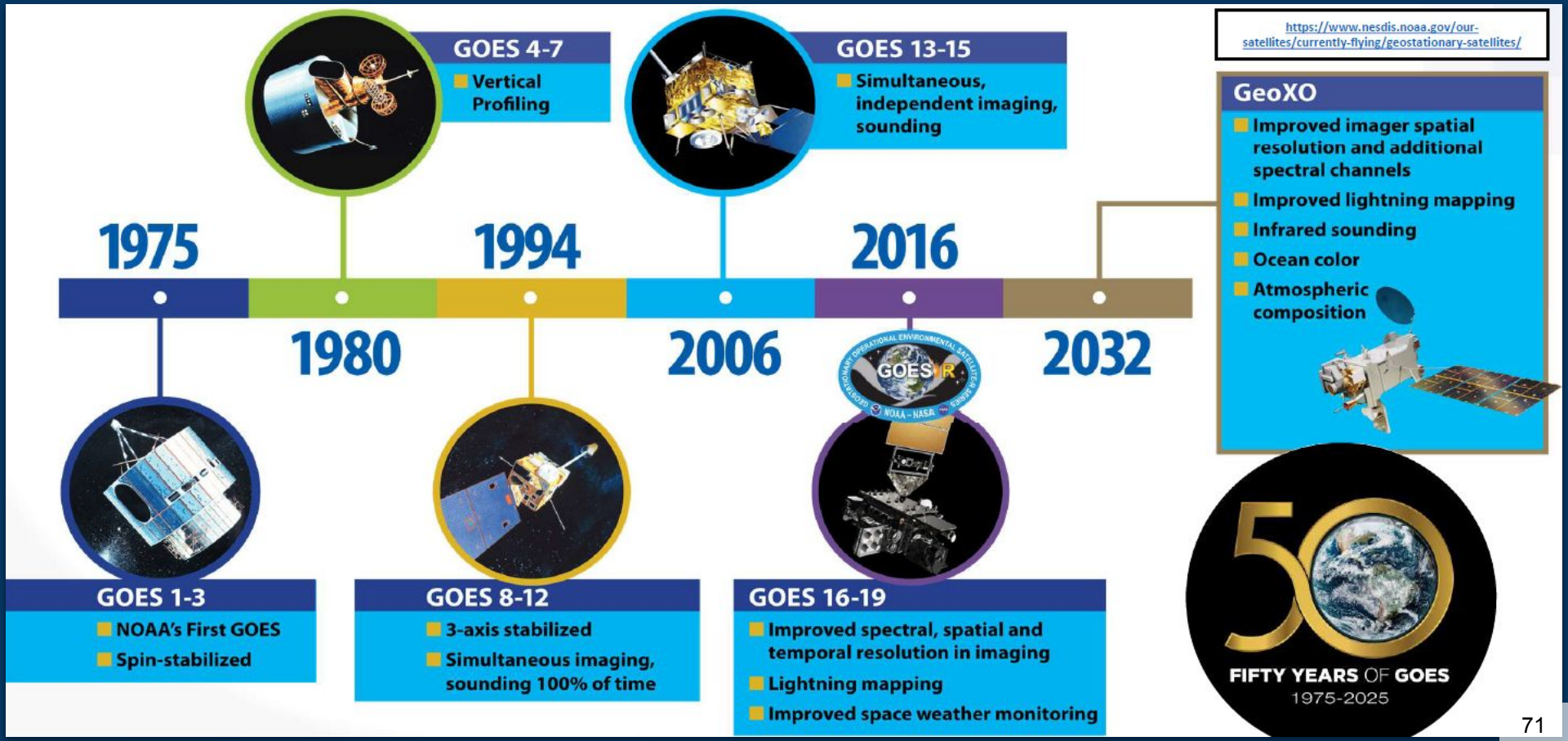


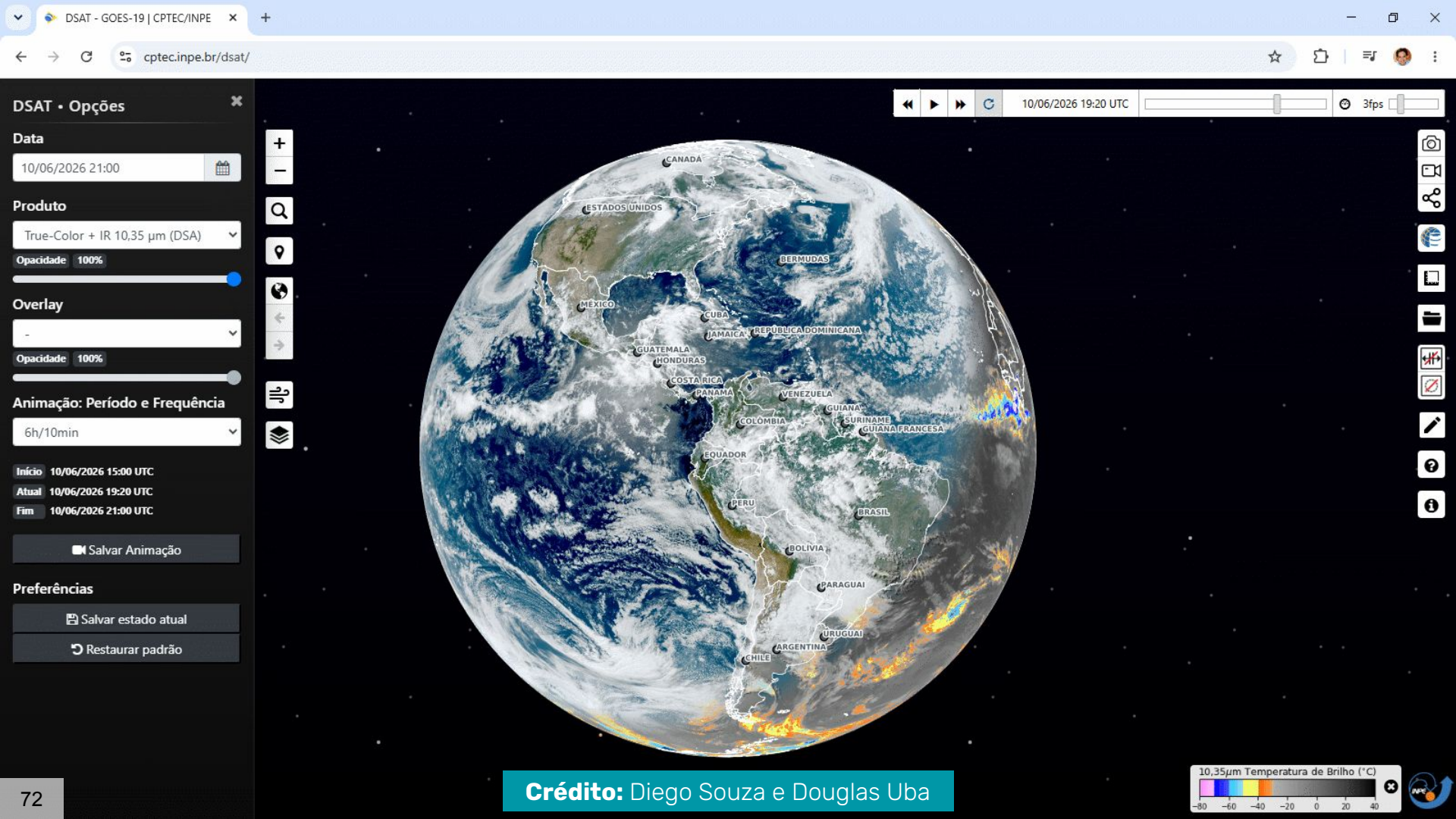
Eta Model - Climate Change - CMIP5 - Day



Radiação Solar – Modelo GL1.2

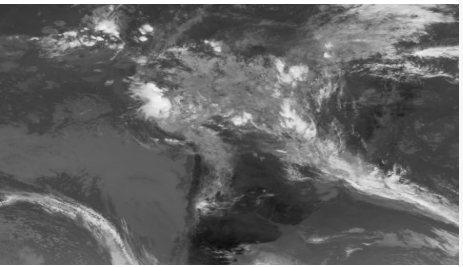
GOES Series

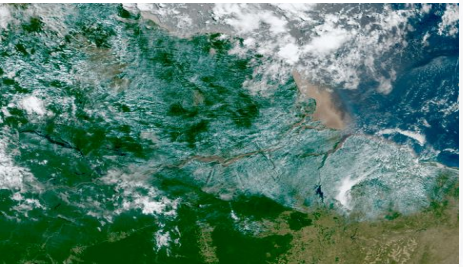

https://www.nesdis.noaa.gov/our-satellites/currently-flying/geostationary-satellites/
GOES 4-7
Vertical Profiling
GOES 13-15
Simultaneous, independent imaging, sounding
GeoXO
Improved imager spatial resolution and additional spectral channels
Improved lightning mapping
Infrared sounding
Ocean color
Atmospheric composition
1975
1980
1994
2006
2016
2032
GOES 1-3
NOAA's First GOES
Spin-stabilized
GOES 8-12
3-axis stabilized
Simultaneous imaging, sounding 100% of time
GOES 16-19
Improved spectral, spatial and temporal resolution in imaging
Lightning mapping
Improved space weather monitoring
FIFTY YEARS OF GOES
1975-2025

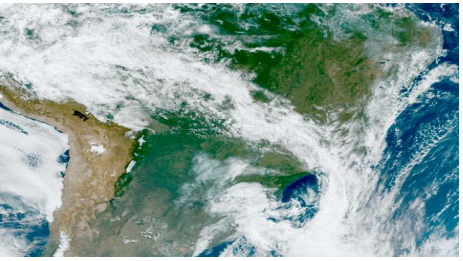


Crédito: Diego Souza e Douglas Uba

GOES: Products Available at BIG







GOES-13/Imager - Level-3 - VIS/IR Imagery (Binary)

GZIP GOES-13 is a satellite of the Geostationary Operational Environmental Satellite (GOES) series operated by the National Oceanic and Atmospheric Administration (NOAA). Launched on May 24, 2006,…

11/25/2011, 3:00:00 PM UTC - 1/8/2018, 3:00:00 PM UTC

GOES-16 Cloud & Moisture Imagery

NetCDF The GOES-16 Advanced Baseline Imager (ABI) L2 Cloud and Moisture Imagery provides 16 spectral bands with high temporal resolution over the American continent. The significance of the GOES…

4/3/2019, 12:00:00 AM UTC - 4/7/2025, 6:30:00 PM UTC

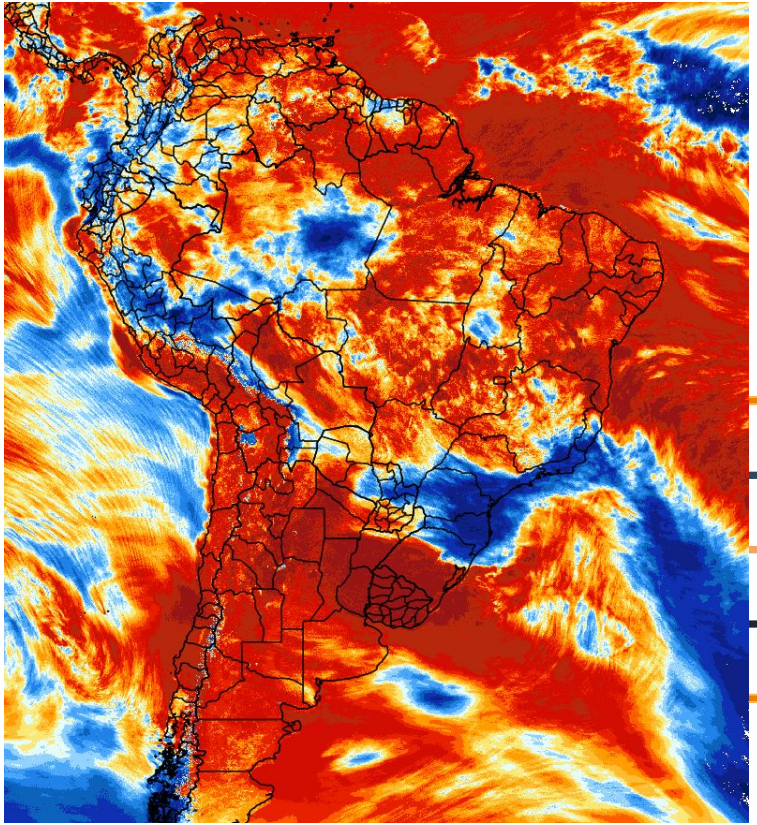
data.inpe.br/dados/goes

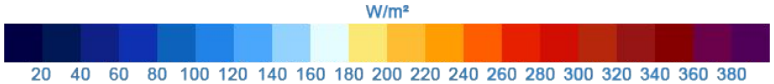
GOES-19 Cloud & Moisture Imagery

NetCDF The GOES-19 Advanced Baseline Imager (ABI) L2 Cloud and Moisture Imagery provides 16 spectral bands with high temporal resolution over the American continent. The significance of the GOES…

4/2/2025, 12:00:00 AM UTC - 6/9/2026, 4:40:00 PM UTC

RADIAÇÃO SOLAR GLOBAL
GL1.2 - GOES





Operational Product
Global Radiation Model
GL versao 1.2

Coverage : SA + adj. Oceans
Period: 1998 – Actual

Resolution
4 km, instantaneous (10 min)
/daily, monthly

Satellites: GOES & MSG

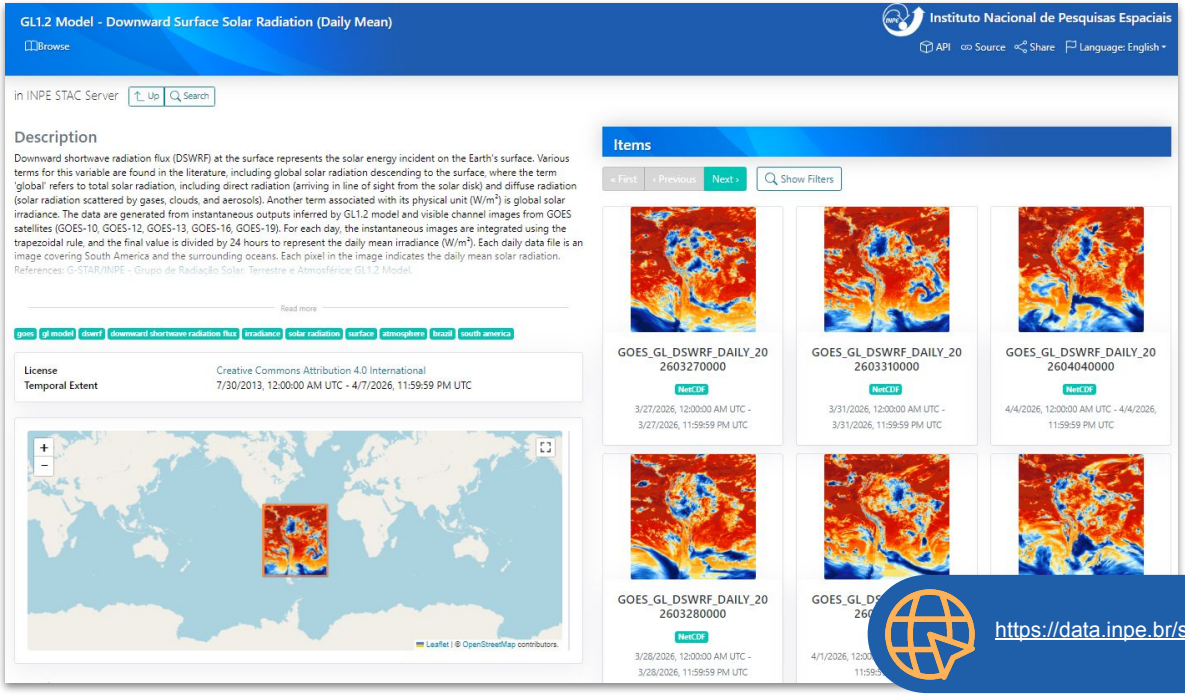
Reference: Ceballos et al., 2004

http://satelite.cptec.inpe.br/radiacao

Crédito: Simone Sievert

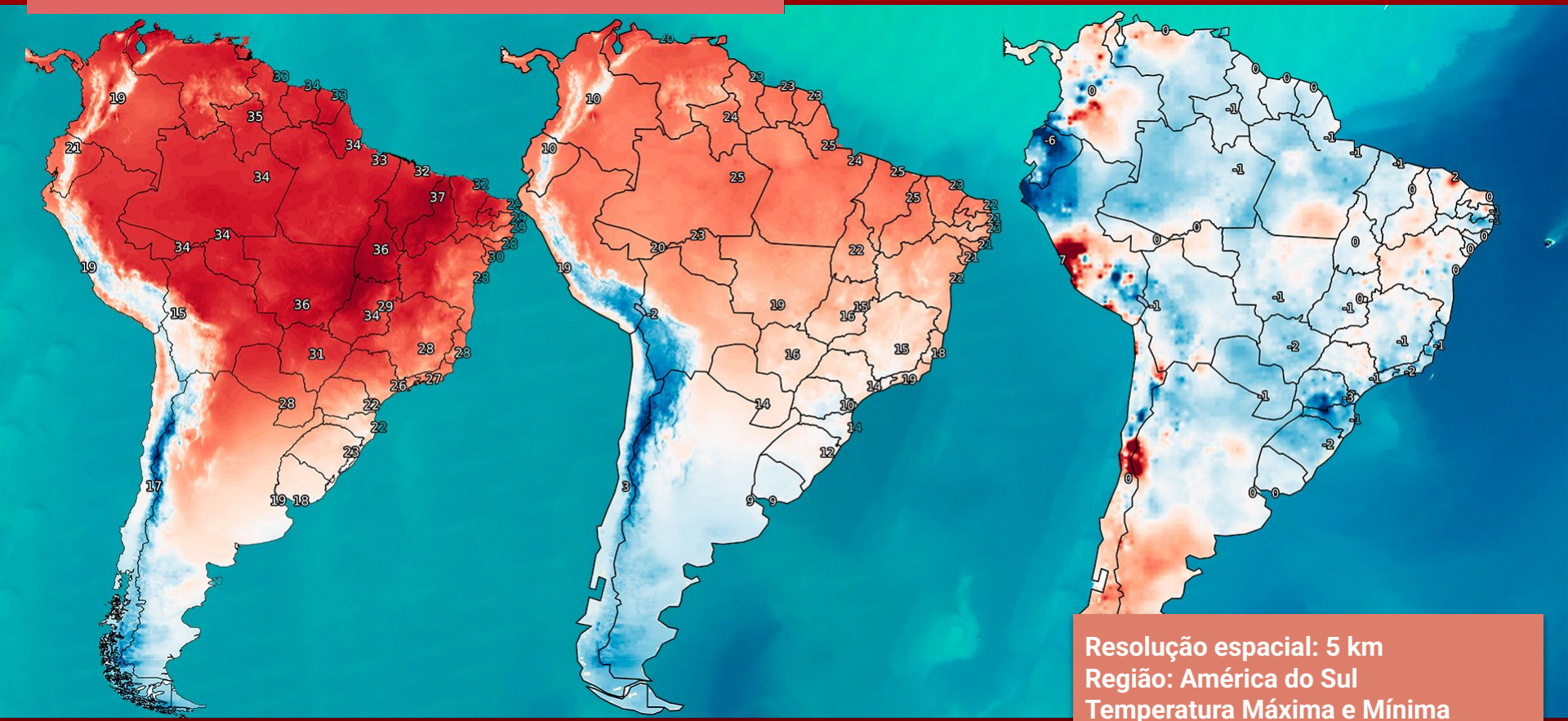
Coleção de Dados GOES-GL1.2

- **Disponível no Catálogo Integrado BIG/BDC**
  - 2013 até o presente - dados diários



https://data.inpe.br/stac/browser/collections/GOES-GL-DSWRF-Daily-1

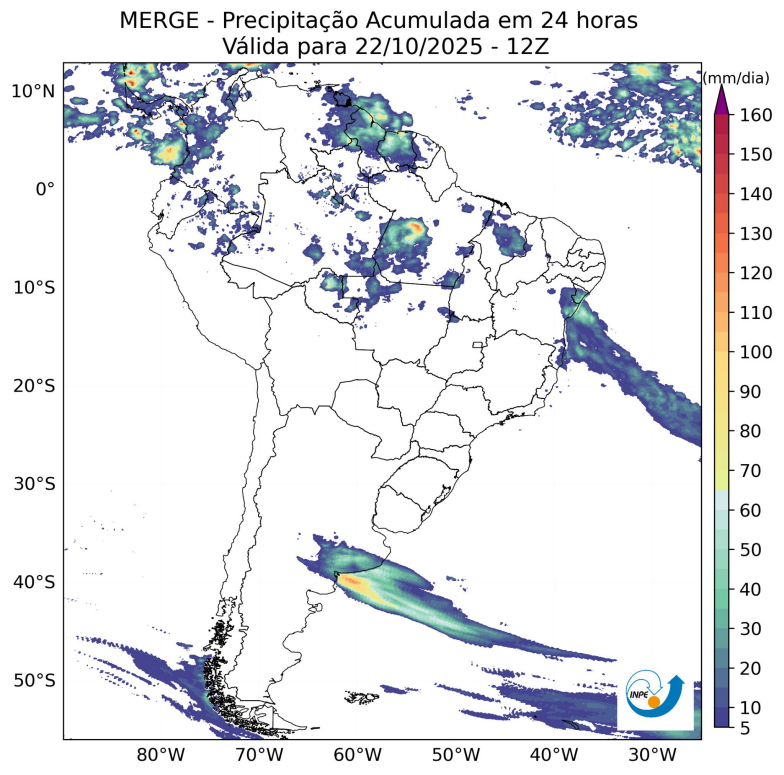
SAMeT / Diário e Horário

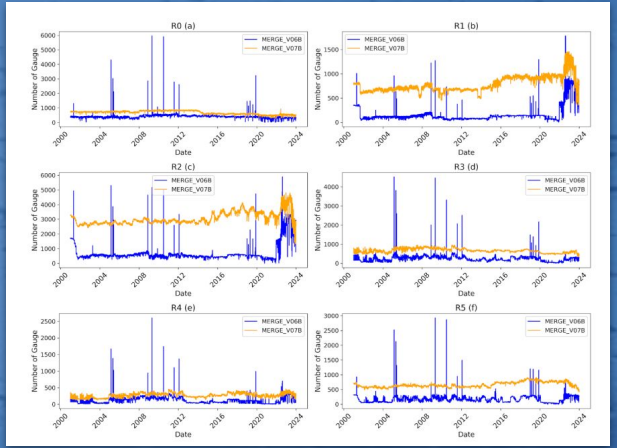


Resolução espacial: 5 km
Região: América do Sul
Temperatura Máxima e Mínima

Rozante, J. R., Ramirez, E., & Fernandes, A. d. A. (2022). A newly developed South American Mapping of Temperature with estimated lapse rate corrections. *International Journal of Climatology*, 42(4), 2135–2152. https://doi.org/10.1002/joc.7356

MERGE / Diário e Horário





Resolução espacial: 11 km
Região: América do Sul
Níveis de Precipitação

Rozante, *et.al* Combining TRMM and Surface Observations of Precipitation: Technique and Validation over South America
https://journals.ametsoc.org/view/journals/wefo/25/3/2010waf2222325_1.xml

SAMeT and Merge: available products

SAMeT - Daily South American Mapping of Temperature

NetCDF South American Mapping of Temperature (SAMeT) is a newly developed product and operationally available for South America through the National Institute for Space Research (CPTEC /…

1/1/2000, 12:00:00 AM UTC - 6/8/2026, 11:59:59 PM UTC

SAMeT - Hourly South American Mapping of Temperature

NetCDF South American Mapping of Temperature (SAMeT) is a newly developed product and operationally available for South America through the National Institute for Space Research (CPTEC /…

4/30/2022, 11:01:00 PM UTC - 6/9/2026, 3:00:00 PM UTC

MERGE - Daily Gridded Precipitation over South America

Text Binary Wmo Grib2 Satellite precipitation estimates combined with surface observations over South America. Satellite estimates for the first few hours are provided by the GPM-Early product; more than 1…

6/1/2000, 12:00:00 PM UTC - 6/8/2026, 12:00:00 PM UTC

MERGE - Hourly Gridded Precipitation over South America

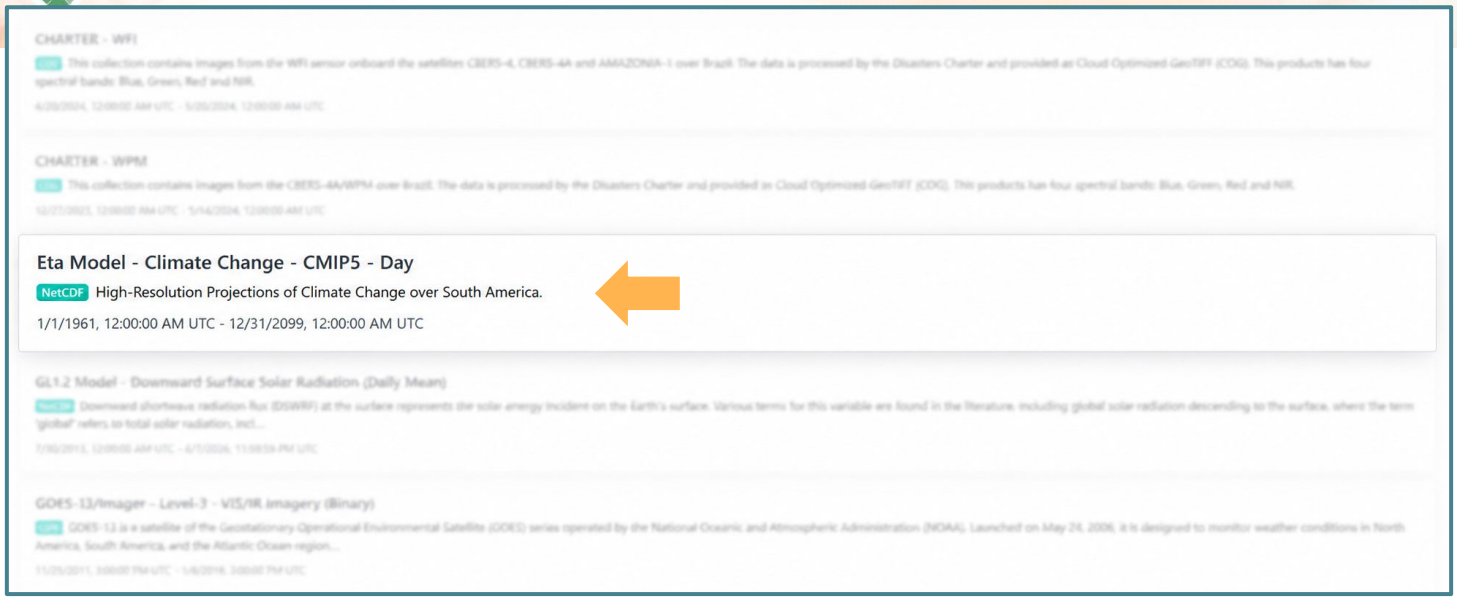
Text Binary Wmo Grib2 Satellite precipitation estimates combined with surface observations over South America. Satellite estimates for the first few hours are provided by the GPM-Early product. This dataset …

12/4/2009, 12:01:00 AM UTC - 6/9/2026, 12:00:00 PM UTC

data.inpe.br/dados/samet

data.inpe.br/dados/merge

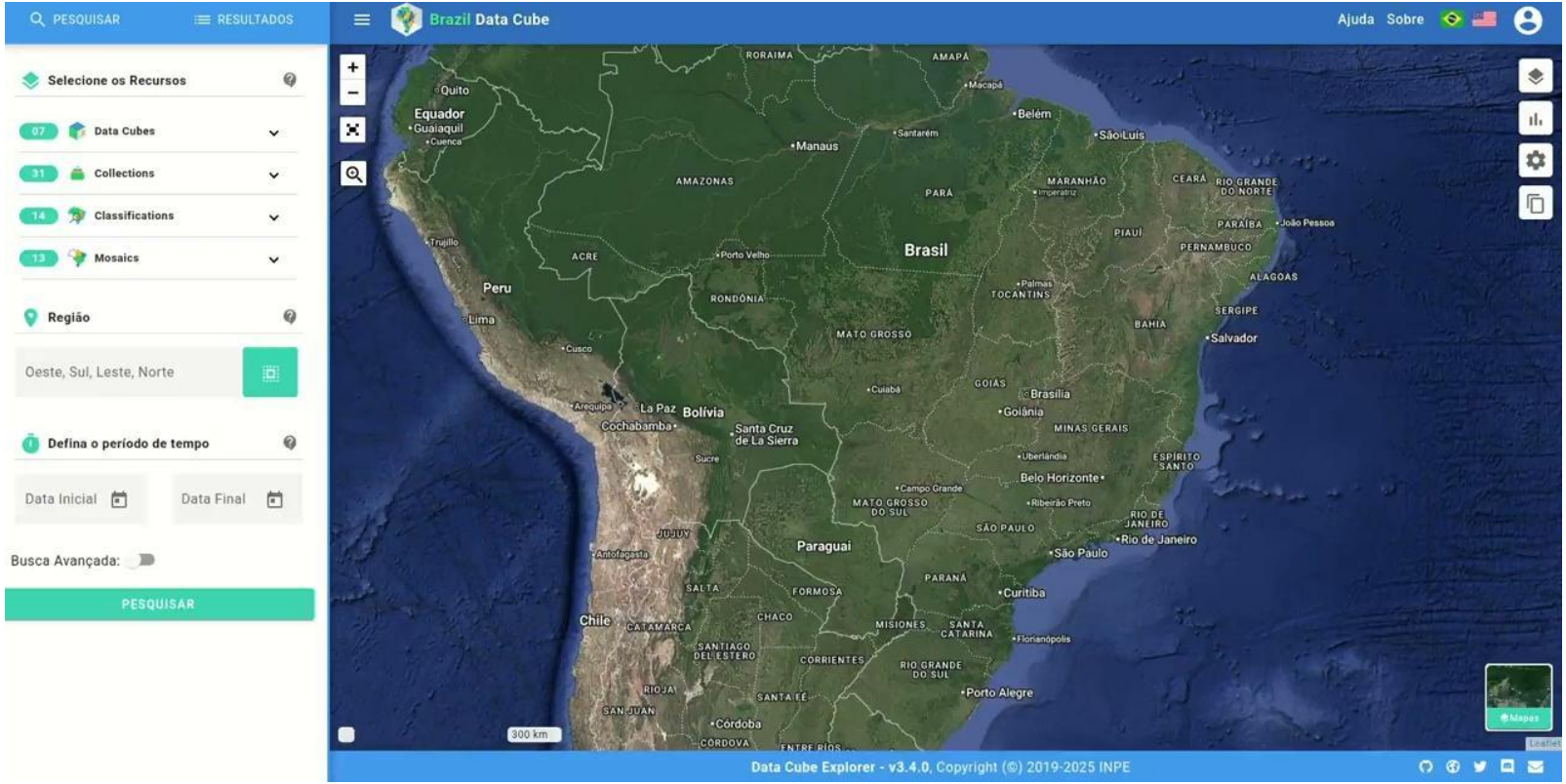
Projeções de Mudanças Climáticas Regionais



Crédito: Diego Chagas

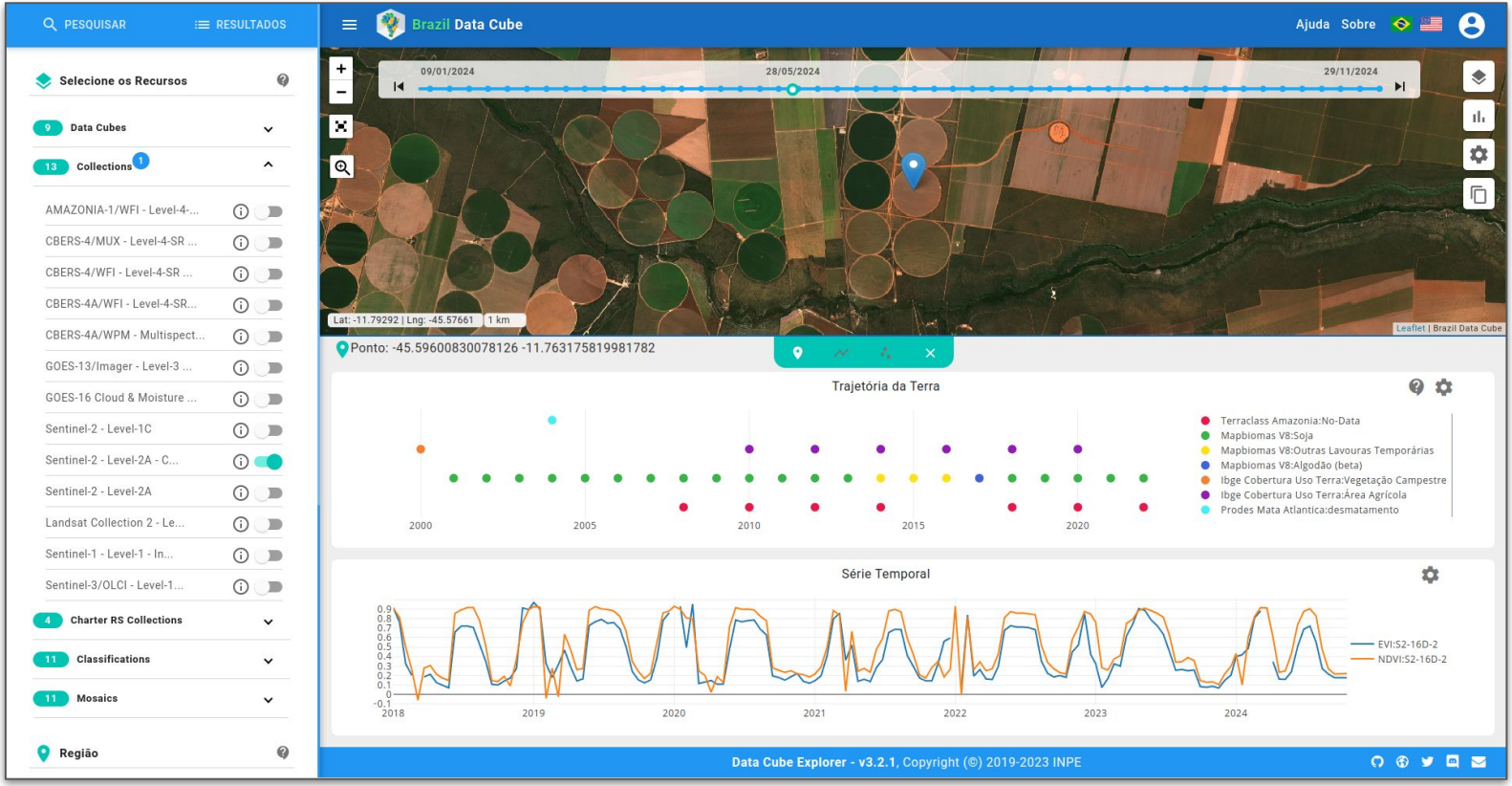
Data Cube Explorer: search and temporal visualization

https://data.inpe.br/bdc/explorer



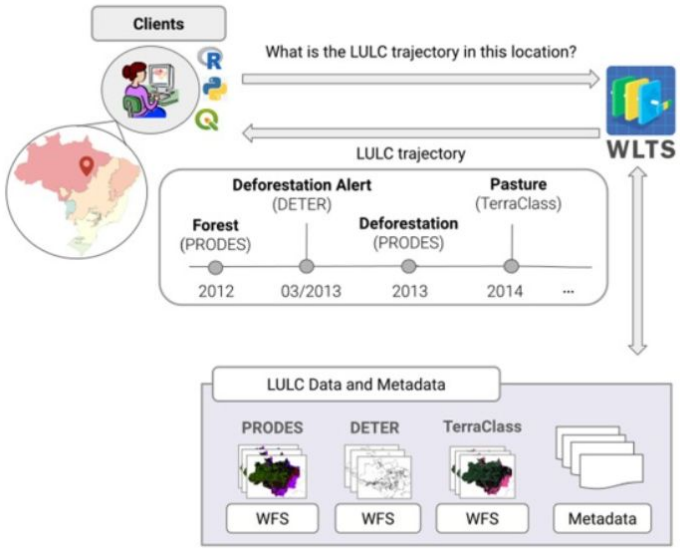
Data Cube Explorer: Time Series and Land Use and Land Cover Trajectories

https://data.inpe.br/bdc/explorer

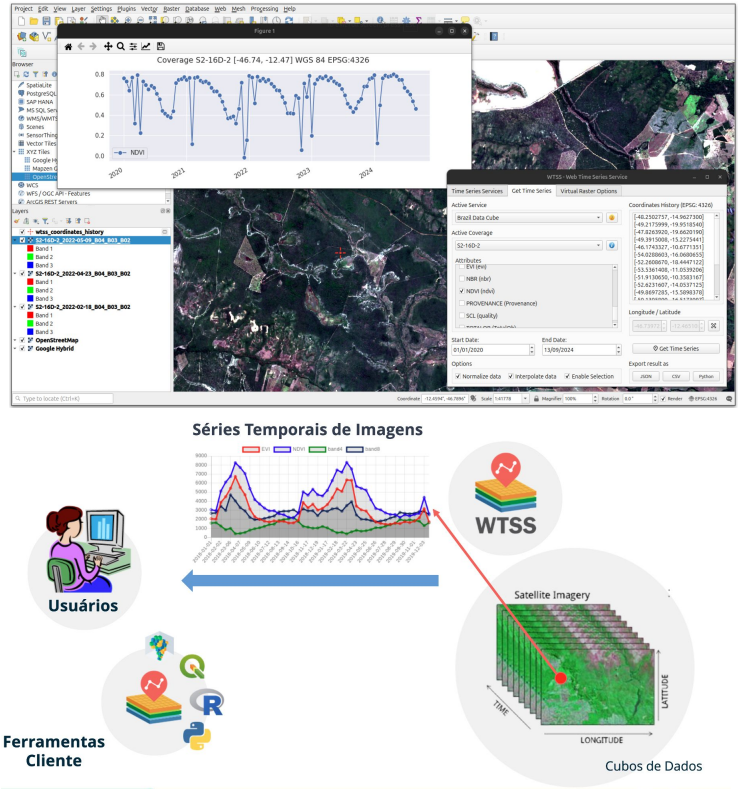


WLTS e WTSS services

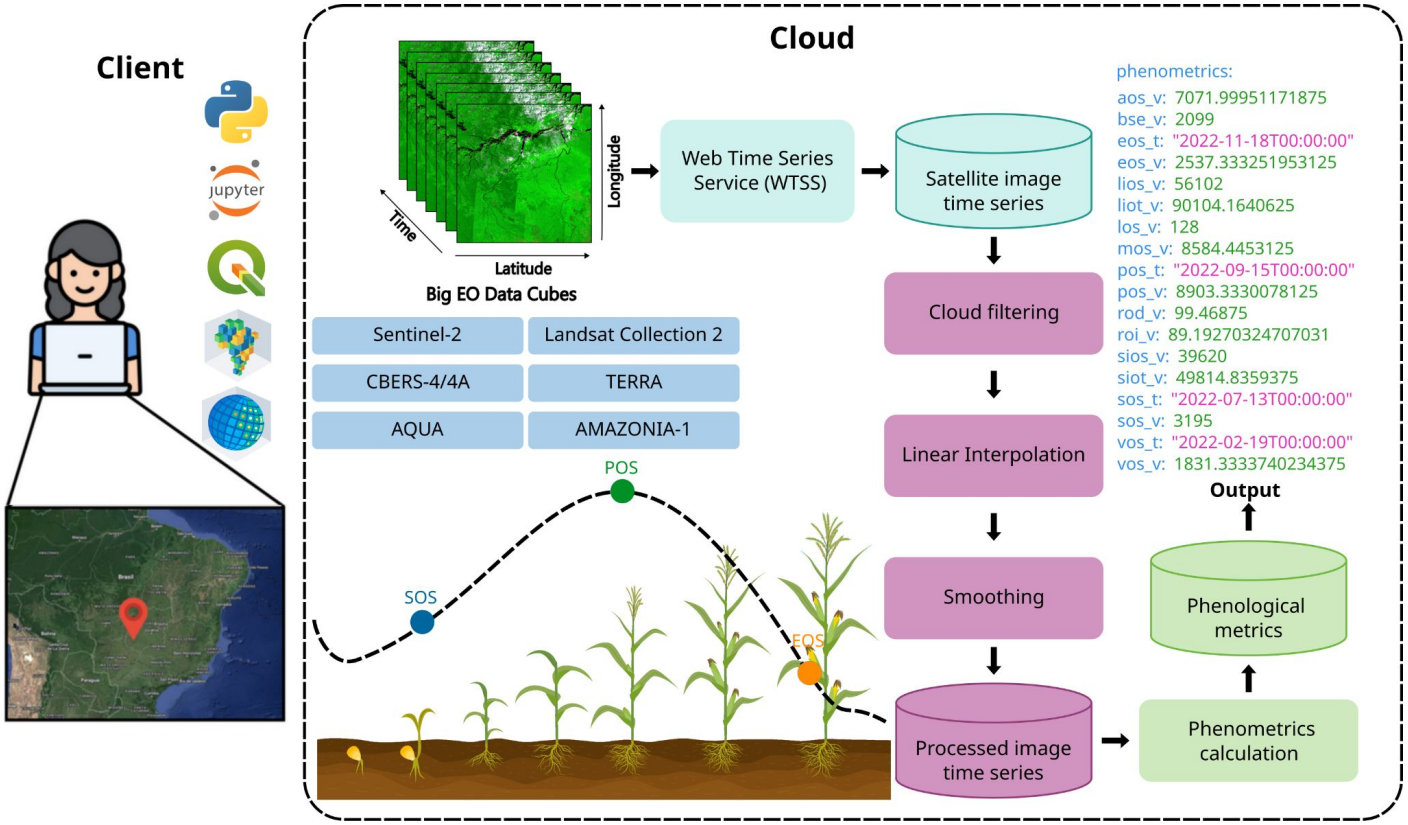
https://data.inpe.br/bdc/wlts



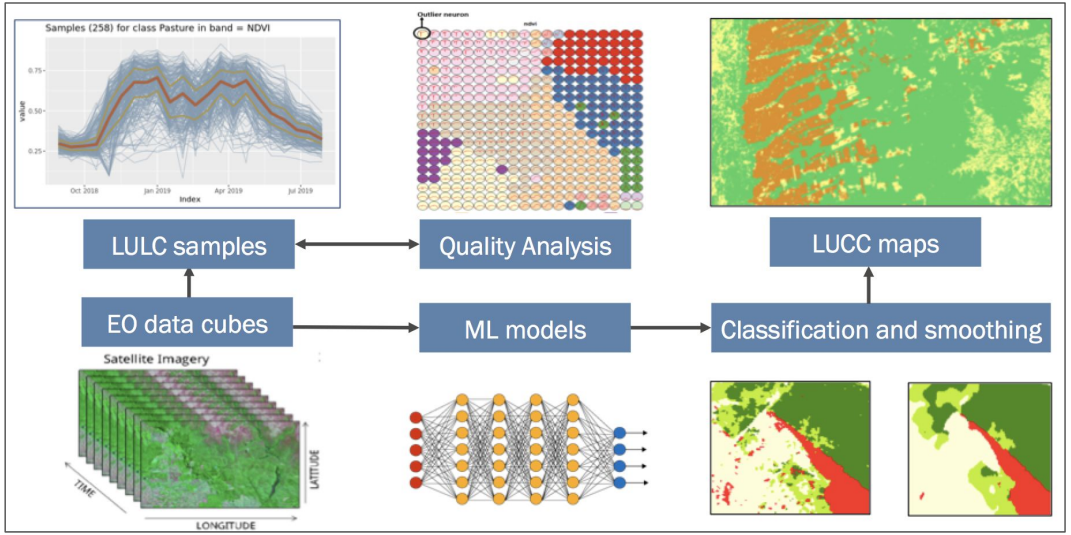
https://data.inpe.br/bdc/wtss



WCPMS: Phenological Metrics Service



SITS (Satellite Image Time Series) R package

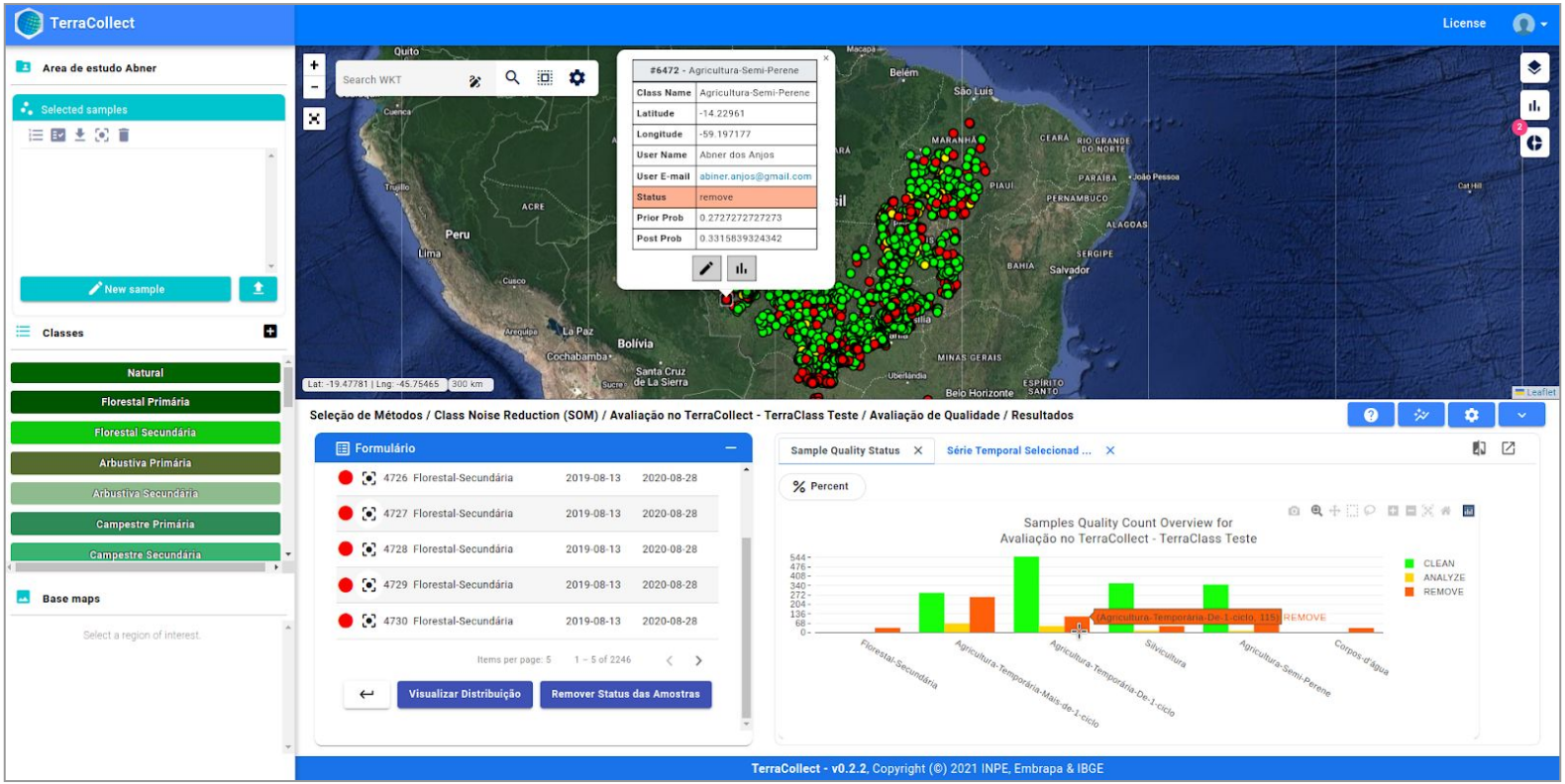



```
cube <- sits_cube(source = "BDC",
                  collection = "S2_10_16D",
                  name = "s2_cube",
                  bands = c("NDVI", "EVI"),
                  tiles = "022024", start_date = "2018-09-01", end_date = "2019-08-28")

samples <- readRDS(url("https://bdc.dpi.inpe.br/rds/S2_10_16D.rds"))

mlp_model <- sits_deeplearning(layers = c(512, 512, 512, 512, 512), activation = "relu")

dl_model <- sits_train(samples, mlp_model)

result <- sits_classify(data = cube, ml_model = dl_model, output_dir = tempdir())
```

https://github.com/e-sensing

TerraCollect: Land Use and Land Cover Sample Collection

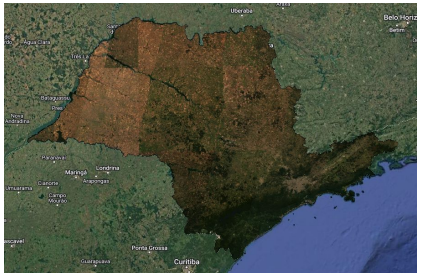
https://data.inpe.br/bdc/terracollect



Interoperability through OGC Services: WMS and TMS

Mosaico Landsat
Estado de São Paulo - 6 meses
(Maio - Outubro)

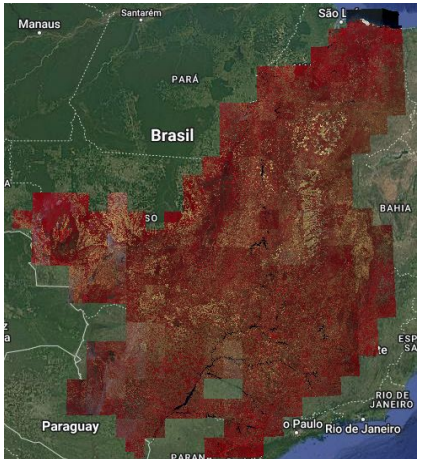
(2013 - 2023)



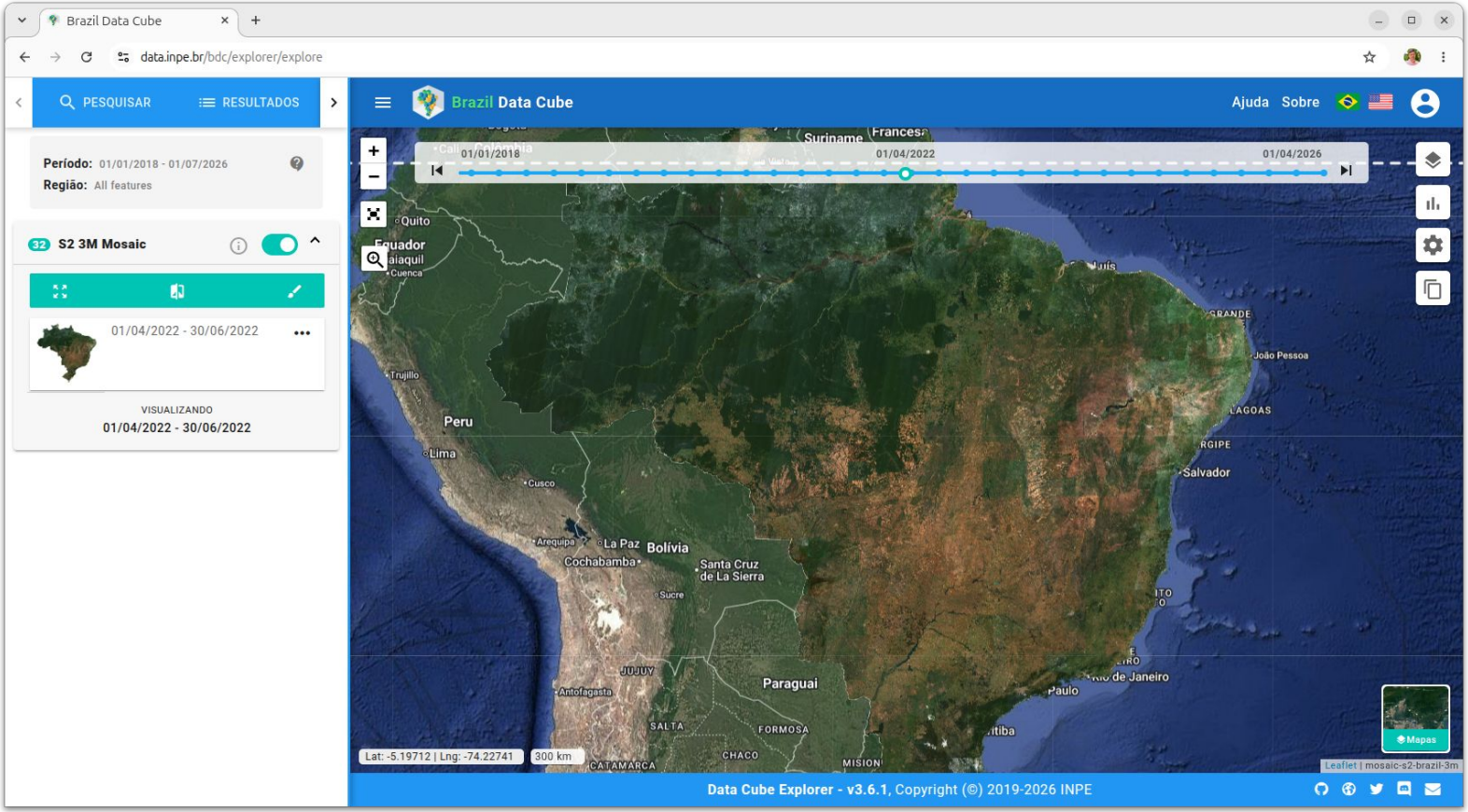
Mosaico Sentinel-2

Cerrado - 2 meses

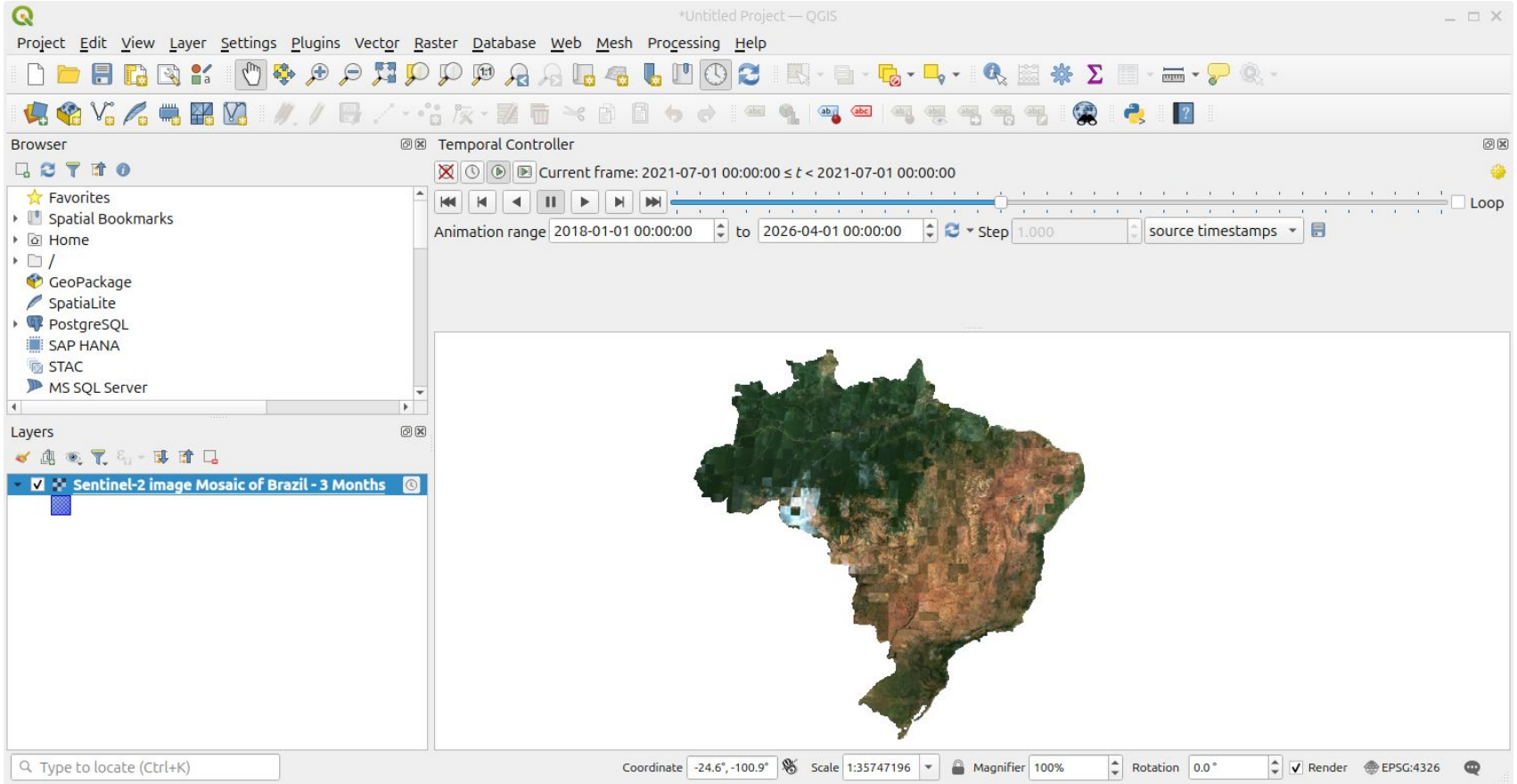
Novembro/2023- Agosto/2024



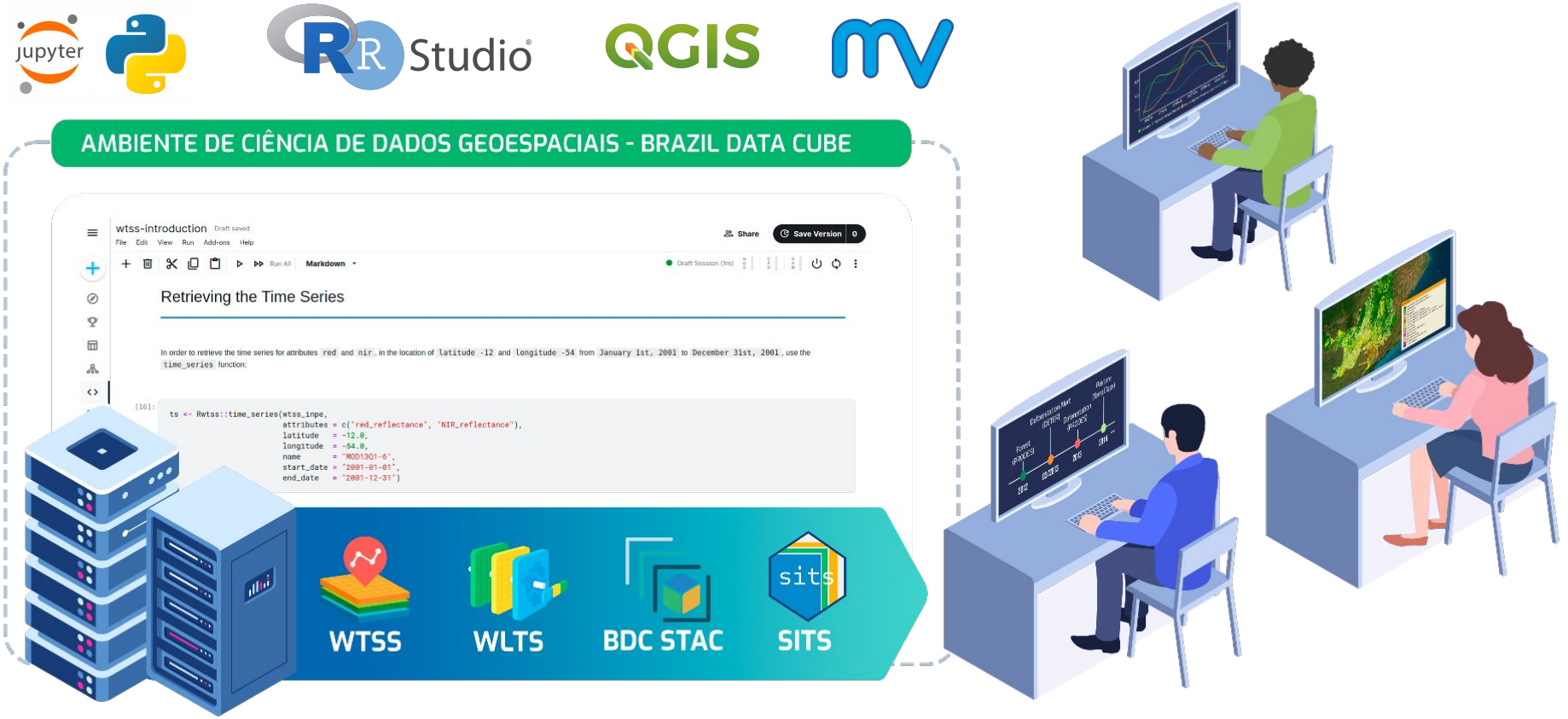
Interoperability through OGC Services: WMS and TMS

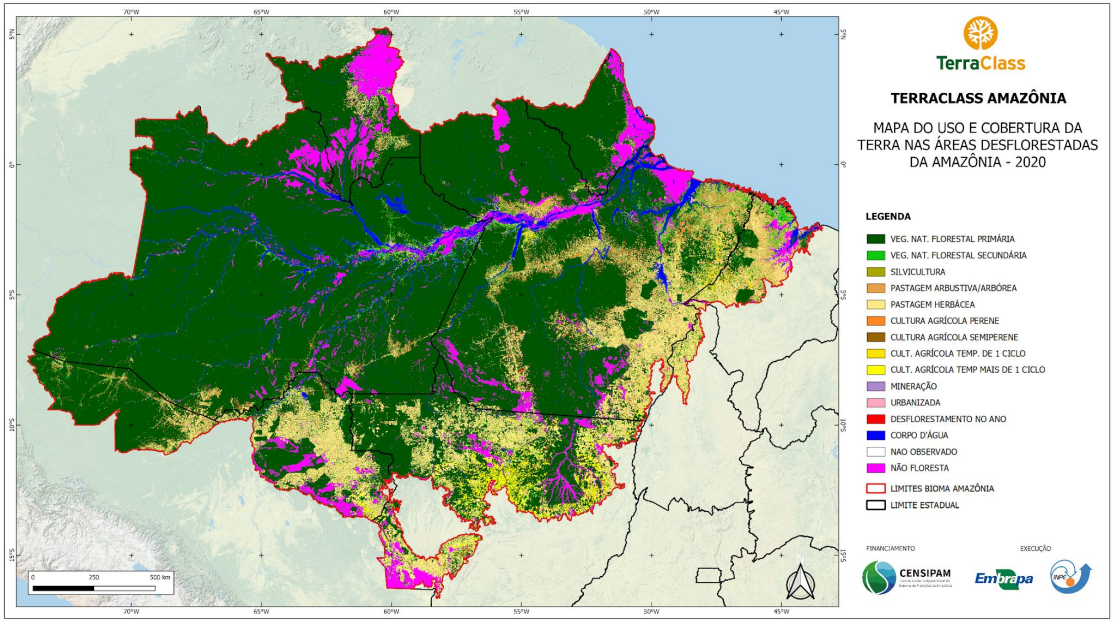


Interoperability through OGC Services: WMS and TMS

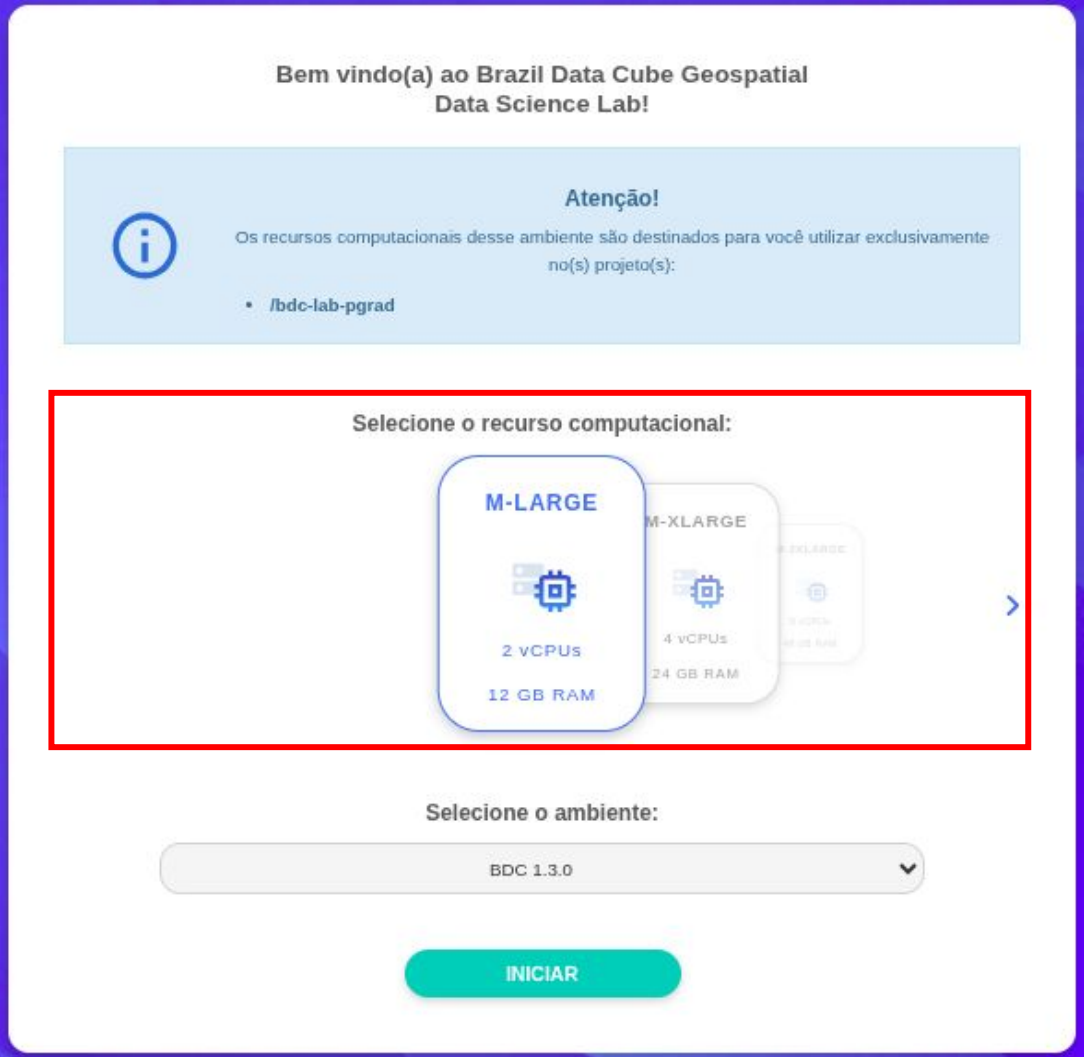


BDC-Lab: Geospatial Data Science Environment



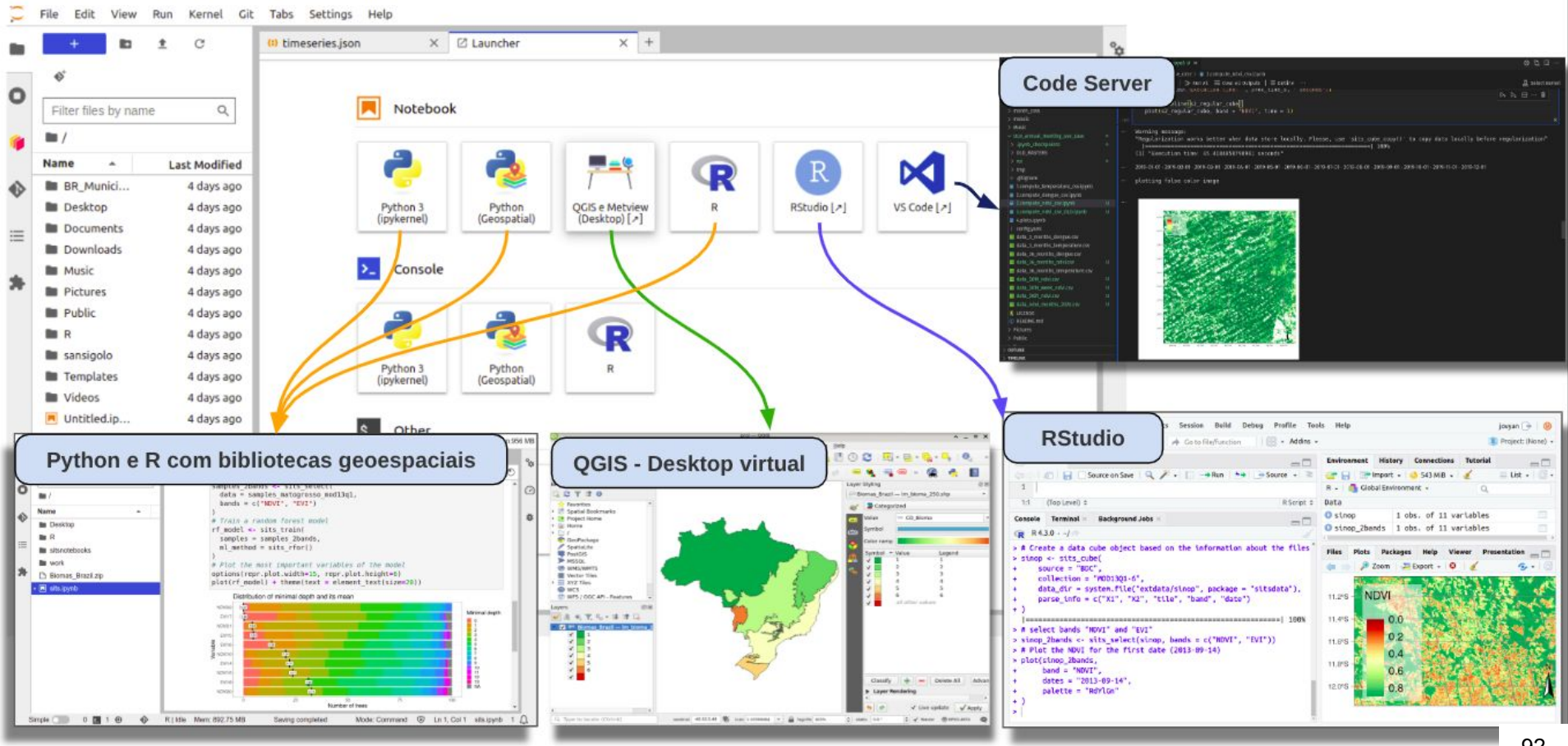


33 Terabytes (292 tiles)
Sentinel-2 data cubes (16-days)
25,000 samples
Random Forest / tempCNN



On-demand instances with different hardware configurations

BDC-Lab 1.4.0 (December/2025)



BDC-Lab: available instances

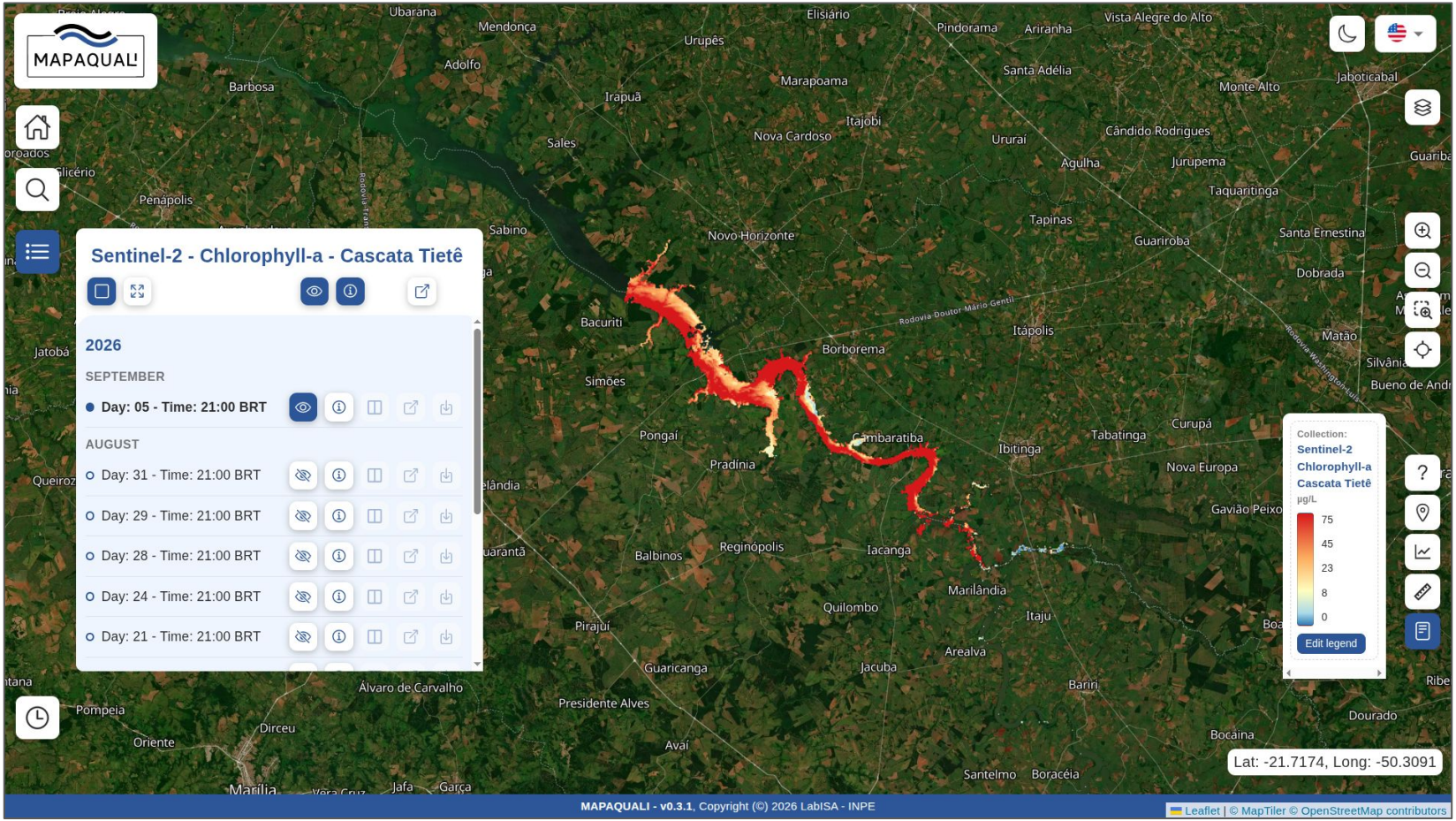
Operational

https://data.inpe.br/bdc/lab

Academic

https://geolab.inpe.br/bdc/lab

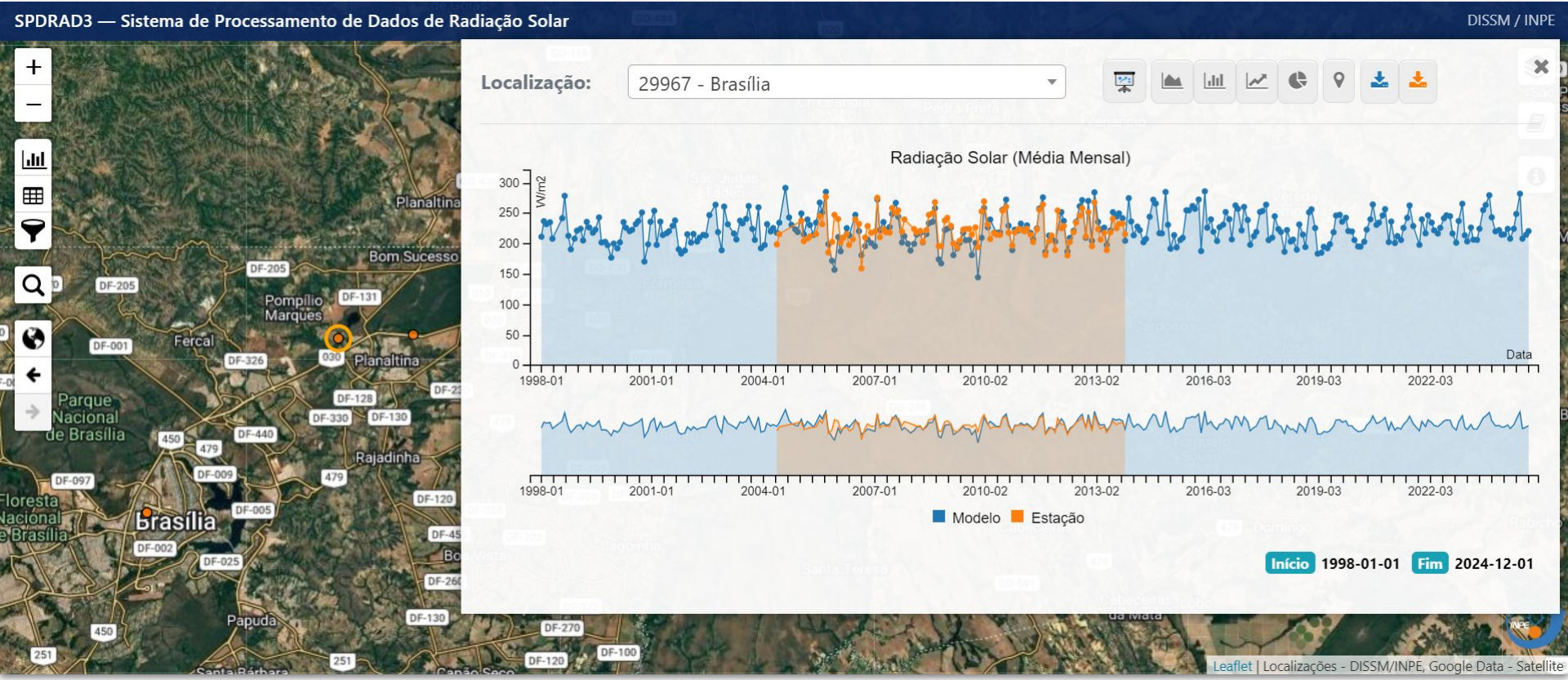
Thematic Platforms: MapaQuali



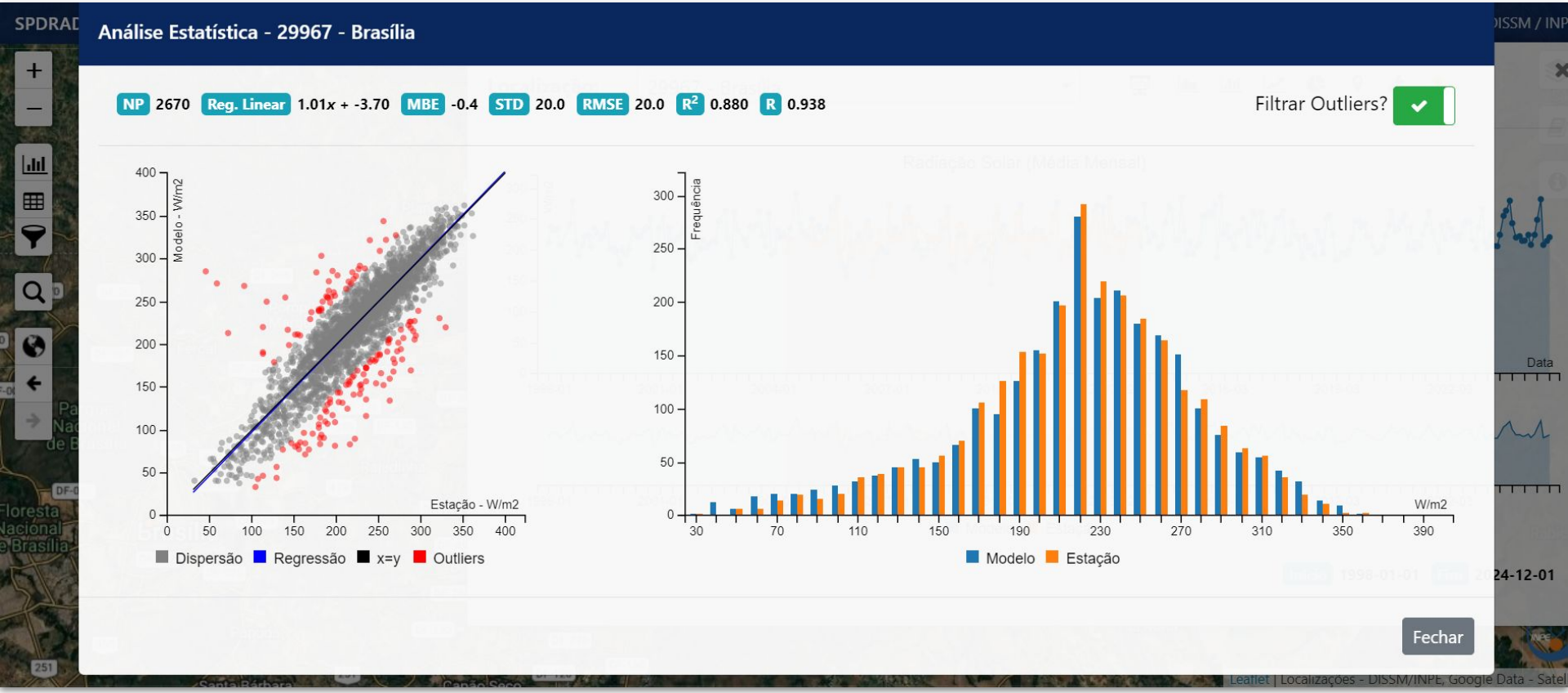
Thematic Platforms: Radiação Solar (SPDRAD3)

data.inpe.br/dissm/spdrad

Thematic Platforms: Solar Radiation (SPDRAD3)



Thematic Platforms: Solar Radiation (SPDRAD3)

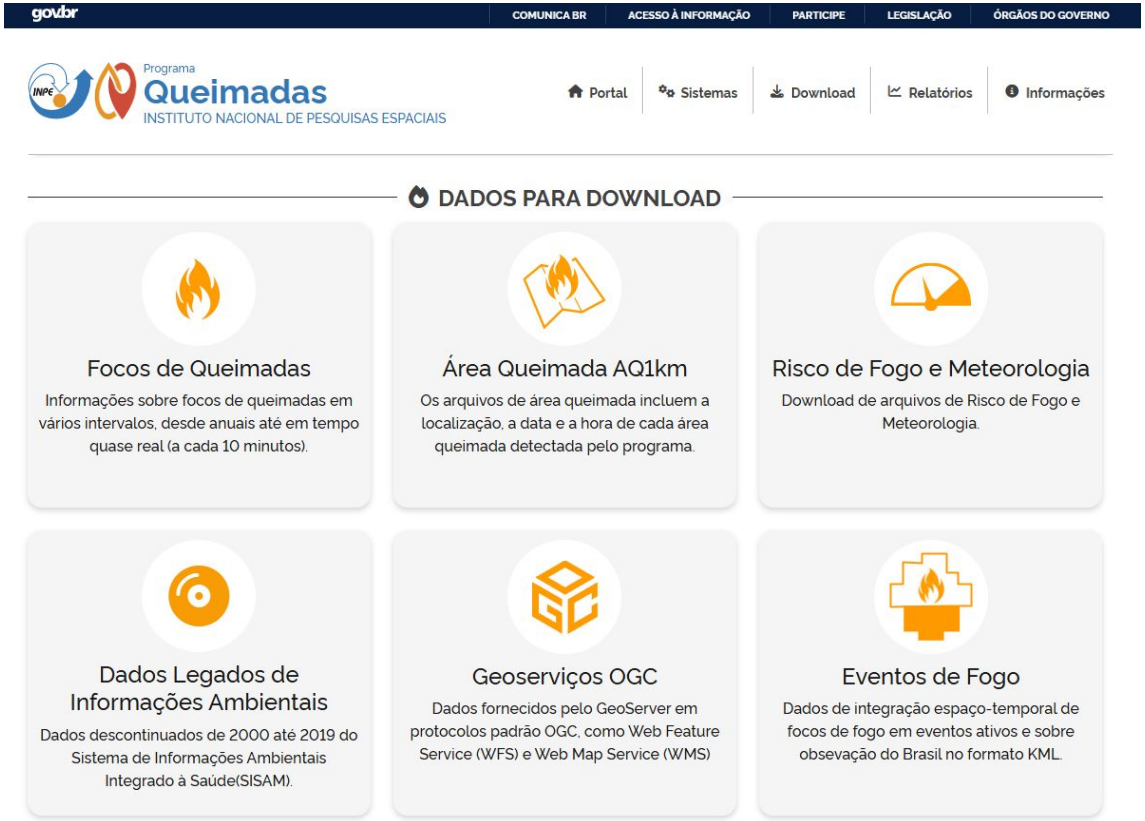


Thematic Platforms: Solar Radiation (SPDRAD3)

Queimadas Program Portal

data.inpe.br/queimadas

Queimadas Program Portal

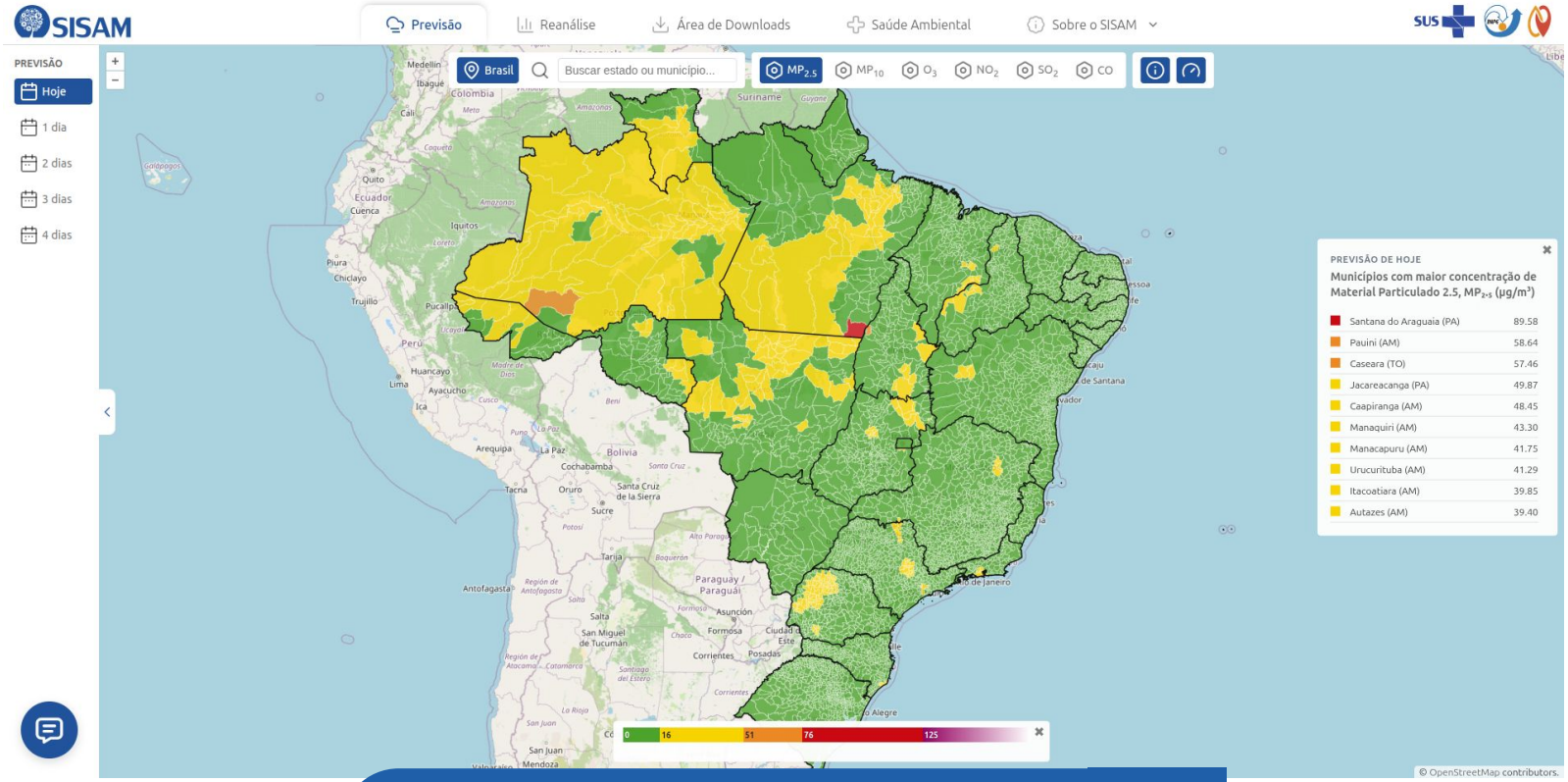


data.inpe.br/queimadas

Queimadas Program: BDQueimadas

data.inpe.br/queimadas/bdqueimadas

Wildfire Program: Environmental Information System Integrated with Health



data.inpe.br/queimadas/sisam

BIG Tech Talks

https://data.inpe.br/bigtechtalks

WEBINAR

USO DE PRODUTOS DE METEOROLOGIA E SENSORIAMENTO REMOTO

21 DE FEVEREIRO DE 2025
13:30 ÀS 16:30H

> ALEX DE ALMEIDA FERNANDES
> Divisão de Previsão de Tempo e Clima (DIPTC)_

Esta palestra abordará a estrutura e os formatos desses dados, além de sua visualização no BDC-Lab com Metview e outras ferramentas.

INSCRIÇÃO GRATUITA AQUI!

3º BIG Tech Talks

WEBINAR

ACESSO, VISUALIZAÇÃO E PROCESSAMENTO DE IMAGENS SENTINEL-2 UTILIZANDO PYTHON

25 DE ABRIL
14:00 ÀS 15:30H

> GILBERTO QUEIROZ
Programa BIG/INPE
&
RENNAN MARUJO
Projeto Brazil Data Cube

2º BIG Tech Talks

WEBINAR

ACESSO E PROCESSAMENTO DE IMAGENS GOES UTILIZANDO STAC E PYTHON

28 DE MARÇO
14:00 ÀS 15:30H

> DOUGLAS UBA & DIEGO SOUZA
> Tecnologistas do INPE
>
> Divisão de Satélites e Sensores Meteorológicos (DISSM)_

5º BIG Tech Talks

WEBINAR

TOPODATA - Acesso ao Banco de Dados Geomorfométricos do Brasil com Python e QGIS

27 DE FEVEREIRO
14:00 ÀS 15:30H

> ÉDIPO HENRIQUE CREMON
&
MÁRCIO DE MORRISON VALERIANO
DIOTG/INPE

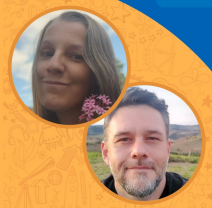
6º BIG Tech Talks

WEBINAR

Modelagem de Radiação Solar com Imagens de Satélite e Acesso aos Dados utilizando STAC e Python

10 DE ABRIL
14:00 ÀS 15:30H

> SIMONE SIEVERT
&
DOUGLAS UBA
DISSM/INPE



4º BIG Tech Talks

WEBINAR

Mapeamento de parâmetros de qualidade de água com Sentinel-2 utilizando STAC, Python e R

23 DE MAIO
14:00 ÀS 15:30H

> Daniel Maciel
Pós-doc Laboratório de Instrumentação de Sistemas Aquáticos
LabISA/INPE

7º BIG Tech Talks

WEBINAR

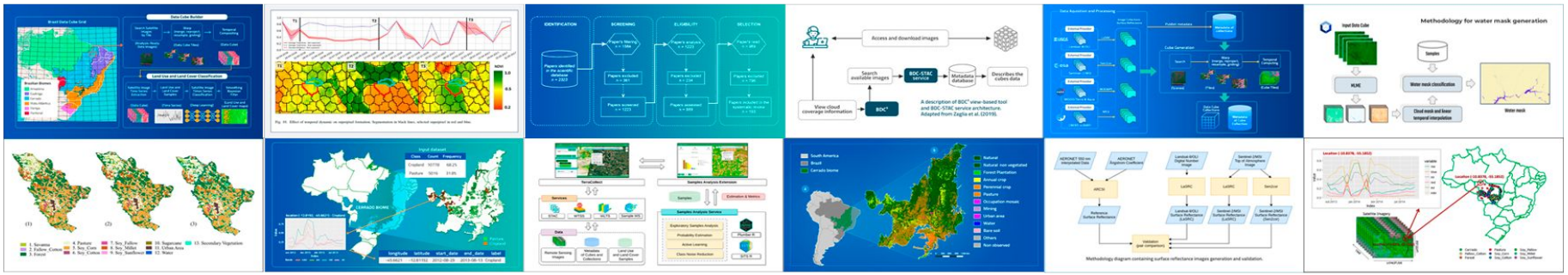
Sentinel-1 com Correção Radiométrica do Terreno: acesso e uso dos dados com Python

24 DE SETEMBRO
14:00 ÀS 15:30H

> CARINA DE MACEDO
&
TÈHRRIE KÖNIG
DIOTG/INPE

+50 PUBLICATIONS

29 – Journals and international events
21 – Journals and national events



https://data.inpe.br/bdc

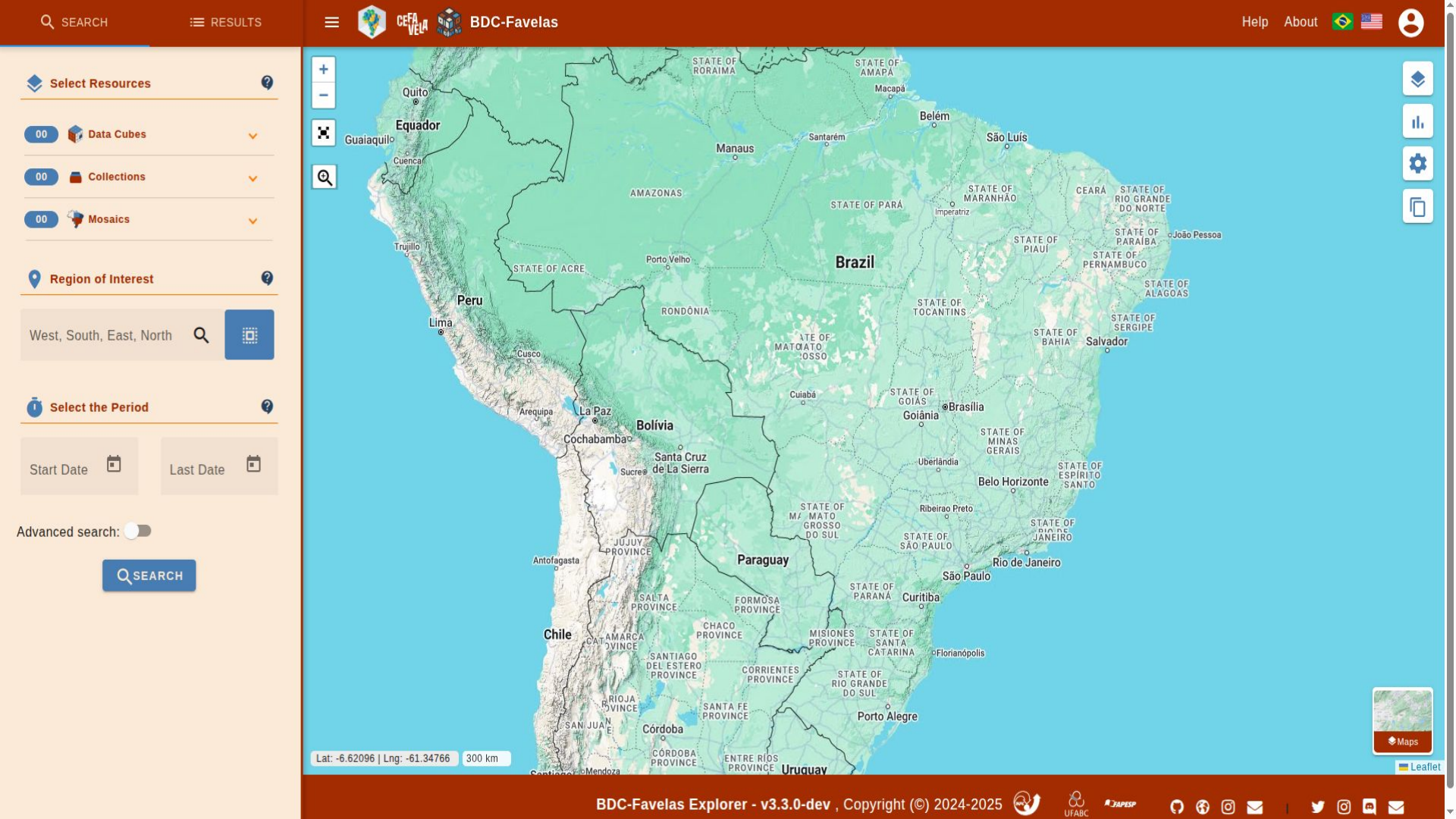
BIG/BDC - constant evolution…

BDC-Favelas

National Technology in the Service of Popular Territories

A **data cube** of products derived from satellite and drone imagery for Brazilian favelas

Public, free, and open access via a web portal, with data collections ready for analysis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BDC-Favelas
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Select Resources
00 Data Cubes
00 Collections
00 Mosaics
Region of Interest
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Last Date
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Cuenca
Lima
Trujillo
Cusco
Arequipa
La Paz
Cochabamba
Santa Cruz de La Sierra
Sucre
Antofagasta
Manaus
Belém
São Luís
Porto Velho
Cuiabá
Goiânia
Brasília
Salvador
Belo Horizonte
Rio de Janeiro
São Paulo
Curitiba
Florianópolis
Porto Alegre
Córdoba
Mendoza
Santiago
Ribeirao Preto
Uberlândia
Imperatriz
Santarém
Macapá
João Pessoa
STATE OF AMAZONAS
STATE OF PARÁ
STATE OF ACRE
RONDÔNIA
STATE OF TOCANTINS
STATE OF MARANHÃO
STATE OF PIAUÍ
CEARA
STATE OF RIO GRANDE DO NORTE
STATE OF PARAÍBA
STATE OF PERNAMBUCO
STATE OF ALAGOAS
STATE OF SERGIPE
STATE OF BAHIA
STATE OF GOIÁS
STATE OF MINAS GERAIS
STATE OF ESPÍRITO SANTO
STATE OF RIO DE JANEIRO
STATE OF SÃO PAULO
STATE OF PARANÁ
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ENTRE RIOS PROVINCE
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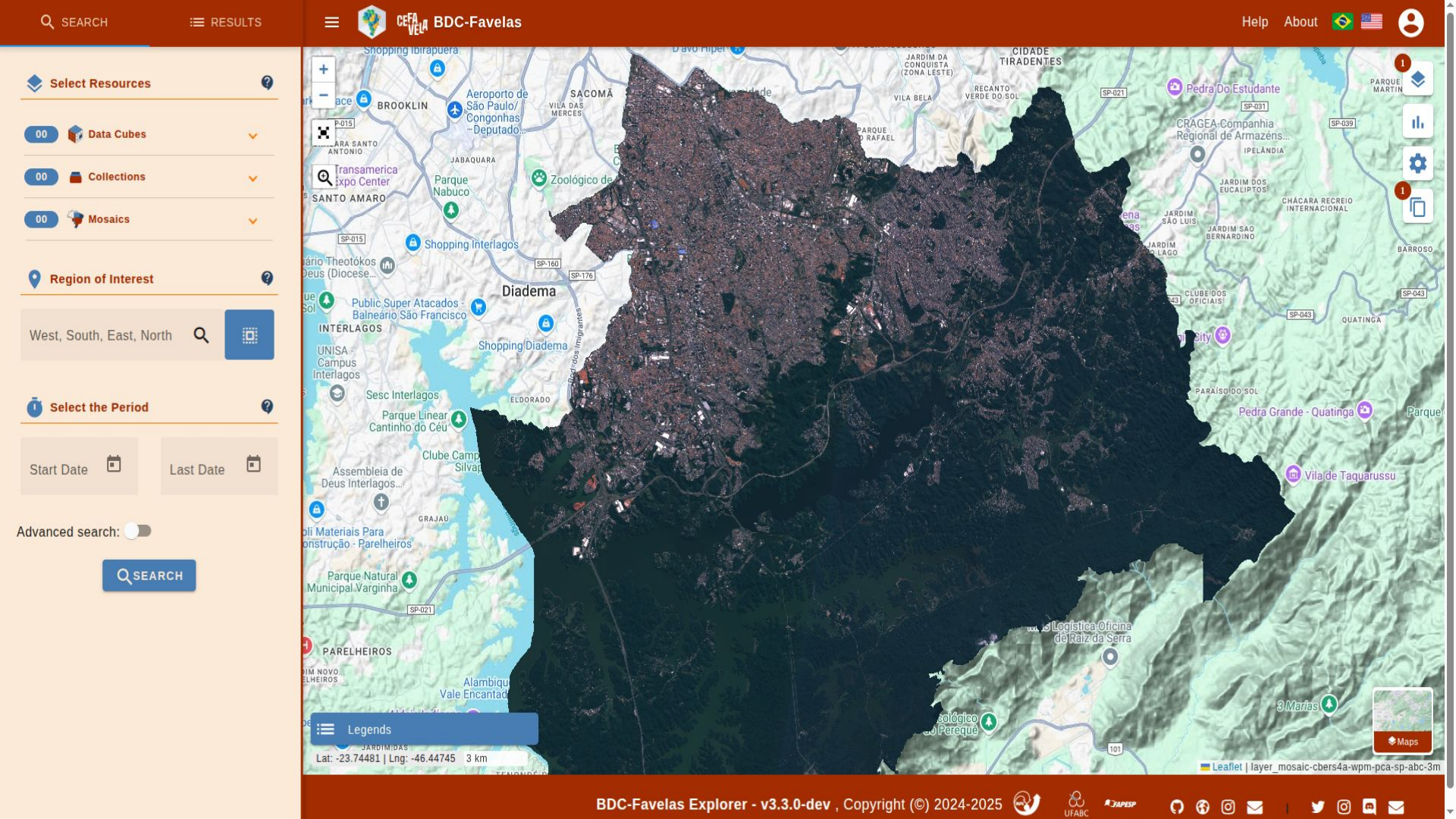
Leaflet
BDC-Favelas Explorer - v3.3.0-dev , Copyright (©) 2024-2025
UFABC
FAPESP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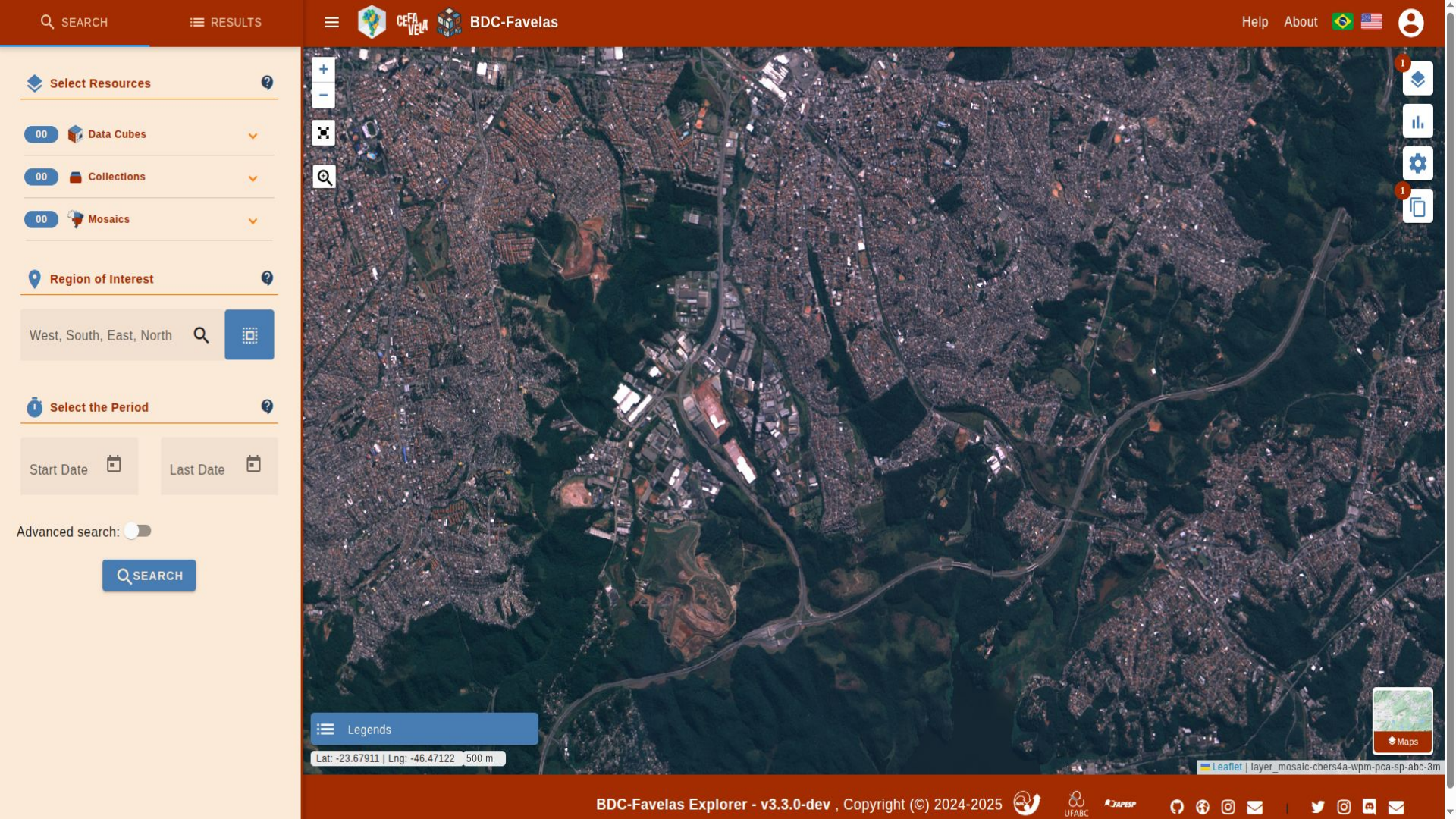
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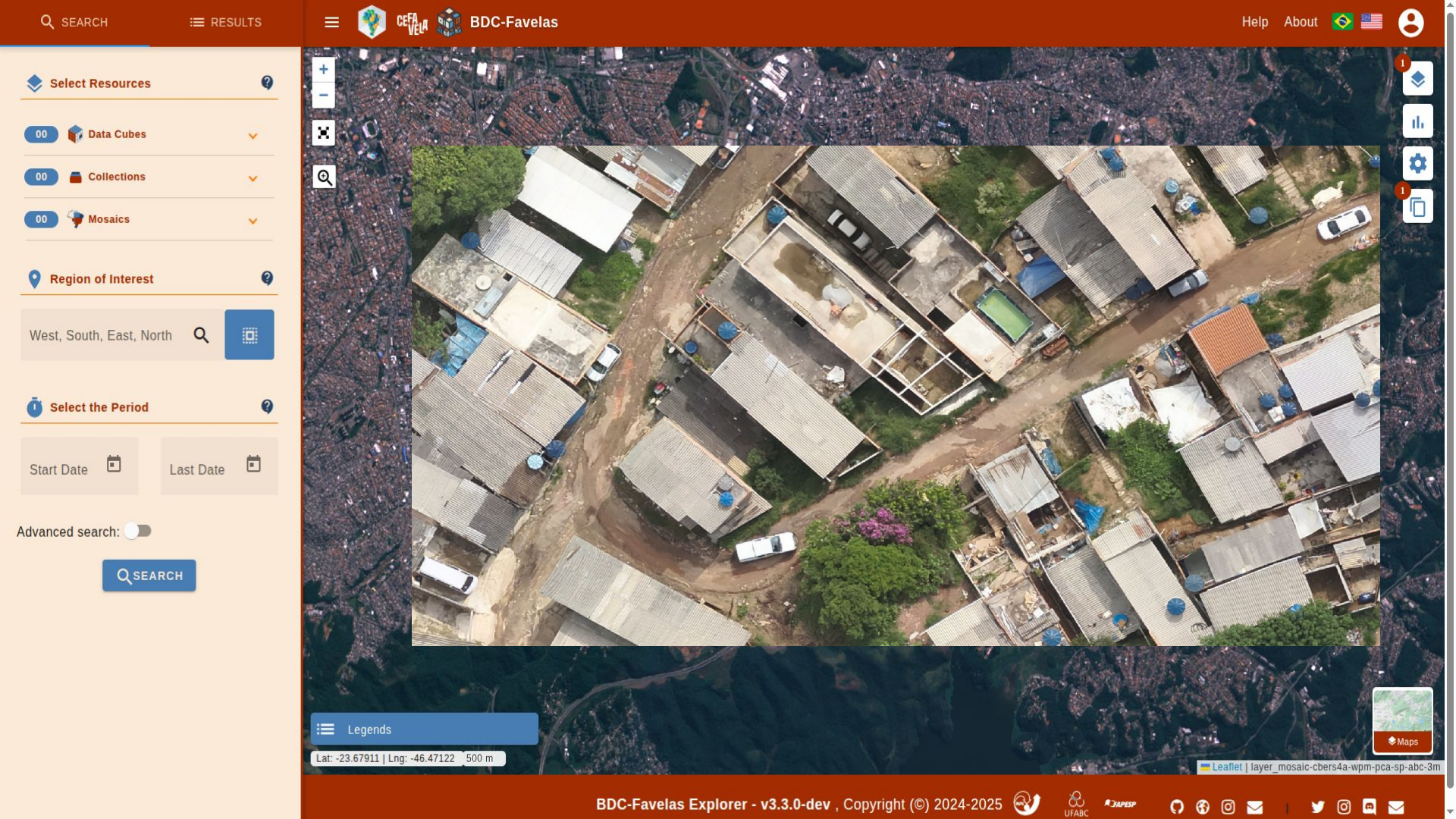
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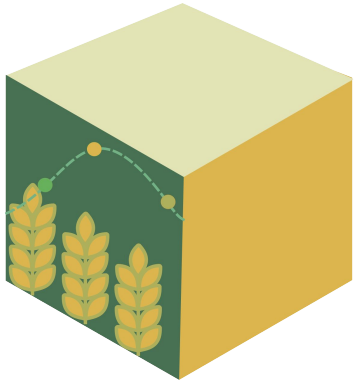
UFABC

BDC-Agri

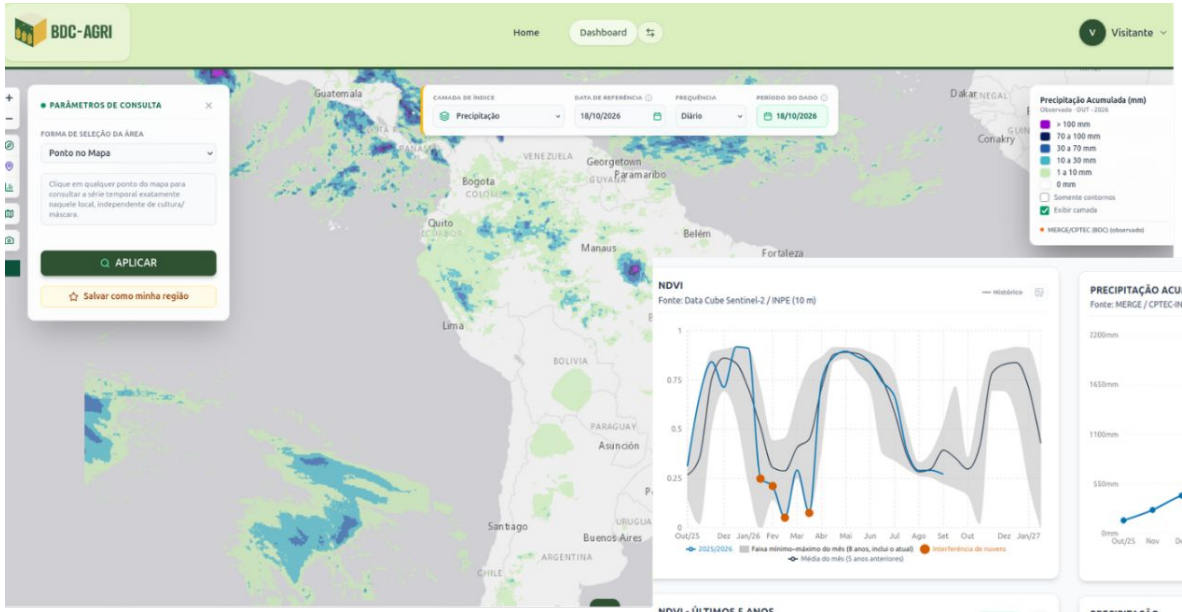
Open platform for Large-scale agricultural monitoring

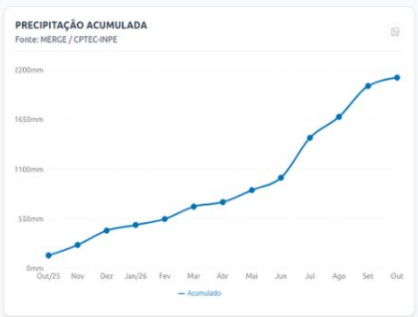


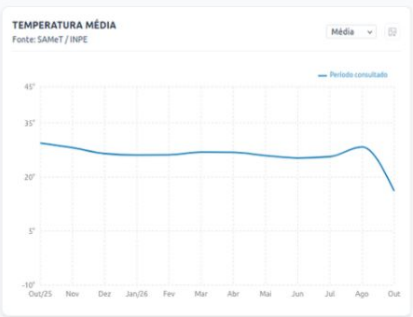
Earth observation data providing consistent, reliable, and timely information on crop conditions

BDC-Agri is based on BDC technologies and incorporates a set of independent modules (phenology metrics, data cubes, meteorological data).

Thematic Platforms: BDC-Agri







Final Remarks

Modernization of Digital Development Infrastructure.

Enhancement of product offerings in cloud-native formats.

Expanded support for applications based on AI pipelines.

Geo Data Lakehouse

Integration of new data products:

- LiDAR database
- Disasters Charter
- MONAN
- Lightning data, Sonda network data, oceanographic products
- Sentinel-1, Sentinel-3, Sentinel-5p

Cubos de Dados de Observação da Terra e Análise de Séries Temporais de Imagens

Karine Reis Ferreira
Rennan Marujo
Gabriel Sansigolo
Felipe Menino Carlos

Brazilian National Institute for Space Research (INPE)

07/11/2026